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**eTable 1.** Search strategies for A) MEDLINE, B) Embase, C) Cochrane CENTRAL, PsycINFO, and ERIC

**A) MEDLINE**

| | |
|---|---|
| 1 | Sex Characteristics/ (54058) |
| 2 | sex distribution/ (55145) |
| 3 | Sex Factors/ (263100) |
| 4 | Sexism/ (1967) |
| 5 | (sex or sexes or sexism).tw,kf. (559154) |
| 6 | gender*.tw,kf. (331420) |
| 7 | wom?n*.tw,kf. (1170111) |
| 8 | female*.tw,kf. (959904) |
| 9 | 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 (2625210) |
| 10 | exp Physicians/ (140699) |
| 11 | physician*.tw,kf. (392562) |
| 12 | (medicine or medical).tw,kf. (1578309) |
| 13 | (surgery or surgical or surgeon*).tw,kf. (1912383) |
| 14 | (doctor or doctors).tw,kf. (122123) |
| 15 | (resident or residents).tw,kf. (156159) |
| 16 | (intern or interne or interns or internes).tw,kf. (5853) |
| 17 | (trainee or trainees).tw,kf. (25596) |
| 18 | clinician*.tw,kf. (224670) |
| 19 | 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 (3873810) |
| 20 | Faculty, Medical/ (13263) |
| 21 | exp Academic Medical Centers/ (93396) |
| 22 | (faculty or faculties).tw,kf. (48393) |
| 23 | (academic or academics).tw,kf. (133076) |
| 24 | 20 or 21 or 22 or 23 (253893) |
| 25 | Achievement/ (15369) |
| 26 | career mobility/ (11441) |
| 27 | achievement*.tw,kf. (61273) |
| 28 | (rank or ranks).tw,kf. (110562) |
| 29 | professor*.tw,kf. (21283) |
| 30 | (career or careers).tw,kf. (42059) |
| 31 | (job or jobs).tw,kf. (61905) |
| 32 | (ladder or ladders).tw,kf. (10401) |
| 33 | advancement*.tw,kf. (54164) |
| 34 | promot*.tw,kf. (1017322) |
| 35 | 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 (1364135) |
| 36 | 9 and 19 and 24 and 35 (3465) |
| 37 | limit 36 to (case reports or comment or editorial or letter or meta analysis or "review" or "systematic review") (207) |
| 38 | 36 not 37 (3383) |

**B) Embase**

| | |
|---|---|
| 1 | exp sex difference/ (400581) |
| 2 | sex ratio/ (72668) |
| 3 | sexism/ or gender bias/ (6087) |
| 4 | (sex or sexes or sexism).tw,kw. (794517) |
| 5 | gender*.tw,kw. (535055) |

| | |
|---|---|
| 6 | wom?n*.tw,kw. (1669994) |
| 7 | female*.tw,kw. (1510495) |
| 8 | 1 or 2 or 3 or 4 or 5 or 6 or 7 (3876952) |
| 9 | exp physician/ (790780) |
| 10 | physician*.tw,kw. (571261) |
| 11 | (medicine or medical).tw,kw. (2386371) |
| 12 | (surgery or surgical or surgeon*).tw,kw. (2693032) |
| 13 | (doctor or doctors).tw,kw. (182709) |
| 14 | (resident or residents).tw,kw. (207836) |
| 15 | (intern or interne or interns or internes).tw,kw. (9879) |
| 16 | (trainee or trainees).tw,kw. (42786) |
| 17 | clinician*.tw,kw. (322509) |
| 18 | 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 (5742530) |
| 19 | medical school/ (63562) |
| 20 | exp teaching hospital/ (173020) |
| 21 | (faculty or faculties).tw,kw. (70075) |
| 22 | (academic or academics).tw,kw. (223412) |
| 23 | 19 or 20 or 21 or 22 (478892) |
| 24 | achievement/ (37218) |
| 25 | job performance/ (16941) |
| 26 | career mobility/ (10384) |
| 27 | achievement*.tw,kw. (81730) |
| 28 | (rank or ranks).tw,kw. (188284) |
| 29 | professor*.tw,kw. (23538) |
| 30 | (career or careers).tw,kw. (52124) |
| 31 | (job or jobs).tw,kw. (80710) |
| 32 | (ladder or ladders).tw,kw. (11600) |
| 33 | advancement*.tw,kw. (70446) |
| 34 | promot*.tw,kw. (1278758) |
| 35 | 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 (1777343) |
| 36 | 8 and 18 and 23 and 35 (6798) |
| 37 | limit 36 to embase (3085) |
| 38 | limit 37 to (editorial or letter or "review") (313) |
| 39 | 37 not 38 (2810) |

**C) Cochrane CENTRAL**

| | |
|---|---|
| 1 | Sex Characteristics/ (1241) |
| 2 | sex distribution/ (746) |
| 3 | Sex Factors/ (5904) |
| 4 | Sexism/ (26) |
| 5 | (sex or sexes or sexism).tw. (43624) |
| 6 | gender*.tw. (32982) |
| 7 | wom?n*.tw. (151619) |
| 8 | female*.tw. (90532) |
| 9 | 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 (281682) |
| 10 | exp Physicians/ (1794) |
| 11 | physician*.tw. (41471) |
| 12 | (medicine or medical).tw. (141009) |
| 13 | (surgery or surgical or surgeon*).tw. (197021) |
| 14 | (doctor or doctors).tw. (10694) |
| 15 | (resident or residents).tw. (10895) |

| | |
|---|---|
| 16 | (intern or interne or interns or internes).tw. (631) |
| 17 | (trainee or trainees).tw. (2553) |
| 18 | clinician*.tw. (22755) |
| 19 | 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 (368556) |
| 20 | Faculty, Medical/ (106) |
| 21 | exp Academic Medical Centers/ (1757) |
| 22 | (faculty or faculties).tw. (3115) |
| 23 | (academic or academics).tw. (12131) |
| 24 | 20 or 21 or 22 or 23 (16538) |
| 25 | Achievement/ (481) |
| 26 | career mobility/ (11) |
| 27 | achievement*.tw. (6235) |
| 28 | (rank or ranks).tw. (15788) |
| 29 | professor*.tw. (570) |
| 30 | (career or careers).tw. (763) |
| 31 | (job or jobs).tw. (2806) |
| 32 | (ladder or ladders).tw. (317) |
| 33 | advancement*.tw. (2232) |
| 34 | promot*.tw. (42698) |
| 35 | 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 (70085) |
| 36 | 9 and 19 and 24 and 35 (317) |

**D) PsycINFO**

| | |
|---|---|
| 1 | human sex differences/ (112551) |
| 2 | exp sex role attitudes/ (12493) |
| 3 | sex discrimination/ (2293) |
| 4 | (sex or sexes or sexism).ti,ab,id. (210417) |
| 5 | gender*.ti,ab,id. (221794) |
| 6 | wom?n*.ti,ab,id. (307718) |
| 7 | female*.ti,ab,id. (365668) |
| 8 | 1 or 2 or 3 or 4 or 5 or 6 or 7 (800890) |
| 9 | exp Physicians/ (43420) |
| 10 | physician*.ti,ab,id. (64370) |
| 11 | (medicine or medical).ti,ab,id. (238071) |
| 12 | (surgery or surgical or surgeon*).ti,ab,id. (43436) |
| 13 | (doctor or doctors).ti,ab,id. (27901) |
| 14 | (resident or residents).ti,ab,id. (47303) |
| 15 | (intern or interne or interns or internes).ti,ab,id. (3474) |
| 16 | (trainee or trainees).ti,ab,id. (16486) |
| 17 | clinician*.ti,ab,id. (90454) |
| 18 | 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 (455043) |
| 19 | educational personnel/ (11565) |
| 20 | (faculty or faculties).ti,ab,id. (35800) |
| 21 | (academic or academics).ti,ab,id. (149957) |
| 22 | 19 or 20 or 21 (182798) |
| 23 | achievement/ (9479) |
| 24 | occupational success/ (2867) |
| 25 | occupational mobility/ (1101) |
| 26 | exp occupational tenure/ (1223) |
| 27 | achievement*.ti,ab,id. (111516) |
| 28 | (rank or ranks).ti,ab,id. (23580) |

| | |
|---|---|
| 29 | professor*.ti,ab,id. (15563) |
| 30 | (career or careers).ti,ab,id. (64031) |
| 31 | (job or jobs).ti,ab,id. (91586) |
| 32 | (ladder or ladders).ti,ab,id. (1646) |
| 33 | advancement*.ti,ab,id. (13461) |
| 34 | promot*.ti,ab,id. (191176) |
| 35 | 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 (476581) |
| 36 | 8 and 18 and 22 and 35 (952) |
| 37 | limit 36 to ("comment/reply" or editorial or letter or reviews) (46) |
| 38 | 36 not 37 (907) |

**E) ERIC**

| |
|---|
| NOFT(sex or sexes or sexism or gender* or wom?n* or female*) AND NOFT(physician* or medicine or medical or surgery or surgical or surgeon* or doctor or doctors or resident or residents or intern or interns or interne or internes or trainee or trainees or clinician*) AND NOFT(faculty or faculties or academic or academics) AND NOFT(achievement* or rank or ranks or professor* or career or careers or job or jobs or ladder or ladders or advancement* or promot*) |

**eTable 2a.** Summary of included studies

| Author | Publication year | Region | Data source | Specialty | Data collection year | Sample size (men) | Sample size (women) | Outcomes |
|---|---|---|---|---|---|---|---|---|
| AAMC[1] | 1992 | US | Database | Multiple | 1990 | 51647 | 13580 | Academic rank |
| Abdellatif[2] | 2019 | North America, Asia, Oceania, Europe | Departmental websites | Multiple | 2016 | 976 | 328 | Publications, h-index |
| Ahmadi[3] | 2018 | US, Canada | Departmental websites | Radiology | 2016 | 358 | 107 | Academic rank |
| Akabas[4] | 2019 | US | Survey | Multiple | 2015 | 2546 | 1105 | Federal research funding |
| Aliano[5] | 2020 | Spain | Survey | Anesthesia | 2018 | 654 | 965 | Academic rank, departmental chair |
| Al-Tamimi[6] | 2004 | Saudi Arabia | Survey | Multiple | 2003 | 169 | 64 | Academic rank |
| Amering[7] | 2011 | Global | Database | Psychiatry | 2007 | 3775 | 3775 | Publications |
| Ash[8] | 2004 | US | Survey | Multiple | 1996 | 941 | 873 | Academic rank, salary |
| Awad[9] | 2017 | US | Departmental websites | Urology | 2015 | 137 | 36 | Academic rank, publications, h-index, federal research funding |
| Baptiste[10] | 2017 | US | Survey | General surgery | 2015 | 91 | 28 | Academic rank, tenure |
| Battaglia[11] | 2019 | US, Canada | Departmental websites | Radiology | 2017 | 77 | 22 | Academic rank, h-index |
| Beeler[12] | 2020 | Global | Survey | Radiation oncology | 2018 | 135 | 55 | Academic rank, publications, h-index |
| Bernardi[13] | 2020 | US | Survey | General surgery | 2019 | 22 | 14 | Academic rank |
| Bickel[14] | 1988 | US | Database | Multiple | 1988 | 48535 | 11673 | Academic rank |

| Author | Publication year | Region | Data source | Specialty | Data collection year | Sample size (men) | Sample size (women) | Outcomes |
|---|---|---|---|---|---|---|---|---|
| Bickel[15] | 1993 | US | Database | Multiple | 1993 | 57049 | 17682 | Academic rank, tenure |
| Bickel[16] | 1991 | US | Database | Multiple | 1991 | 2918 | 669 | Academic rank |
| Bissing[17] | 2019 | US | Database | Anesthesia | 2015 | 5236 | 2945 | Academic rank |
| Breyer[18] | 2020 | US | Survey | Urology | 2017 | 2597 | 329 | Academic rank |
| Bruggmann[19] | 2017 | Germany | Database | Obstetrics and gynecology, otolaryngology | 2013 | 12591 | 12857 | Departmental chair |
| Buckley[20] | 2000 | US | Survey | Multiple | 1997 | 215 | 96 | Academic rank, tenure |
| Burns[21] | 2019 | Canada | Survey | Anesthesia | 2018 | 192 | 73 | Academic rank, department chair |
| Camacci[22] | 2020 | US | Departmental websites | Ophthalmology | 2018 | 158 | 60 | Publications, h-index |
| Carnevale[23] | 2020 | US | Departmental websites | Vascular surgery | 2019 | 774 | 177 | Academic rank, publications, h-index, federal research funding, departmental chair |
| Carpenter[24] | 2018 | US | Departmental websites | General surgery | 2017 | 117 | 29 | Academic rank |
| Carr[25] | 1992 | US | Survey | Internal medicine | 1986 | 435 | 145 | Tenure, department chair |
| Carr[26] | 1998 | US | Survey | Multiple | 1995 | 1010 | 953 | Publications |
| Caturegli[27] | 2020 | US | Database | General surgery | 2020 | 50 | 76 | Academic rank, publications, h-index, federal research funding |
| Chaiyachati[28] | 2019 | US | Survey | Internal medicine, pediatrics, family medicine | 2016 | 58 | 104 | Publications |

| Author | Publication year | Region | Data source | Specialty | Data collection year | Sample size (men) | Sample size (women) | Outcomes |
|---|---|---|---|---|---|---|---|---|
| Chandrasekar [29] | 2020 | US, Canada | Departmental websites | Urology | 2019 | 1894 | 320 | Academic rank |
| Chang[30] | 2016 | US | Database | Multiple | 2008 | 43189 | 18642 | Academic rank |
| Chaudhary[31] | 2020 | US | Database | Psychiatry | 2018 | 5283 | 5807 | Academic rank |
| Chauvin[32] | 2019 | Canada | Departmental websites | Psychiatry | 2016 | 1923 | 1456 | Academic rank, publications |
| Cheng[33] | 2019 | US | Departmental websites | Vascular surgery | 2018 | 534 | 108 | Academic rank |
| Cheng[34] | 2019 | US | Survey | Multiple | 2018 | 37 | 35 | Salary |
| Chevreul[35] | 2018 | France | Database | Multiple | 2015 | 4737 | 1679 | Academic rank |
| Chokshi[36] | 2009 | US | Survey | General surgery | 2007 | 275 | 36 | Federal research funding |
| Colletti[37] | 2000 | US | Survey | General surgery | 2000 | 45 | 9 | Tenure |
| Cuoco[38] | 2020 | US | Database | Neurosurgery | 2018 | 74 | 7 | Academic rank, salary |
| Cydulka[39] | 2000 | US | Survey | Emergency medicine | 1997 | 923 | 274 | Academic rank |
| Danic[40] | 2003 | Croatia | Database | Multiple | 2000 | 333 | 168 | Academic rank |
| Day[41] | 2010 | US | Database | Orthopedic surgery | 2006 | 2119 | 327 | Academic rank |
| Deipolyi[42] | 2020 | US | Departmental websites | Interventional radiology | 2019 | 734 | 108 | Academic rank, publications, h-index |
| Dennis[43] | 1990 | US | Survey | Multiple | 1987 | 85 | 51 | Salary |
| Dermody[44] | 2019 | US | Database | Otolaryngology | 2016 | 161 | 49 | Academic rank, h-index, salary |
| Desai[45] | 2018 | US | Departmental websites | General surgery | 2017 | 325 | 105 | H-index |
| Diehl[46] | 2008 | US | Survey | Family medicine | 2006 | 276 | 49 | Salary |

| Author | Publication year | Region | Data source | Specialty | Data collection year | Sample size (men) | Sample size (women) | Outcomes |
|---|---|---|---|---|---|---|---|---|
| Dorismond[47] | 2020 | US | Departmental websites | Otolaryngology | 2019 | 47 | 23 | Academic rank, h-index, federal research funding |
| Dossani[48] | 2019 | US | Database | Neurosurgery | 2019 | 761 | 80 | Academic rank |
| Dubosh[49] | 2019 | US | Departmental websites | Emergency medicine | 2018 | 77 | 32 | Academic rank, federal research funding |
| Duke[50] | 2019 | US | Survey | Multiple | 2020 | 75 | 86 | Academic rank |
| Eloy[51] | 2013 | US | Departmental websites | Multiple | 2012 | 6819 | 3133 | Academic rank, h-index, departmental chair |
| Eloy[52] | 2013 | US | Departmental websites | Otolaryngology | 2012 | 149 | 55 | Federal research funding |
| Epperson[53] | 2020 | US, Canada | Departmental websites | Otolaryngology | 2018 | 343 | 62 | Academic rank, h-index, departmental chair |
| Fassiotto[54] | 2018 | US | Database | Multiple | 2015 | 599 | 467 | Academic rank |
| Florence[55] | 2011 | US | Survey | General surgery | 2006 | 154 | 17 | Academic rank, departmental chair |
| Fritz[56] | 1991 | US | Survey | Pediatrics | 1989 | 22 | 45 | Academic rank |
| Gambhir[57] | 2020 | US | Database | General surgery | 2016 | 120 | 50 | Academic rank, salary |
| Garstka[58] | 2019 | US | Departmental websites | Otolaryngology | 2017 | 579 | 153 | Academic rank, h-index, departmental chair |
| Gauthier[59] | 2013 | Germany, Switzerland, Austria | Survey | Multiple | 2011 | 67 | 62 | Department chair |
| Gawad[60] | 2020 | Canada | Departmental websites | General surgery | 2018 | 294 | 111 | Academic rank, publications |
| Geltzeiler[61] | 2017 | US | Departmental websites | General surgery | 2015 | 281 | 77 | Academic rank, publications |

| Author | Publication year | Region | Data source | Specialty | Data collection year | Sample size (men) | Sample size (women) | Outcomes |
|---|---|---|---|---|---|---|---|---|
| Gerety[62] | 2013 | US, Canada, Asia, South America, Europe | Survey | Plastic surgery | 2012 | 104 | 9 | Academic rank |
| Gerson[63] | 2007 | US | Survey | Gastroenterology | 2006 | 195 | 262 | Academic rank, departmental chair |
| Girod[64] | 2011 | US | Survey | Multiple | 2007 | 1397 | 1077 | Academic rank, publications |
| Gjerde[65] | 1994 | US | Survey | Family medicine | 1989 | 29 | 11 | Academic rank |
| Grimm[66] | 2016 | US | Departmental websites | Radiology | 2015 | 2502 | 1262 | Departmental chairs |
| Guss[67] | 2019 | US | Departmental websites | Radiation oncology | 2017 | 166 | 81 | Academic rank, federal research funding, salary, departmental chair |
| Hafeez[68] | 2019 | Global | Departmental websites | Psychiatry | 2017 | 579 | 257 | Publications, h-index |
| Harrington[69] | 2018 | US | Departmental websites | Radiology | 2017 | 144 | 71 | Academic rank |
| Harris[70] | 2019 | Global | Departmental websites | General surgery | 2017 | 426 | 102 | Academic rank, h-index |
| Herrick-Reynolds[71] | 2019 | US | Database | General surgery | 2017 | 914 | 149 | Academic rank |
| Herzke[72] | 2020 | US | Survey | Family medicine | 2019 | 52 | 14 | Academic rank |
| Higgins[73] | 2020 | US | Survey | Orthopedic surgery | 2020 | 287 | 90 | Academic rank |
| Higgins[74] | 1979 | US | Database | Multiple | 1978 | 23561 | 2648 | Academic rank |
| Hill[75] | 2018 | US | Survey | Obstetrics and gynecology | 2015 | 420 | 257 | Academic rank |
| Holliday[76] | 2015 | US | Survey | Multiple | 2009 | 685 | 582 | Academic rank |

| Author | Publication year | Region | Data source | Specialty | Data collection year | Sample size (men) | Sample size (women) | Outcomes |
|---|---|---|---|---|---|---|---|---|
| Holliday[77] | 2014 | US | Departmental websites | Radiation oncology | 2012 | 738 | 293 | Academic rank, federal research funding |
| Hoof[78] | 2020 | US | Departmental websites | Orthopedic surgery | 2018 | 3889 | 434 | Academic rank |
| Hoops[79] | 2018 | US | Database | General surgery | 2017 | 42 | 16 | Academic rank |
| Hoover[80] | 2005 | US | Database | Multiple | 2004 | 77947 | 35471 | Academic rank |
| Houtrow[81] | 2020 | US | Survey | Physical medicine and rehabilitation | 2017 | 68 | 130 | Academic rank, salary, departmental chair |
| Houtrow[82] | 2011 | US | Survey | Physical medicine and rehabilitation | 2009 | 25 | 61 | Salary |
| Hurley[83] | 2014 | US | Database | Anesthesia | 2008 | 4360 | 1783 | Publications |
| Ibn Auf[84] | 2019 | Sudan | Survey | Multiple | 2017 | 100 | 53 | Academic rank |
| Ilin[85] | 2020 | Canada | Departmental websites | Urology | 2019 | 164 | 22 | Academic rank |
| Inam[86] | 2020 | Pakistan | Survey | General surgery | 2019 | 67 | 32 | Academic rank |
| Ingram[87] | 2015 | US | Departmental websites | Respirology | 2015 | 50 | 35 | Academic rank, h-index |
| Jacquet[88] | 2014 | US | Survey | Emergency medicine | 2012 | 81 | 35 | Academic rank |
| Jagsi[89] | 2013 | US | Survey | Multiple | 2011 | 593 | 419 | Academic rank, salary |
| Jagsi[90] | 2012 | US | Survey | Multiple | 2003 | 553 | 247 | Academic rank, publications, salary |
| Jaque[91] | 2020 | Germany | Database | Urology | 2015 | 1944 | 499 | Departmental chair |
| Jena[92] | 2016 | US | Database | Multiple | 2013 | 6692 | 3549 | Academic rank, publications, federal research funding, salary |

| Author | Publication year | Region | Data source | Specialty | Data collection year | Sample size (men) | Sample size (women) | Outcomes |
|---|---|---|---|---|---|---|---|---|
| Jena[93] | 2015 | US | Database | Multiple | 2014 | 60609 | 30464 | Academic rank, publications, federal research funding |
| John[94] | 2016 | US | Departmental websites | Dermatology | 2015 | 493 | 474 | Academic rank, h-index, departmental chair |
| Johnson[95] | 1992 | US | Database | Multiple | 1992 | 53891 | 15568 | Academic rank |
| Jolly[96] | 2014 | US | Survey | Multiple | 2009 | 612 | 437 | Academic rank |
| Kaderli[97] | 2015 | Switzerland | Survey | General surgery | 2011 | 448 | 64 | Department chair |
| Karimian[98] | 2012 | Iran | Survey | Multiple | 2008 | 121 | 106 | Academic rank |
| Kenevan[99] | 2018 | US | Departmental websites | Anesthesia | 2017 | 101 | 46 | Academic rank, department chair |
| Khan[100] | 2019 | US | Departmental websites | Cardiology | 2017 | 349 | 63 | Academic rank, publications, h-index, federal research funding |
| Khan[101] | 2014 | US | Departmental websites | Neurosurgery | 2012 | 1144 | 81 | H-index |
| Khurshid[102] | 2018 | US, Canada | Departmental websites | Radiology | 2017 | 112 | 258 | Academic rank |
| Kilic[103] | 2020 | US | Survey | Pathology | 2018 | 2436 | 1755 | Academic rank |
| Killaspy[104] | 2003 | UK | Survey | Psychiatry | 2001 | 265 | 66 | Academic rank |
| Kim[105] | 2019 | US | Departmental websites | Urology | 2017 | 327 | 118 | Publications |
| Kirchner[106] | 1984 | US | Survey | Radiology | 1981 | 2407 | 333 | Academic rank |
| Klingensmith[107] | 2006 | US | Survey | General surgery | 2003 | 267 | 57 | Academic rank |
| Kloosterboer[108] | 2020 | US | Departmental websites | Ophthalmology | 2019 | 84 | 32 | Publications, h-index |

| Author | Publication year | Region | Data source | Specialty | Data collection year | Sample size (men) | Sample size (women) | Outcomes |
|---|---|---|---|---|---|---|---|---|
| Komlenac[109] | 2019 | Sweden, Netherlands, Austria | Survey | Multiple | 2012 | 468 | 550 | Departmental chair |
| Larese[110] | 2019 | Italy | Database | Multiple | 2015 | 6044 | 2680 | Academic rank |
| LaRocca[111] | 2018 | US | Departmental websites | General surgery | 2017 | 216 | 92 | Academic rank |
| Levey[112] | 1990 | US | Survey | Internal medicine | 1983 | 7015 | 932 | Academic rank, federal research funding |
| Lewis-Stevenson[113] | 2001 | US | Survey | Family medicine | 1999 | 747 | 395 | Academic rank |
| Long[114] | 2010 | US | Departmental websites | Anesthesia | 2008 | 94 | 37 | Academic rank, department chair |
| Madsen[115] | 2017 | US | Survey | Emergency medicine | 2015 | 924 | 447 | Academic rank, salary, departmental chair |
| Mascarenhas[116] | 2017 | Canada | Database | Multiple | 2014 | 54 | 16 | Academic rank |
| Maxwell[117] | 2020 | US | Database | General surgery | 2016 | 1543 | 450 | Academic rank, h-index, salary |
| Mayer[118] | 2017 | US | Departmental websites | Urology | 2016 | 1686 | 236 | Academic rank, departmental chair, federal research funding |
| McAlister[119] | 2014 | Canada | Survey | Ophthalmology | 2012 | 283 | 102 | Academic rank |
| McDermott[120] | 2018 | US | Departmental websites | Neurology | 2016 | 1184 | 528 | Academic rank, publications, h-index |
| McDonald[121] | 2017 | US | Departmental websites | Radiology | 2016 | 519 | 192 | Academic rank, h-index |
| Mendoza[122] | 2015 | US | Survey | Pediatrics | 2015 | 5104 | 6064 | Departmental chair |
| Modena[123] | 1999 | Italy | Survey | Cardiology | 1996 | 1407 | 294 | Publications |

| Author | Publication year | Region | Data source | Specialty | Data collection year | Sample size (men) | Sample size (women) | Outcomes |
|---|---|---|---|---|---|---|---|---|
| Moghimi[124] | 2019 | US, Canada | Departmental websites | Radiology | 2016 | 179 | 58 | Academic rank |
| Monroe[125] | 2015 | US | Survey | Multiple | 2012 | 293 | 181 | Academic rank, department chair |
| Morgan[126] | 2019 | US | Survey | Obstetrics and gynecology | 2018 | 25 | 87 | Academic rank |
| Morris[127] | 2018 | US | Database | General surgery | 2017 | 61 | 18 | Academic rank |
| Morrison[128] | 2014 | Canada | Database | Internal medicine | 2010 | 398 | 213 | Academic rank |
| Mueller[129] | 2017 | US | Departmental websites | Surgery | 2017 | 160 | 52 | Publications, h-index |
| Muffly[130] | 2016 | US | Database | Anesthesia | 2015 | 760 | 575 | Academic rank |
| Nakata[131] | 2017 | Japan | Database | General surgery | 2015 | 121 | 13 | Academic rank |
| Ngaage[132] | 2019 | US | Departmental websites | Neurosurgery | 2018 | 1350 | 131 | Academic rank, h-index |
| Nguyen[133] | 2018 | US | Database | General surgery | 2017 | 331 | 213 | H-index |
| Nickerson[134] | 1990 | US | Database | Multiple | 1988 | 663 | 278 | Academic rank |
| Niu[135] | 2020 | US | Database | Radiology | 2017 | 5795 | 2256 | Academic rank, departmental chair |
| Nowling[136] | 2018 | US | Database | Multiple | 2012 | 1898 | 1195 | Academic rank |
| Odell[137] | 2019 | US, Canada | Departmental websites | Neurosurgery | 2017 | 1592 | 219 | Academic rank |
| Okafor[138] | 2020 | US | Departmental websites | Otolaryngology | 2020 | 1342 | 412 | Academic rank |
| Okoshi[139] | 2014 | Japan | Database | General surgery | 2013 | 326 | 40 | Academic rank |
| O'Neill[140] | 2019 | US, Canada | Departmental websites | Radiology | 2018 | 281 | 129 | Academic rank |
| Orhurhu[141] | 2020 | US | Departmental websites | Anesthesia | 2018 | 520 | 176 | Academic rank |

| Author | Publication year | Region | Data source | Specialty | Data collection year | Sample size (men) | Sample size (women) | Outcomes |
|---|---|---|---|---|---|---|---|---|
| Pace[142] | 2020 | US | Departmental websites | Urology | 2018 | 1433 | 224 | Academic rank, publications, h-index |
| Pagel[143] | 2015 | US | Database | Anesthesia | 2015 | 315 | 82 | Publications, h-index, federal research funding |
| Paik[144] | 2014 | US | Departmental websites | Plastic surgery | 2012 | 426 | 79 | Academic rank, departmental chair |
| Paulus[145] | 2016 | US | Survey | Multiple | 2012 | 130 | 102 | Academic rank |
| Penfold[146] | 1987 | Canada | Survey | Psychiatry | 1985 | 1348 | 225 | Academic rank, departmental chair |
| Pfeiffer[147] | 2016 | Germany | Database | Multiple | 2013 | 385 | 539 | Publications |
| Pololi[148] | 2013 | US | Survey | Multiple | 2009 | 1046 | 1172 | Academic rank |
| Qamar[149] | 2020 | Canada | Departmental websites | Radiology | 2017 | 597 | 335 | Academic rank |
| Qamar[150] | 2018 | US, Canada | Departmental websites | Radiology | 2016 | 190 | 84 | Academic rank, departmental chair |
| Rao[151] | 2018 | US | Database | Multiple | 2005 | 1029 | 452 | Academic rank |
| Raptis[152] | 2012 | North America, Europe | Survey | General surgery | 2011 | 365 | 74 | Publications |
| Rayburn[153] | 2012 | US | Database | Multiple | 2009 | 2181 | 2077 | Tenure |
| Reed[154] | 2011 | US | Database | Multiple | 2007 | 50 | 25 | Academic rank, publications |
| Reiser[155] | 1993 | US | Survey | Psychiatry | 1984 | 235 | 45 | Publications |
| Riaz[156] | 2020 | US | Database | Hematology | 2017 | 1310 | 726 | Academic rank, publications, h-index, federal research funding, departmental chair |

| Author | Publication year | Region | Data source | Specialty | Data collection year | Sample size (men) | Sample size (women) | Outcomes |
|---|---|---|---|---|---|---|---|---|
| Robboy[157] | 2017 | US | Database | Pathology | 2016 | 35 | 24 | Publications, tenure |
| Rosso[158] | 2019 | France | Database | Multiple | 2019 | 172 | 150 | Academic rank |
| Rotbart[159] | 2012 | US | Database | Pediatrics | 2009 | 145 | 118 | Academic rank, departmental chair, tenure |
| Sadeghpour[160] | 2012 | US | Survey | Dermatology | 2009 | 159 | 100 | Academic rank, salary, tenure |
| Sandhu[161] | 2007 | UK | Survey | Multiple | 2005 | 2649 | 716 | Academic rank |
| Santosa[162] | 2019 | US | Departmental websites | Plastic surgery | 2016 | 241 | 41 | Academic rank, publications |
| Sasor[163] | 2018 | US | Departmental websites | Plastic surgery | 2017 | 678 | 136 | Academic rank, federal research funding |
| Schnapp[164] | 2003 | US | Survey | Respirology | 1999 | 3364 | 674 | Academic rank |
| Schroen[165] | 2004 | US | Survey | General surgery | 1999 | 168 | 149 | Academic rank, publications, tenure |
| Schwarz[166] | 2016 | Germany | Survey | General surgery | 2014 | 200 | 93 | Academic rank, department chair |
| Selter[167] | 2020 | US | Departmental websites | Obstetrics and gynecology | 2018 | 129 | 134 | Academic rank, departmental chair |
| Shah[168] | 2018 | US, Canada | Departmental websites | Dermatology | 2017 | 852 | 784 | Academic rank |
| Shaikh[169] | 2019 | Global | Database | Neurosurgery | 2019 | 294 | 34 | Academic rank |
| Sheikh[170] | 2018 | US | Departmental websites | Psychiatry | 2018 | 3053 | 2181 | Academic rank |
| Shiber[171] | 2020 | US | Survey | Obstetrics and gynecology | 2019 | 41 | 88 | Salary |
| Shollen[172] | 2009 | US | Survey | Multiple | 2005 | 244 | 110 | Academic rank |
| Singh[173] | 2008 | US | Survey | Gastroenterology | 2015 | 168 | 25 | Salary |

| Author | Publication year | Region | Data source | Specialty | Data collection year | Sample size (men) | Sample size (women) | Outcomes |
|---|---|---|---|---|---|---|---|---|
| Smith[174] | 2019 | US | Departmental websites | Plastics | 2018 | 663 | 159 | Academic rank, departmental chair |
| Soothill[175] | 2010 | UK | Survey | Psychiatry | 2008 | 102 | 37 | Academic rank |
| Speck[176] | 2012 | US | Database | Multiple | 2007 | 566 | 335 | Tenure |
| Spencer[177] | 2016 | US | Survey | Urology | 2017 | 660 | 73 | Salary |
| Sperling[178] | 2019 | US | Departmental websites | Obstetrics and gynecology | 2017 | 312 | 347 | Academic rank, federal research funding |
| Stadler[179] | 2017 | Global | Survey | Multiple | 2014 | 149 | 33 | Academic rank |
| Steger[180] | 2009 | Austria | Survey | Multiple | 2005 | 220 | 173 | Publications, departmental chair |
| Stewart[181] | 2020 | US | Departmental websites | Dermatology | 2019 | 178 | 206 | Academic rank, publications, federal research funding |
| Svider[182] | 2014 | US | Database | Ophthalmology | 2013 | 433 | 157 | H-index, federal research funding |
| Tekerek[183] | 2016 | Turkey | Survey | Multiple | 2016 | 12 | 5 | Academic rank |
| Therattil[184] | 2016 | US | Departmental websites | Plastic surgery | 2013 | 491 | 101 | H-index |
| Tracy[185] | 2019 | US | Departmental websites | Otolaryngology | 2019 | 1086 | 274 | Academic rank |
| Tsukada[186] | 2009 | Japan | Survey | Cardiology | 2007 | 58 | 25 | Academic rank |
| Tulunay-Ugur[187] | 2019 | US | Survey | Otolaryngology | 2018 | 152 | 66 | Academic rank |
| Turaga[188] | 2013 | US | Departmental websites | General surgery | 2012 | 87 | 3 | Publications, h-index |
| Valsangkar[189] | 2016 | US | Departmental websites | General surgery | 2016 | 3087 | 928 | Academic rank, publications, federal research |

| Author | Publication year | Region | Data source | Specialty | Data collection year | Sample size (men) | Sample size (women) | Outcomes |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | funding, departmental chair |
| Villablanca[190] | 2017 | US | Survey | Multiple | 2010 | 552 | 322 | Academic rank |
| Vydareny[191] | 2000 | US | Survey | Radiology | 1997 | 494 | 213 | Academic rank, tenure |
| Wagner[192] | 2007 | US | Survey | Physical rehabilitation and medicine | 2007 | 164 | 107 | Academic rank, federal research funding, tenure |
| Waisbren[193] | 2008 | US | Database | Multiple | 2003 | 3557 | 1884 | Academic rank |
| Waljee[194] | 2015 | US | Survey | Multiple | 2012 | 420 | 283 | Academic rank |
| Wallis[195] | 1981 | US | Survey | Multiple | 1978 | 949 | 112 | Academic rank |
| Warner[196] | 2017 | US | Database | Multiple | 2015 | 3010 | 2435 | Publications, h-index |
| Waseem[197] | 2019 | Global | Departmental websites | Internal medicine | 2018 | 221 | 89 | Academic rank |
| Wasserstein[198] | 2007 | US | Survey | Multiple | 2003 | 1062 | 370 | Academic rank |
| Weaver[199] | 2015 | US | Survey | Family medicine | 2010 | 513 | 263 | Salary |
| Whitley[200] | 1987 | US | Survey | Radiology | 1986 | 999 | 200 | Academic rank, publications |
| Wiler[201] | 2019 | US | Survey | Emergency medicine | 2017 | 1430 | 797 | Salary |
| Wilkes[202] | 2015 | UK, Ireland | Departmental websites | Neurosurgery | 2014 | 290 | 25 | H-index |
| Wilkinson[203] | 1986 | US | Database | Anesthesia | 1984 | 1388 | 374 | Academic rank |
| Willett[204] | 2015 | US | Survey | Internal medicine | 2012 | 169 | 72 | Academic rank |
| Wilson[205] | 2013 | US | Departmental websites | Radiation oncology | 2012 | 66 | 21 | Academic rank |
| Winch[206] | 2017 | UK | Survey | Pediatrics | 2015 | 60 | 26 | Academic rank |

| Author | Publication year | Region | Data source | Specialty | Data collection year | Sample size (men) | Sample size (women) | Outcomes |
|---|---|---|---|---|---|---|---|---|
| Wise[207] | 2004 | Canada | Survey | Obstetrics and gynecology | 2002 | 187 | 112 | Academic rank |
| Woods[208] | 2018 | US | Database | Multiple | 2017 | 1163 | 988 | Academic rank |
| Woodward[209] | 2017 | US | Departmental websites | Gastroenterology | 2015 | 118 | 27 | Academic rank |
| Wright[210] | 2007 | US | Database | Multiple | 2000 | 283 | 110 | Academic rank |
| Wu[211] | 2019 | North America, Europe, Oceania | Departmental websites | General surgery | 2017 | 220 | 48 | Academic rank |
| Xierali[212] | 2020 | US | Database | Family medicine | 2017 | 2887 | 2865 | Tenure |
| Yang[213] | 2019 | US, Canada | Departmental websites | Physical medicine and rehabilitation | 2017 | 653 | 392 | Academic rank |
| Yasukawa[214] | 2014 | Japan | Survey | Multiple | 2011 | 195 | 381 | Academic rank |
| Yu[215] | 2013 | US | Database | Multiple | 2008 | 82491 | 42804 | Academic rank |
| Yue[216] | 2020 | Canada | Departmental websites | Orthopedic surgery | 2019 | 618 | 95 | Academic rank |
| Zelnik[217] | 2018 | Israel | Survey | Internal medicine | 2016 | 58 | 8 | Academic rank |
| Zivin[218] | 2020 | US | Survey | Multiple | 2017 | 683 | 627 | Academic rank |

* AAMC (American Association of Medical Colleges), US (United States), UK (United Kingdom)

**eTable 2b.** Number of included studies by country of publication

| Country | Number of included studies |
|---|---|
| Austria | 1 |
| Canada | 11 |
| Croatia | 1 |
| France | 2 |
| Germany | 5 |
| International | 8 |
| Iran | 1 |
| Israel | 1 |
| Italy | 2 |
| Japan | 4 |
| North America | 14 |
| Pakistan | 1 |
| Saudi Arabia | 1 |
| Sudan | 1 |
| Sweden | 1 |
| Switzerland | 1 |
| Turkey | 1 |
| UK | 5 |
| US | 157 |

*UK (United Kingdom), US (United States)

**eTable 3.** Demographic data reported by 90 studies

| Author | | Age (years) | Race (white) | Years in practice | Hours worked per week | % time conducting research | Married | Have children | Additional PhD |
|---|---|---|---|---|---|---|---|---|---|
| Aliano 2020 | Men | - | - | - | - | - | 71.5% | 82.2% | 21.3% |
| | Women | - | - | - | - | - | 62.2% | 63.0% | 14.5% |
| Ash 2004 | Men | 47.0 | 80.0% | - | 58.0 | 28.0% | - | - | - |
| | Women | 45.0 | 82.0% | - | 56.0 | 29.0% | - | - | - |
| Awad 2017 | Men | - | - | 9.8 | - | - | - | - | - |
| | Women | - | - | 9.5 | - | - | - | - | - |
| Baptiste 2017 | Men | - | - | - | - | - | 92.3% | 94.1% | - |
| | Women | - | - | - | - | - | 71.4% | 90.0% | - |
| Beeler 2020 | Men | - | 65.9% | - | - | 21.5% | - | - | - |
| | Women | - | 63.6% | - | - | 12.7% | - | - | - |
| Bernardi 2020 | Men | - | - | - | - | - | 91.0% | 77.0% | - |
| | Women | - | - | - | - | - | 71.0% | 71.0% | - |
| Buckley 2000 | Men | 48.0 | - | 12.0 | - | - | - | - | - |
| | Women | 41.0 | - | 7.0 | - | - | - | - | - |
| Burns 2019 | Men | 45.4 | - | 12.9 | - | - | 89.1% | 57.1% | - |
| | Women | 43.6 | - | 11.1 | - | - | 76.4% | 43.8% | - |
| Camacci 2020 | Men | - | - | - | - | - | - | - | 7.0% |
| | Women | - | - | - | - | - | - | - | 15.0% |

| Author | | Age (years) | Race (white) | Years in practice | Hours worked per week | % time conducting research | Married | Have children | Additional PhD |
|---|---|---|---|---|---|---|---|---|---|
| Carr 1992 | Men | 40.0 | 97.0% | - | 58.0 | 18.0% | - | - | - |
| | Women | 37.0 | 93.0% | - | 55.0 | 14.0% | - | - | - |
| Carr 1998 | Men | 47.0 | 87.0% | 13.0 | 59.0 | 28.0% | 88.0% | 84.0% | - |
| | Women | 45.0 | 91.0% | 11.0 | 55.0 | 28.1% | 74.0% | 69.0% | - |
| Chaiyachati 2019 | Men | - | 91.0% | - | - | - | - | - | - |
| | Women | - | 89.0% | - | - | - | - | - | - |
| Chauvin 2019 | Men | - | - | 20.0 | - | - | - | - | - |
| | Women | - | - | 15.3 | - | - | - | - | - |
| Colletti 2000 | Men | - | - | - | - | - | 100.0% | 99.0% | 11.0% |
| | Women | - | - | - | - | - | 89.0% | 44.0% | 0.0% |
| Cydulka 2000 | Men | 40.7 | 86.0% | - | - | - | - | - | - |
| | Women | 38.5 | 84.0% | - | - | - | - | - | - |
| Deipolyi 2020 | Men | - | - | 20.0 | - | - | - | - | - |
| | Women | - | - | 16.0 | - | - | - | - | - |
| Dennis 1990 | Men | - | - | - | - | - | 65.0% | - | - |
| | Women | - | - | - | - | - | 86.0% | - | - |
| Dermody 2019 | Men | - | - | 24.6 | - | - | - | - | - |
| | Women | - | - | 18.4 | - | - | - | - | - |
| Dorismond 2020 | Men | - | - | - | - | - | - | - | 23.4% |
| | Women | - | - | - | - | - | - | - | 26.1% |

| Author | | Age (years) | Race (white) | Years in practice | Hours worked per week | % time conducting research | Married | Have children | Additional PhD |
|---|---|---|---|---|---|---|---|---|---|
| Dossani 2019 | Men | - | - | - | - | - | - | - | 13.9% |
| | Women | - | - | - | - | - | - | - | 18.6% |
| Epperson 2020 | Men | - | - | 20.3 | - | - | - | - | - |
| | Women | - | - | 13.9 | - | - | - | - | - |
| Fassiotto 2018 | Men | - | 63.6% | - | - | - | - | - | - |
| | Women | - | 54.2% | - | - | - | - | - | - |
| Florence 2011 | Men | 48.7 | - | 15.8 | - | - | 88.0% | 90.5% | - |
| | Women | 43.7 | - | 6.8 | - | - | 88.0% | 64.7% | - |
| Fritz 1991 | Men | - | - | - | 52.0 | 25.0% | 95.0% | - | - |
| | Women | - | - | - | 48.0 | 15.0% | 70.0% | - | - |
| Gauthier 2013 | Men | 45.6 | - | - | - | - | 68.6% | 65.2% | - |
| | Women | 36.6 | - | - | - | - | 77.4% | 35.5% | - |
| Gawad 2020 | Men | - | - | 18.5 | - | - | - | - | 9.5% |
| | Women | - | - | 13.5 | - | - | - | - | 12.6% |
| Geltzeiler 2017 | Men | - | - | 18.0 | - | - | - | - | - |
| | Women | - | - | 11.0 | - | - | - | - | - |
| Gerety 2013 | Men | 53.6 | - | - | - | - | - | - | - |
| | Women | 45.1 | - | - | - | - | - | - | - |
| Gerson 2007 | Men | 46.0 | 66.0% | 8.6 | - | - | - | - | - |
| | Women | 42.0 | 62.0% | 6.5 | - | - | - | - | - |

| Author | | Age (years) | Race (white) | Years in practice | Hours worked per week | % time conducting research | Married | Have children | Additional PhD |
|---|---|---|---|---|---|---|---|---|---|
| Girod 2011 | Men | 49.9 | 96.6% | 21.3 | - | - | - | - | - |
| | Women | 47.2 | 92.9% | 16.8 | - | - | - | - | - |
| Guss 2019 | Men | - | - | 12.0 | - | - | - | - | 26.5% |
| | Women | - | - | 10.0 | - | - | - | - | 18.5% |
| Harris 2019 | Men | - | - | 23.0 | - | - | - | - | 21.9% |
| | Women | - | - | 17.0 | - | - | - | - | 36.1% |
| Higgins 2020 | Men | - | - | - | - | - | 92.7% | 94.4% | - |
| | Women | - | - | - | - | - | 68.9% | 71.1% | - |
| Higgins 1979 | Men | - | - | - | - | - | - | - | 7.8% |
| | Women | - | - | - | - | - | - | - | 3.9% |
| Hill 2018 | Men | - | 89.0% | - | - | - | 93.0% | 91.0% | - |
| | Women | - | 78.0% | - | - | - | 87.0% | 72.0% | - |
| Holliday 2015 | Men | 40.0 | 70.4% | - | - | - | 92.2% | 82.0% | 30.4% |
| | Women | 40.5 | 70.6% | - | - | - | 87.4% | 76.6% | 15.5% |
| Holliday 2014 | Men | - | - | 17.0 | - | - | - | - | 17.2% |
| | Women | - | - | 12.0 | - | - | - | - | 9.2% |
| Houtrow 2020 | Men | - | - | - | - | - | - | - | 23.9% |
| | Women | - | - | - | - | - | - | - | 14.6% |
| Ibn 2019 | Men | 42.6 | - | 9.1 | - | - | 86.9% | - | 10.2% |
| | Women | 38.8 | - | 6.2 | - | - | 64.2% | - | 11.3% |

| Author | | Age (years) | Race (white) | Years in practice | Hours worked per week | % time conducting research | Married | Have children | Additional PhD |
|---|---|---|---|---|---|---|---|---|---|
| Inam 2020 | Men | - | - | - | - | - | 74.6% | - | - |
| | Women | - | - | - | - | - | 43.8% | - | - |
| Jagsi 2013 | Men | 40.5 | 68.8% | 4.6 | 59.4 | 65.7% | 92.9% | 83.0% | 34.4% |
| | Women | 39.9 | 68.8% | 4.7 | 54.0 | 65.4% | 89.0% | 79.5% | 21.0% |
| Jagsi 2012 | Men | 45.0 | 75.1% | - | 63.2 | 58.0% | 91.0% | 89.3% | 49.2% |
| | Women | 45.3 | 76.5% | - | 58.1 | 57.0% | 87.0% | 84.2% | 48.6% |
| Jena 2016 | Men | 51.5 | - | 19.4 | - | - | - | - | - |
| | Women | 46.4 | - | 14.6 | - | - | - | - | - |
| Jena 2015 | Men | 52.3 | - | 20.8 | - | - | - | - | - |
| | Women | 46.8 | - | 15.1 | - | - | - | - | - |
| Jolly 2014 | Men | 40.6 | 68.3% | - | 59.0 | - | 92.5% | 83.2% | - |
| | Women | 40.0 | 66.4% | - | 54.0 | - | 88.8% | 79.2% | - |
| Kaderli 2015 | Men | 50.0 | - | - | - | - | 83.3% | 84.8% | - |
| | Women | 43.0 | - | - | - | - | 76.6% | 42.2% | - |
| Kenevan 2018 | Men | 52.0 | - | - | - | - | - | - | - |
| | Women | 52.5 | - | - | - | - | - | - | - |
| Khan 2019 | Men | - | - | 24.0 | - | - | - | - | 8.6% |
| | Women | - | - | 21.0 | - | - | - | - | 7.9% |
| Klingensmith 2006 | Men | 49.9 | - | - | - | - | - | - | - |
| | Women | 43.5 | - | - | - | - | - | - | - |

| Author | | Age (years) | Race (white) | Years in practice | Hours worked per week | % time conducting research | Married | Have children | Additional PhD |
|---|---|---|---|---|---|---|---|---|---|
| Komlenac 2019 | Men | - | - | - | 54.0 | - | - | 64.0% | - |
| | Women | - | - | - | 43.0 | - | - | 39.0% | - |
| Long 2010 | Men | 53.0 | - | - | - | - | - | - | - |
| | Women | 51.0 | - | - | - | - | - | - | - |
| Madsen 2017 | Men | - | 77.8% | 12.0 | - | - | - | - | - |
| | Women | - | 77.8% | 8.4 | - | - | - | - | - |
| Maxwell 2020 | Men | - | - | 26.3 | - | - | - | - | - |
| | Women | - | - | 18.1 | - | - | - | - | - |
| Mayer 2017 | Men | - | - | 22.0 | - | - | - | - | 2.8% |
| | Women | - | - | 13.0 | - | - | - | - | 0.3% |
| McAlister 2014 | Men | - | - | - | - | - | - | 90.0% | - |
| | Women | - | - | - | - | - | - | 81.0% | - |
| Modena 1999 | Men | - | - | - | 39.0 | - | 87.0% | - | - |
| | Women | - | - | - | 37.7 | - | 66.0% | - | - |
| Monroe 2015 | Men | 49.0 | - | - | - | - | - | - | - |
| | Women | 43.0 | - | - | - | - | - | - | - |
| Mueller 2017 | Men | - | - | 14.3 | - | - | - | - | - |
| | Women | - | - | 12.6 | - | - | - | - | - |
| Muffly 2016 | Men | 50.9 | - | - | - | - | - | - | - |
| | Women | 47.5 | - | - | - | - | - | - | - |

| Author | | Age (years) | Race (white) | Years in practice | Hours worked per week | % time conducting research | Married | Have children | Additional PhD |
|---|---|---|---|---|---|---|---|---|---|
| Ngaage 2019 | Men | - | - | 10.0 | - | - | - | - | - |
| | Women | - | - | 4.0 | - | - | - | - | - |
| Pagel 2015 | Men | - | - | 20.0 | - | - | - | - | - |
| | Women | - | - | 16.0 | - | - | - | - | - |
| Paulus 2016 | Men | 51.0 | - | 19.0 | - | - | - | - | - |
| | Women | 45.0 | - | 14.0 | - | - | - | - | - |
| Reed 2011 | Men | - | - | 26.6 | - | - | - | - | - |
| | Women | - | - | 26.6 | - | - | - | - | - |
| Riaz 2020 | Men | - | - | 20.0 | - | - | - | - | - |
| | Women | - | - | 14.0 | - | - | - | - | - |
| Robboy 2017 | Men | - | 71.4% | - | - | - | - | - | - |
| | Women | - | 87.5% | - | - | - | - | - | - |
| Sadeghpour 2012 | Men | 49.9 | - | - | 52.9 | - | - | - | - |
| | Women | 42.6 | - | - | 50.2 | - | - | - | - |
| Santosa 2019 | Men | - | - | - | - | - | - | - | 11.6% |
| | Women | - | - | - | - | - | - | - | 29.3% |
| Sasor 2018 | Men | - | - | 16.4 | - | - | - | - | 4.3% |
| | Women | - | - | 9.9 | - | - | - | - | 4.4% |
| Schnapp 2003 | Men | 47.0 | - | - | - | - | - | - | - |
| | Women | 42.0 | - | - | - | - | - | - | - |

| Author | | Age (years) | Race (white) | Years in practice | Hours worked per week | % time conducting research | Married | Have children | Additional PhD |
|---|---|---|---|---|---|---|---|---|---|
| Schroen 2004 | Men | 48.1 | - | 15.5 | - | - | 96.0% | 90.0% | - |
| | Women | 42.7 | - | 9.3 | - | - | 77.0% | 62.0% | - |
| Schwarz 2016 | Men | 47.1 | - | - | 65.5 | 9.1% | 93.0% | 86.2% | - |
| | Women | 47.5 | - | - | 59.7 | 15.1% | 74.6% | 77.4% | - |
| Shaikh 2019 | Men | - | - | - | - | - | - | - | 20.4% |
| | Women | - | - | - | - | - | - | - | 17.6% |
| Shollen 2009 | Men | - | - | - | 57.0 | 42.0% | - | - | - |
| | Women | - | - | - | 57.0 | 42.0% | - | - | - |
| Singh 2008 | Men | 45.0 | - | - | 58.0 | - | 95.0% | 97.0% | - |
| | Women | 46.0 | - | - | 53.0 | - | 92.0% | 87.0% | - |
| Smith 2019 | Men | 51.6 | 75.7% | 14.0 | - | - | - | - | 4.3% |
| | Women | 44.6 | 73.4% | 9.0 | - | - | - | - | 3.8% |
| Speck 2012 | Men | 37.3 | 71.9% | - | - | - | 54.4% | - | 15.7% |
| | Women | 36.9 | 66.0% | - | - | - | 44.5% | - | 7.2% |
| Spencer 2016 | Men | 50.0 | - | 19.0 | 60.0 | - | - | - | - |
| | Women | 42.0 | - | 10.0 | 55.0 | - | - | - | - |
| Stadler 2017 | Men | - | - | 9.3 | - | - | - | - | - |
| | Women | - | - | 9.2 | - | - | - | - | - |
| Stewart 2020 | Men | - | 70.2% | 22.4 | - | - | - | - | 27.0% |
| | Women | - | 61.7% | 16.1 | - | - | - | - | 22.8% |

| Author | | Age (years) | Race (white) | Years in practice | Hours worked per week | % time conducting research | Married | Have children | Additional PhD |
|---|---|---|---|---|---|---|---|---|---|
| Tracy 2019 | Men | - | - | 17.2 | - | - | - | - | - |
| | Women | - | - | 10.8 | - | - | - | - | - |
| Tsukada 2009 | Men | - | - | 18.1 | - | - | 85.4% | 69.2% | - |
| | Women | - | - | 10.8 | - | - | 66.7% | 50.0% | - |
| Tulunay-Ugur 2019 | Men | - | 72.0% | - | - | - | 96.8% | - | - |
| | Women | - | 64.2% | - | - | - | 75.0% | - | - |
| Valsangkar 2016 | Men | - | - | - | - | - | - | - | 4.9% |
| | Women | - | - | - | - | - | - | - | 11.3% |
| Wagner 2007 | Men | 47.3 | 81.3% | 15.2 | 59.8 | 26.1% | - | - | 9.8% |
| | Women | 45.1 | 79.0% | 12.4 | 54.3 | 22.2% | - | - | 0.9% |
| Warner 2017 | Men | - | 71.7% | - | - | - | - | - | 9.3% |
| | Women | - | 71.8% | - | - | - | - | - | 4.5% |
| Weaver 2015 | Men | 45.0 | - | 6.0 | - | - | 86.0% | 46.0% | - |
| | Women | 42.0 | - | 5.0 | - | - | 80.0% | 45.0% | - |
| Wise 2004 | Men | 52.8 | - | - | - | - | 96.0% | - | 18.0% |
| | Women | 43.4 | - | - | - | - | 74.0% | - | 28.0% |
| Yasukawa 2014 | Men | - | - | - | - | - | 90.0% | 83.0% | 53.0% |
| | Women | - | - | - | - | - | 72.0% | 78.0% | 40.0% |
| Median across studies | Men | 47.4 | 74.0% | 18.0 | 58.0 | 27.1% | 91.0% | 84.0% | 14.8% |
| | Women | 43.5 | 75.0% | 12.0 | 54.7 | 25.2% | 75.0% | 71.0% | 14.6% |

A dashed line (-) represents values that were not reported, could not be calculated from reported data, and authors were not able to provide.

**eTable 4.** Quality assessment of included studies

| Author | Representativeness of cohort | Adequacy of sample size | Characterization of non-respondents | Ascertainment of exposure | Comparability of groups | Outcome assessment | Statistical tests | NOS Score |
|---|---|---|---|---|---|---|---|---|
| AAMC 1992 | 1 | 1 | 1 | 2 | 0 | 2 | 0 | 7 |
| Abdellatif 2019 | 0 | 1 | 1 | 2 | 1 | 2 | 1 | 8 |
| Ahmadi 2018 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Akabas 2019 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Aliano 2020 | 1 | 1 | 0 | 2 | 1 | 1 | 1 | 7 |
| Al-Tamimi 2004 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Amering 2011 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Ash 2004 | 1 | 1 | 0 | 2 | 2 | 1 | 1 | 8 |
| Awad 2017 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Baptiste 2017 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Battaglia 2019 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Beeler 2020 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Bernardi 2020 | 0 | 0 | 0 | 2 | 0 | 1 | 0 | 3 |
| Bickel 1988 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Bickel 1991 | 1 | 1 | 1 | 2 | 0 | 2 | 0 | 7 |
| Bickel 1993 | 1 | 1 | 1 | 2 | 0 | 2 | 0 | 7 |
| Bissing 2019 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Breyer 2020 | 1 | 1 | 1 | 2 | 0 | 1 | 1 | 7 |

| Author | Representativeness of cohort | Adequacy of sample size | Characterization of non-respondents | Ascertainment of exposure | Comparability of groups | Outcome assessment | Statistical tests | NOS Score |
|---|---|---|---|---|---|---|---|---|
| Bruggmann 2017 | 1 | 1 | 1 | 2 | 0 | 2 | 0 | 7 |
| Buckley 2000 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Burns 2019 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Camacci 2020 | 0 | 1 | 1 | 2 | 0 | 2 | 1 | 7 |
| Carnevale 2020 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Carpenter 2018 | 0 | 1 | 1 | 2 | 0 | 2 | 1 | 7 |
| Carr 1992 | 1 | 1 | 0 | 2 | 2 | 1 | 1 | 8 |
| Carr 1998 | 1 | 1 | 1 | 2 | 2 | 1 | 1 | 9 |
| Caturegli 2020 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Chaiyachati 2019 | 1 | 1 | 0 | 2 | 2 | 1 | 1 | 8 |
| Chandrasekar 2020 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Chang 2016 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Chaudhary 2020 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Chauvin 2019 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 9 |
| Cheng 2019 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Cheng 2019 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Chevreul 2018 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Chokshi 2009 | 0 | 1 | 0 | 2 | 2 | 1 | 1 | 7 |
| Colletti 2000 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |

| Author | Representativeness of cohort | Adequacy of sample size | Characterization of non-respondents | Ascertainment of exposure | Comparability of groups | Outcome assessment | Statistical tests | NOS Score |
|---|---|---|---|---|---|---|---|---|
| Cuoco 2020 | 0 | 1 | 1 | 2 | 0 | 2 | 1 | 7 |
| Cydulka 2000 | 1 | 1 | 0 | 2 | 2 | 1 | 1 | 8 |
| Danic 2003 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Day 2010 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Deipolyi 2020 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Dennis 1990 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Dermody 2019 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Desai 2018 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 9 |
| Diehl 2008 | 0 | 1 | 0 | 2 | 2 | 1 | 1 | 7 |
| Dorismond 2020 | 0 | 1 | 1 | 2 | 0 | 2 | 1 | 7 |
| Dossani 2019 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Dubosh 2019 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Duke 2019 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Eloy 2013 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 9 |
| Eloy 2013 | 0 | 1 | 1 | 2 | 2 | 2 | 1 | 9 |
| Epperson 2020 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Fassiotto 2018 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Florence 2011 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Fritz 1991 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Gambhir 2020 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |

| Author | Representativeness of cohort | Adequacy of sample size | Characterization of non-respondents | Ascertainment of exposure | Comparability of groups | Outcome assessment | Statistical tests | NOS Score |
|---|---|---|---|---|---|---|---|---|
| Garstka 2019 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 9 |
| Gauthier 2013 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Gawad 2020 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Geltzeiler 2017 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 9 |
| Gerety 2013 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Gerson 2007 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Girod 2011 | 0 | 1 | 0 | 2 | 2 | 1 | 1 | 7 |
| Gjerde 1994 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Grimm 2016 | 1 | 1 | 1 | 2 | 0 | 2 | 0 | 7 |
| Guss 2019 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Hafeez 2019 | 0 | 1 | 1 | 2 | 0 | 2 | 1 | 7 |
| Harrington 2018 | 0 | 1 | 1 | 2 | 0 | 2 | 1 | 7 |
| Harris 2019 | 0 | 1 | 1 | 2 | 2 | 2 | 1 | 9 |
| Herrick-Reynolds 2019 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Herzke 2020 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Higgins 1979 | 1 | 1 | 1 | 2 | 1 | 2 | 0 | 8 |
| Higgins 2020 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Hill 2018 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Holliday 2014 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Holliday 2015 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |

| Author | Representativeness of cohort | Adequacy of sample size | Characterization of non-respondents | Ascertainment of exposure | Comparability of groups | Outcome assessment | Statistical tests | NOS Score |
|---|---|---|---|---|---|---|---|---|
| Hoof 2020 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Hoops 2018 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Hoover 2005 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Houtrow 2011 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Houtrow 2020 | 1 | 1 | 1 | 2 | 2 | 1 | 1 | 9 |
| Hurley 2014 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 9 |
| Ibn 2019 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Ilin 2020 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Inam 2020 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Ingram 2015 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Jacquet 2014 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Jagsi 2012 | 0 | 1 | 1 | 2 | 2 | 1 | 1 | 8 |
| Jagsi 2013 | 0 | 1 | 1 | 2 | 2 | 2 | 1 | 9 |
| Jaque 2020 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Jena 2015 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Jena 2016 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| John 2016 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 9 |
| Johnson 1992 | 1 | 1 | 1 | 2 | 0 | 2 | 0 | 7 |
| Jolly 2014 | 0 | 1 | 0 | 2 | 1 | 1 | 1 | 6 |
| Kaderli 2015 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |

| Author | Representativeness of cohort | Adequacy of sample size | Characterization of non-respondents | Ascertainment of exposure | Comparability of groups | Outcome assessment | Statistical tests | NOS Score |
|---|---|---|---|---|---|---|---|---|
| Karimian 2012 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Kenevan 2018 | 0 | 1 | 1 | 2 | 0 | 2 | 1 | 7 |
| Khan 2014 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 9 |
| Khan 2019 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 9 |
| Khurshid 2018 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Kilic 2020 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Killaspy 2003 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Kim 2019 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Kirchner 1984 | 0 | 1 | 0 | 2 | 0 | 1 | 0 | 4 |
| Klingensmith 2006 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Kloosterboer 2020 | 0 | 1 | 1 | 2 | 0 | 2 | 1 | 7 |
| Komlenac 2019 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Larese 2019 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| LaRocca 2018 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Levey 1990 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Lewis-Stevenson 2001 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Long 2010 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Madsen 2017 | 1 | 1 | 0 | 2 | 2 | 1 | 1 | 8 |
| Mascarenhas 2017 | 1 | 1 | 1 | 2 | 0 | 2 | 0 | 7 |

| Author | Representativeness of cohort | Adequacy of sample size | Characterization of non-respondents | Ascertainment of exposure | Comparability of groups | Outcome assessment | Statistical tests | NOS Score |
|---|---|---|---|---|---|---|---|---|
| Maxwell 2020 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Mayer 2017 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| McAlister 2014 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| McDermott 2018 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 9 |
| McDonald 2017 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Mendoza 2015 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Modena 1999 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Moghimi 2019 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Monroe 2015 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Morgan 2019 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Morris 2018 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Mueller 2017 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 9 |
| Muffly 2016 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Nakata 2017 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Ngaage 2019 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Nguyen 2018 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Nickerson 1990 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Niu 2020 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Nowling 2018 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |

| Author | Representativeness of cohort | Adequacy of sample size | Characterization of non-respondents | Ascertainment of exposure | Comparability of groups | Outcome assessment | Statistical tests | NOS Score |
|---|---|---|---|---|---|---|---|---|
| Odell 2019 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Okafor 2020 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Okoshi 2014 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| O'Neill 2019 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Orhurhu 2020 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Pace 2020 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Pagel 2015 | 0 | 1 | 1 | 2 | 1 | 2 | 1 | 8 |
| Paik 2014 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 9 |
| Paulus 2016 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Penfold 1987 | 0 | 1 | 0 | 2 | 0 | 1 | 0 | 4 |
| Pfeiffer 2016 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Pololi 2013 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Qamar 2018 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 9 |
| Qamar 2020 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Rao 2018 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Raptis 2012 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Rayburn 2012 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 9 |
| Reed 2011 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Reiser 1993 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Riaz 2020 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |

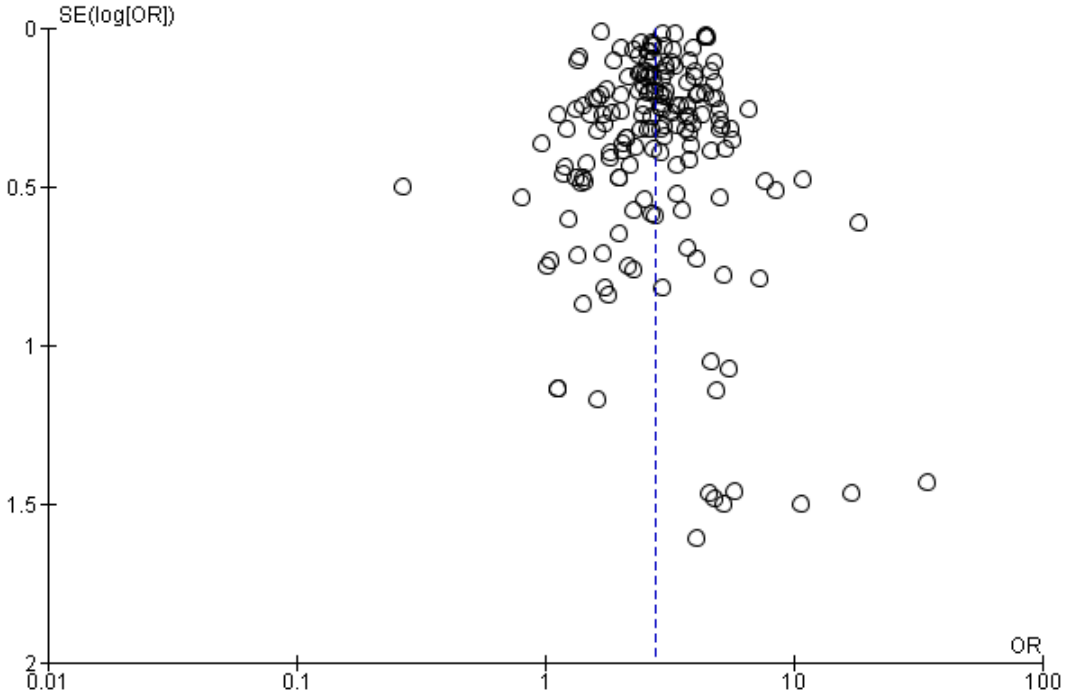
| Author | Representativeness of cohort | Adequacy of sample size | Characterization of non-respondents | Ascertainment of exposure | Comparability of groups | Outcome assessment | Statistical tests | NOS Score |
|---|---|---|---|---|---|---|---|---|
| Robboy 2017 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Rose 2015 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Rosso 2019 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Rotbart 2012 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 9 |
| Sadeghpour 2012 | 0 | 1 | 0 | 2 | 2 | 1 | 1 | 7 |
| Sandhu 2007 | 1 | 1 | 1 | 2 | 0 | 1 | 1 | 7 |
| Santosa 2019 | 0 | 1 | 1 | 2 | 2 | 2 | 1 | 9 |
| Sasor 2018 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Schnapp 2003 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Schroen 2004 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Schwarz 2016 | 1 | 1 | 0 | 2 | 0 | 2 | 1 | 7 |
| Selter 2020 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Shah 2018 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Shaikh 2019 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Sheikh 2018 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Shiber 2020 | 1 | 1 | 0 | 2 | 2 | 1 | 1 | 8 |
| Shollen 2009 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Singh 2008 | 0 | 1 | 0 | 2 | 2 | 1 | 1 | 7 |
| Smith 2019 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 9 |
| Soothill 2010 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |

| Author | Representativeness of cohort | Adequacy of sample size | Characterization of non-respondents | Ascertainment of exposure | Comparability of groups | Outcome assessment | Statistical tests | NOS Score |
|---|---|---|---|---|---|---|---|---|
| Speck 2012 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Spencer 2016 | 0 | 1 | 0 | 2 | 2 | 1 | 1 | 7 |
| Sperling 2019 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Stadler 2017 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Steger 2009 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Stewart 2020 | 0 | 1 | 1 | 2 | 2 | 2 | 1 | 9 |
| Svider 2014 | 0 | 1 | 1 | 2 | 2 | 2 | 1 | 9 |
| Tekerek 2016 | 0 | 0 | 0 | 2 | 0 | 1 | 1 | 4 |
| Therattil 2016 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Tracy 2019 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Tsukada 2009 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Tulunay-Ugur 2019 | 0 | 0 | 0 | 2 | 0 | 1 | 1 | 4 |
| Turaga 2013 | 0 | 1 | 1 | 2 | 1 | 2 | 1 | 8 |
| Valsangkar 2016 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 9 |
| Villablanca 2017 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Vydareny 2000 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Wagner 2007 | 0 | 1 | 0 | 2 | 2 | 1 | 1 | 7 |
| Waisbren 2008 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Waljee 2015 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |

| Author | Representativeness of cohort | Adequacy of sample size | Characterization of non-respondents | Ascertainment of exposure | Comparability of groups | Outcome assessment | Statistical tests | NOS Score |
|---|---|---|---|---|---|---|---|---|
| Wallis 1981 | 0 | 1 | 0 | 2 | 0 | 1 | 0 | 4 |
| Warner 2017 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Waseem 2019 | 1 | 1 | 1 | 2 | 0 | 1 | 1 | 7 |
| Wasserstein 2007 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Weaver 2015 | 0 | 1 | 0 | 2 | 2 | 1 | 1 | 7 |
| Whitley 1987 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Wiler 2019 | 1 | 1 | 0 | 2 | 2 | 1 | 1 | 8 |
| Wilkes 2015 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Wilkinson 1986 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Willett 2015 | 1 | 1 | 0 | 2 | 2 | 1 | 1 | 8 |
| Wilson 2013 | 0 | 1 | 1 | 2 | 0 | 2 | 1 | 7 |
| Winch 2017 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Wise 2004 | 1 | 1 | 0 | 2 | 0 | 1 | 1 | 6 |
| Woods 2018 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Woodward 2017 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Wright 2007 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 10 |
| Wu 2019 | 1 | 1 | 1 | 2 | 0 | 1 | 1 | 7 |
| Xierali 2020 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Yang 2019 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |

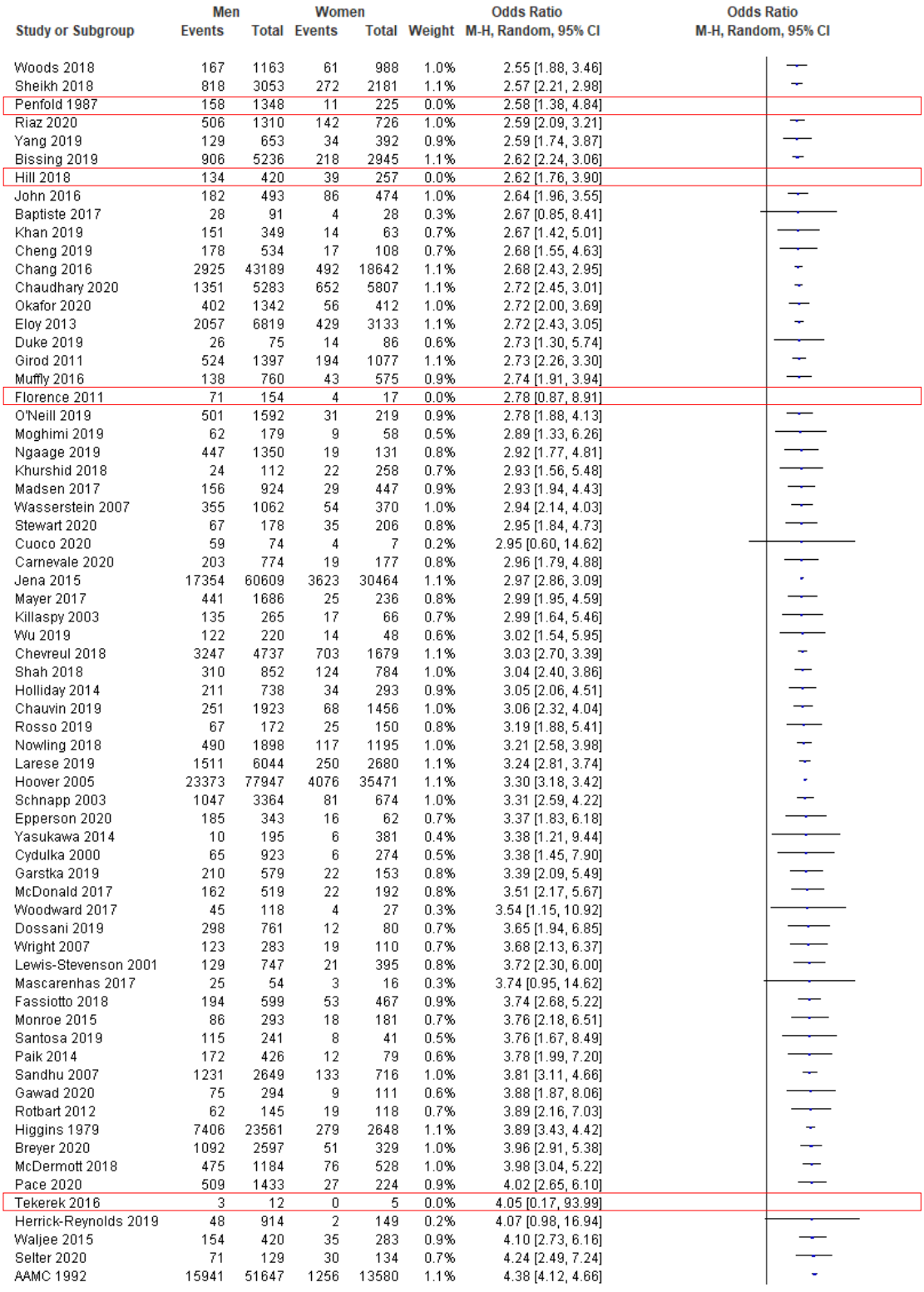
| Author | Representativeness of cohort | Adequacy of sample size | Characterization of non-respondents | Ascertainment of exposure | Comparability of groups | Outcome assessment | Statistical tests | NOS Score |
|---|---|---|---|---|---|---|---|---|
| Yasukawa 2014 | 1 | 1 | 0 | 2 | 2 | 1 | 1 | 8 |
| Yu 2013 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 9 |
| Yue 2020 | 1 | 1 | 1 | 2 | 0 | 2 | 1 | 8 |
| Zelnik 2018 | 0 | 1 | 0 | 2 | 0 | 1 | 1 | 5 |
| Zivin 2020 | 1 | 1 | 0 | 2 | 2 | 1 | 1 | 8 |

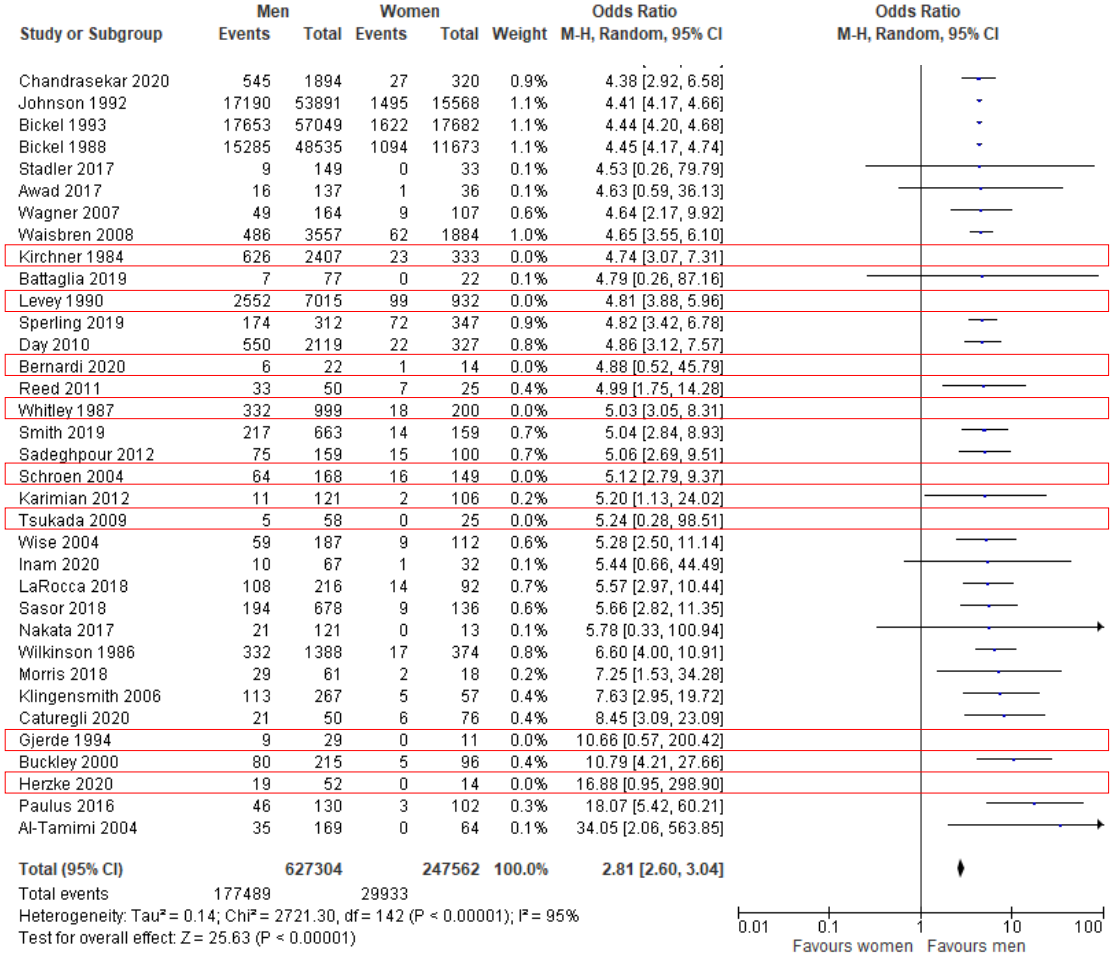
*Quality assessment was performed using Newcastle-Ottawa Scale (NOS) for cross-sectional studies. A maximum of one point is awarded for representativeness of cohort, adequacy of sample size, characterization of non-respondents, and statistical tests. A maximum of two points is awarded for characterization of non-respondents, ascertainment of exposure, and comparability of groups. An overall maximum score of 10 can be awarded per study. NOS score ≥ 8 is low risk, 6 – 7 is moderate risk, and ≤ 5 is high risk.



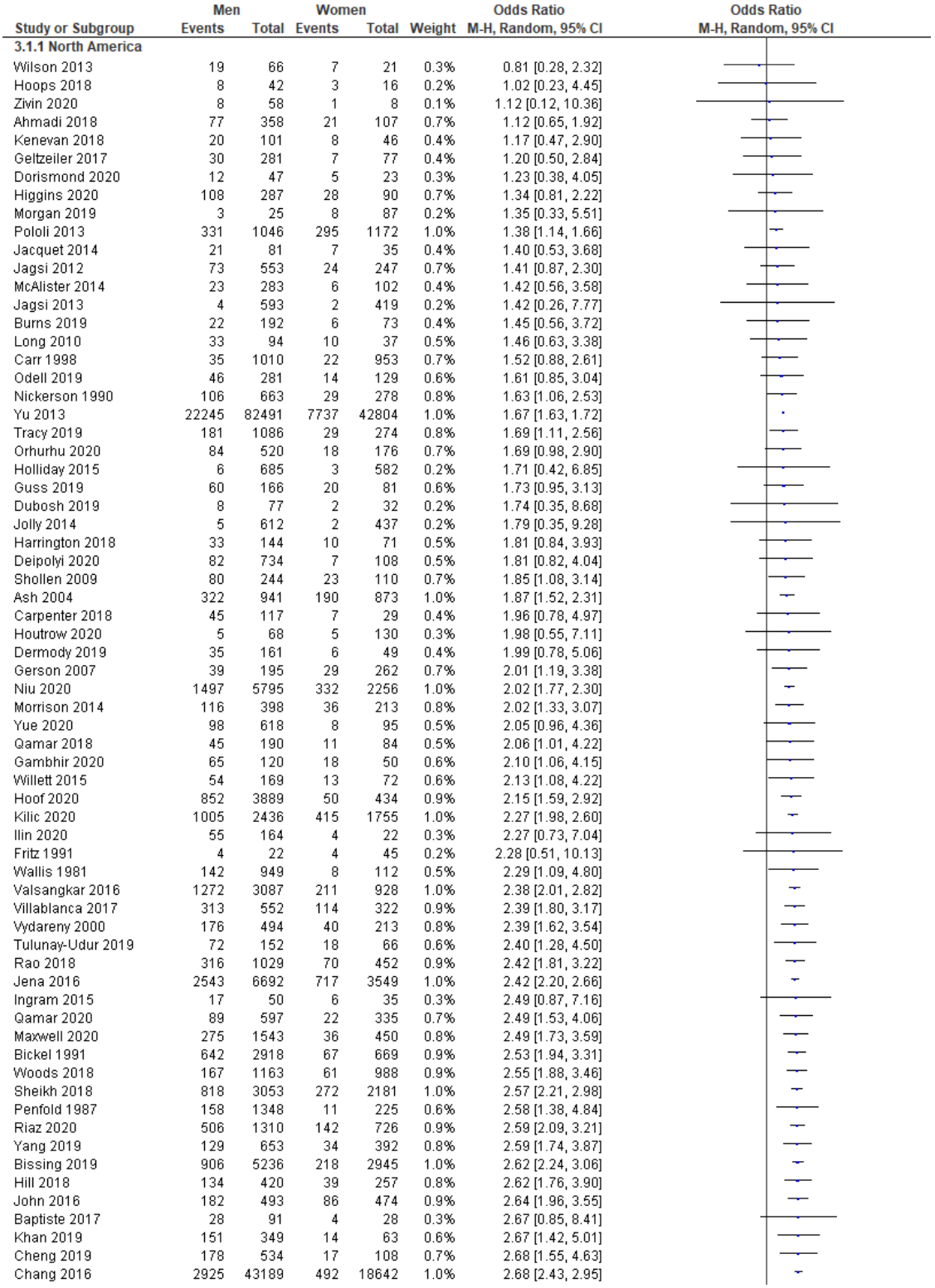

**eFigure 1.** Funnel plot of full professorship for men vs. women. SE (standard error), OR (odds ratio).

| Study or Subgroup | Men Events | Men Total | Women Events | Women Total | Weight | Odds Ratio M-H, Random, 95% CI |
|---|---|---|---|---|---|---|
| Schwarz 2016 | 7 | 200 | 11 | 93 | 0.4% | 0.27 [0.10, 0.72] |
| Wilson 2013 | 19 | 66 | 7 | 21 | 0.4% | 0.81 [0.28, 2.32] |
| Shaikh 2019 | 162 | 294 | 19 | 34 | 0.6% | 0.97 [0.47, 1.98] |
| Hoops 2018 | 8 | 42 | 3 | 16 | 0.2% | 1.02 [0.23, 4.45] |
| Gerety 2013 | 36 | 104 | 3 | 9 | 0.0% | 1.06 [0.25, 4.49] |
| Zelnik Yovel 2018 | 8 | 58 | 1 | 8 | 0.0% | 1.12 [0.12, 10.36] |
| Zivin 2020 | 8 | 58 | 1 | 8 | 0.1% | 1.12 [0.12, 10.36] |
| Ahmadi 2018 | 77 | 358 | 21 | 107 | 0.7% | 1.12 [0.65, 1.92] |
| Kenevan 2018 | 20 | 101 | 8 | 46 | 0.5% | 1.17 [0.47, 2.90] |
| Geltzeiler 2017 | 30 | 281 | 7 | 77 | 0.5% | 1.20 [0.50, 2.84] |
| Beeler 2020 | 73 | 135 | 27 | 55 | 0.0% | 1.22 [0.65, 2.29] |
| Dorismond 2020 | 12 | 47 | 5 | 23 | 0.3% | 1.23 [0.38, 4.05] |
| Winch 2017 | 32 | 60 | 12 | 26 | 0.0% | 1.33 [0.53, 3.36] |
| Higgins 2020 | 108 | 287 | 28 | 90 | 0.0% | 1.34 [0.81, 2.22] |
| Aliano 2020 | 272 | 654 | 334 | 965 | 1.0% | 1.35 [1.10, 1.65] |
| Morgan 2019 | 3 | 25 | 8 | 87 | 0.0% | 1.35 [0.33, 5.51] |
| Pololi 2013 | 331 | 1046 | 295 | 1172 | 1.1% | 1.38 [1.14, 1.66] |
| Jacquet 2014 | 21 | 81 | 7 | 35 | 0.0% | 1.40 [0.53, 3.68] |
| Jagsi 2012 | 73 | 553 | 24 | 247 | 0.8% | 1.41 [0.87, 2.30] |
| McAlister 2014 | 23 | 283 | 6 | 102 | 0.0% | 1.42 [0.56, 3.58] |
| Jagsi 2013 | 4 | 593 | 2 | 419 | 0.2% | 1.42 [0.26, 7.77] |
| Burns 2019 | 22 | 192 | 6 | 73 | 0.0% | 1.45 [0.56, 3.72] |
| Long 2010 | 33 | 94 | 10 | 37 | 0.5% | 1.46 [0.63, 3.38] |
| Carr 1998 | 35 | 1010 | 22 | 953 | 0.7% | 1.52 [0.88, 2.61] |
| Harris 2019 | 275 | 426 | 55 | 102 | 0.8% | 1.56 [1.01, 2.41] |
| Odell 2019 | 46 | 281 | 14 | 129 | 0.6% | 1.61 [0.85, 3.04] |
| Ibn Auf 2019 | 3 | 100 | 1 | 53 | 0.1% | 1.61 [0.16, 15.85] |
| Nickerson 1990 | 106 | 663 | 29 | 278 | 0.8% | 1.63 [1.06, 2.53] |
| Yu 2013 | 22245 | 82491 | 7737 | 42804 | 1.1% | 1.67 [1.63, 1.72] |
| Tracy 2019 | 181 | 1086 | 29 | 274 | 0.9% | 1.69 [1.11, 2.56] |
| Orhurhu 2020 | 84 | 520 | 18 | 176 | 0.7% | 1.69 [0.98, 2.90] |
| Holliday 2015 | 6 | 685 | 3 | 582 | 0.0% | 1.71 [0.42, 6.85] |
| Guss 2019 | 60 | 166 | 20 | 81 | 0.7% | 1.73 [0.95, 3.13] |
| Dubosh 2019 | 8 | 77 | 2 | 32 | 0.2% | 1.74 [0.35, 8.68] |
| Danic 2003 | 165 | 333 | 60 | 168 | 0.9% | 1.77 [1.21, 2.59] |
| Jolly 2014 | 5 | 612 | 2 | 437 | 0.2% | 1.79 [0.35, 9.28] |
| Harrington 2018 | 33 | 144 | 10 | 71 | 0.5% | 1.81 [0.84, 3.93] |
| Deipolyi 2020 | 82 | 734 | 7 | 108 | 0.5% | 1.81 [0.82, 4.04] |
| Shollen 2009 | 80 | 244 | 23 | 110 | 0.7% | 1.85 [1.08, 3.14] |
| Ash 2004 | 322 | 941 | 190 | 873 | 1.0% | 1.87 [1.52, 2.31] |
| Carpenter 2018 | 45 | 117 | 7 | 29 | 0.4% | 1.96 [0.78, 4.97] |
| Houtrow 2020 | 5 | 68 | 5 | 130 | 0.3% | 1.98 [0.55, 7.11] |
| Dermody 2019 | 35 | 161 | 6 | 49 | 0.4% | 1.99 [0.78, 5.06] |
| Gerson 2007 | 39 | 195 | 29 | 262 | 0.0% | 2.01 [1.19, 3.38] |
| Niu 2020 | 1497 | 5795 | 332 | 2256 | 1.1% | 2.02 [1.77, 2.30] |
| Morrison 2014 | 116 | 398 | 36 | 213 | 0.9% | 2.02 [1.33, 3.07] |
| Yue 2020 | 98 | 618 | 8 | 95 | 0.6% | 2.05 [0.96, 4.36] |
| Qamar 2018 | 45 | 190 | 11 | 84 | 0.6% | 2.06 [1.01, 4.22] |
| Gambhir 2020 | 65 | 120 | 18 | 50 | 0.6% | 2.10 [1.06, 4.15] |
| Willett 2015 | 54 | 169 | 13 | 72 | 0.6% | 2.13 [1.08, 4.22] |
| Okoshi 2014 | 33 | 326 | 2 | 40 | 0.2% | 2.14 [0.49, 9.28] |
| Hoof 2020 | 852 | 3889 | 50 | 434 | 1.0% | 2.15 [1.59, 2.92] |
| Soothill 2010 | 42 | 102 | 9 | 37 | 0.0% | 2.18 [0.93, 5.09] |
| Kilic 2020 | 1005 | 2436 | 415 | 1755 | 1.1% | 2.27 [1.98, 2.60] |
| Ilin 2020 | 55 | 164 | 4 | 22 | 0.3% | 2.27 [0.73, 7.04] |
| Fritz 1991 | 4 | 22 | 4 | 45 | 0.2% | 2.28 [0.51, 10.13] |
| Wallis 1981 | 142 | 949 | 8 | 112 | 0.0% | 2.29 [1.09, 4.80] |
| Valsangkar 2016 | 1272 | 3087 | 211 | 928 | 1.1% | 2.38 [2.01, 2.82] |
| Villablanca 2017 | 313 | 552 | 114 | 322 | 1.0% | 2.39 [1.80, 3.17] |
| Vydareny 2000 | 176 | 494 | 40 | 213 | 0.0% | 2.39 [1.62, 3.54] |
| Tulunay-Ugur 2019 | 72 | 152 | 18 | 66 | 0.0% | 2.40 [1.28, 4.50] |
| Rao 2018 | 316 | 1029 | 70 | 452 | 1.0% | 2.42 [1.81, 3.22] |
| Jena 2016 | 2543 | 6692 | 717 | 3549 | 1.1% | 2.42 [2.20, 2.66] |
| Waseem 2019 | 175 | 221 | 54 | 89 | 0.7% | 2.47 [1.44, 4.21] |
| Ingram 2015 | 17 | 50 | 6 | 35 | 0.4% | 2.49 [0.87, 7.16] |
| Qamar 2020 | 89 | 597 | 22 | 335 | 0.8% | 2.49 [1.53, 4.06] |
| Maxwell 2020 | 275 | 1543 | 36 | 450 | 0.9% | 2.49 [1.73, 3.59] |
| Bickel 1991 | 642 | 2918 | 67 | 669 | 1.0% | 2.53 [1.94, 3.31] |


Men
Women
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
Woods 2018 167 1163 61 988 1.0% 2.55 [1.88, 3.46]
Sheikh 2018 818 3053 272 2181 1.1% 2.57 [2.21, 2.98]
Penfold 1987 158 1348 11 225 0.0% 2.58 [1.38, 4.84]
Riaz 2020 506 1310 142 726 1.0% 2.59 [2.09, 3.21]
Yang 2019 129 653 34 392 0.9% 2.59 [1.74, 3.87]
Bissing 2019 906 5236 218 2945 1.1% 2.62 [2.24, 3.06]
Hill 2018 134 420 39 257 0.0% 2.62 [1.76, 3.90]
John 2016 182 493 86 474 1.0% 2.64 [1.96, 3.55]
Baptiste 2017 28 91 4 28 0.3% 2.67 [0.85, 8.41]
Khan 2019 151 349 14 63 0.7% 2.67 [1.42, 5.01]
Cheng 2019 178 534 17 108 0.7% 2.68 [1.55, 4.63]
Chang 2016 2925 43189 492 18642 1.1% 2.68 [2.43, 2.95]
Chaudhary 2020 1351 5283 652 5807 1.1% 2.72 [2.45, 3.01]
Okafor 2020 402 1342 56 412 1.0% 2.72 [2.00, 3.69]
Eloy 2013 2057 6819 429 3133 1.1% 2.72 [2.43, 3.05]
Duke 2019 26 75 14 86 0.6% 2.73 [1.30, 5.74]
Girod 2011 524 1397 194 1077 1.1% 2.73 [2.26, 3.30]
Muffly 2016 138 760 43 575 0.9% 2.74 [1.91, 3.94]
Florence 2011 71 154 4 17 0.0% 2.78 [0.87, 8.91]
O'Neill 2019 501 1592 31 219 0.9% 2.78 [1.88, 4.13]
Moghimi 2019 62 179 9 58 0.5% 2.89 [1.33, 6.26]
Ngaage 2019 447 1350 19 131 0.8% 2.92 [1.77, 4.81]
Khurshid 2018 24 112 22 258 0.7% 2.93 [1.56, 5.48]
Madsen 2017 156 924 29 447 0.9% 2.93 [1.94, 4.43]
Wasserstein 2007 355 1062 54 370 1.0% 2.94 [2.14, 4.03]
Stewart 2020 67 178 35 206 0.8% 2.95 [1.84, 4.73]
Cuoco 2020 59 74 4 7 0.2% 2.95 [0.60, 14.62]
Carnevale 2020 203 774 19 177 0.8% 2.96 [1.79, 4.88]
Jena 2015 17354 60609 3623 30464 1.1% 2.97 [2.86, 3.09]
Mayer 2017 441 1686 25 236 0.8% 2.99 [1.95, 4.59]
Killaspy 2003 135 265 17 66 0.7% 2.99 [1.64, 5.46]
Wu 2019 122 220 14 48 0.6% 3.02 [1.54, 5.95]
Chevreul 2018 3247 4737 703 1679 1.1% 3.03 [2.70, 3.39]
Shah 2018 310 852 124 784 1.0% 3.04 [2.40, 3.86]
Holliday 2014 211 738 34 293 0.9% 3.05 [2.06, 4.51]
Chauvin 2019 251 1923 68 1456 1.0% 3.06 [2.32, 4.04]
Rosso 2019 67 172 25 150 0.8% 3.19 [1.88, 5.41]
Nowling 2018 490 1898 117 1195 1.0% 3.21 [2.58, 3.98]
Larese 2019 1511 6044 250 2680 1.1% 3.24 [2.81, 3.74]
Hoover 2005 23373 77947 4076 35471 1.1% 3.30 [3.18, 3.42]
Schnapp 2003 1047 3364 81 674 1.0% 3.31 [2.59, 4.22]
Epperson 2020 185 343 16 62 0.7% 3.37 [1.83, 6.18]
Yasukawa 2014 10 195 6 381 0.4% 3.38 [1.21, 9.44]
Cydulka 2000 65 923 6 274 0.5% 3.38 [1.45, 7.90]
Garstka 2019 210 579 22 153 0.8% 3.39 [2.09, 5.49]
McDonald 2017 162 519 22 192 0.8% 3.51 [2.17, 5.67]
Woodward 2017 45 118 4 27 0.3% 3.54 [1.15, 10.92]
Dossani 2019 298 761 12 80 0.7% 3.65 [1.94, 6.85]
Wright 2007 123 283 19 110 0.7% 3.68 [2.13, 6.37]
Lewis-Stevenson 2001 129 747 21 395 0.8% 3.72 [2.30, 6.00]
Mascarenhas 2017 25 54 3 16 0.3% 3.74 [0.95, 14.62]
Fassiotto 2018 194 599 53 467 0.9% 3.74 [2.68, 5.22]
Monroe 2015 86 293 18 181 0.7% 3.76 [2.18, 6.51]
Santosa 2019 115 241 8 41 0.5% 3.76 [1.67, 8.49]
Paik 2014 172 426 12 79 0.6% 3.78 [1.99, 7.20]
Sandhu 2007 1231 2649 133 716 1.0% 3.81 [3.11, 4.66]
Gawad 2020 75 294 9 111 0.6% 3.88 [1.87, 8.06]
Rotbart 2012 62 145 19 118 0.7% 3.89 [2.16, 7.03]
Higgins 1979 7406 23561 279 2648 1.1% 3.89 [3.43, 4.42]
Breyer 2020 1092 2597 51 329 1.0% 3.96 [2.91, 5.38]
McDermott 2018 475 1184 76 528 1.0% 3.98 [3.04, 5.22]
Pace 2020 509 1433 27 224 0.9% 4.02 [2.65, 6.10]
Tekerek 2016 3 12 0 5 0.0% 4.05 [0.17, 93.99]
Herrick-Reynolds 2019 48 914 2 149 0.2% 4.07 [0.98, 16.94]
Waljee 2015 154 420 35 283 0.9% 4.10 [2.73, 6.16]
Selter 2020 71 129 30 134 0.7% 4.24 [2.49, 7.24]
AAMC 1992 15941 51647 1256 13580 1.1% 4.38 [4.12, 4.66]

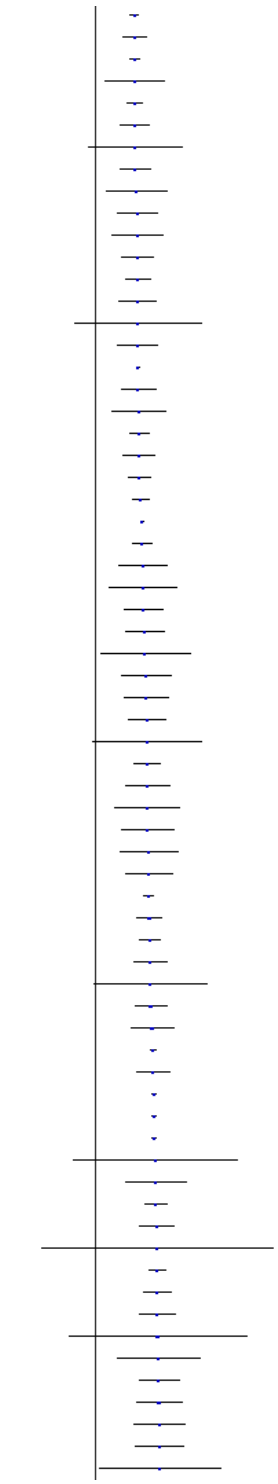


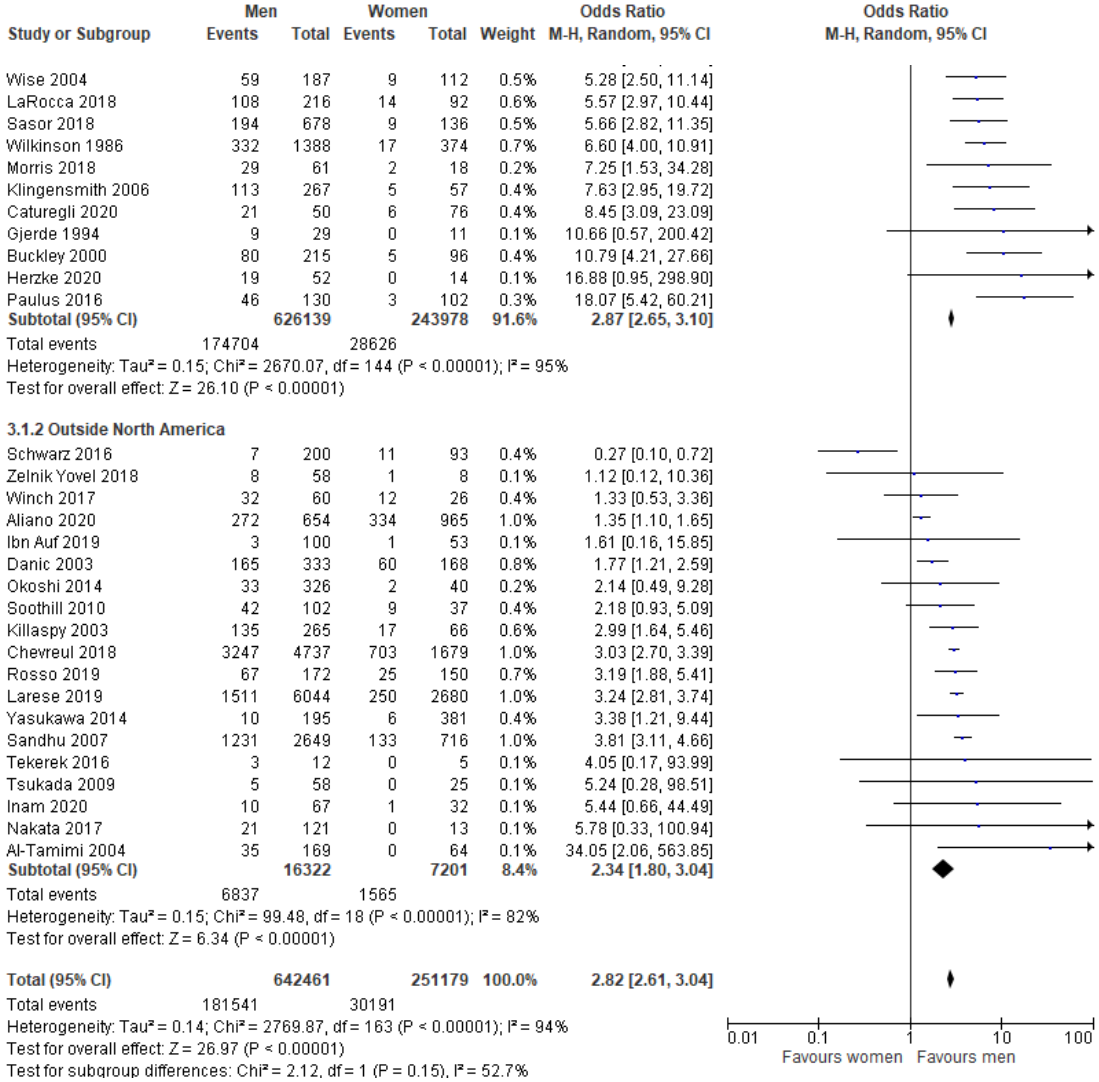

**eFigure 2.** Sensitivity analysis of full professorship removing 27 poor quality studies (red box, NOS score ≤ 5). M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).



| Study or Subgroup | Men Events | Men Total | Women Events | Women Total | Weight | Odds Ratio M-H, Random, 95% CI |
|---|---|---|---|---|---|---|
| **3.1.1 North America** | | | | | | |
| Wilson 2013 | 19 | 66 | 7 | 21 | 0.3% | 0.81 [0.28, 2.32] |
| Hoops 2018 | 8 | 42 | 3 | 16 | 0.2% | 1.02 [0.23, 4.45] |
| Zivin 2020 | 8 | 58 | 1 | 8 | 0.1% | 1.12 [0.12, 10.36] |
| Ahmadi 2018 | 77 | 358 | 21 | 107 | 0.7% | 1.12 [0.65, 1.92] |
| Kenevan 2018 | 20 | 101 | 8 | 46 | 0.4% | 1.17 [0.47, 2.90] |
| Geltzeiler 2017 | 30 | 281 | 7 | 77 | 0.4% | 1.20 [0.50, 2.84] |
| Dorismond 2020 | 12 | 47 | 5 | 23 | 0.3% | 1.23 [0.38, 4.05] |
| Higgins 2020 | 108 | 287 | 28 | 90 | 0.7% | 1.34 [0.81, 2.22] |
| Morgan 2019 | 3 | 25 | 8 | 87 | 0.2% | 1.35 [0.33, 5.51] |
| Pololi 2013 | 331 | 1046 | 295 | 1172 | 1.0% | 1.38 [1.14, 1.66] |
| Jacquet 2014 | 21 | 81 | 7 | 35 | 0.4% | 1.40 [0.53, 3.68] |
| Jagsi 2012 | 73 | 553 | 24 | 247 | 0.7% | 1.41 [0.87, 2.30] |
| McAlister 2014 | 23 | 283 | 6 | 102 | 0.4% | 1.42 [0.56, 3.58] |
| Jagsi 2013 | 4 | 593 | 2 | 419 | 0.2% | 1.42 [0.26, 7.77] |
| Burns 2019 | 22 | 192 | 6 | 73 | 0.4% | 1.45 [0.56, 3.72] |
| Long 2010 | 33 | 94 | 10 | 37 | 0.5% | 1.46 [0.63, 3.38] |
| Carr 1998 | 35 | 1010 | 22 | 953 | 0.7% | 1.52 [0.88, 2.61] |
| Odell 2019 | 46 | 281 | 14 | 129 | 0.6% | 1.61 [0.85, 3.04] |
| Nickerson 1990 | 106 | 663 | 29 | 278 | 0.8% | 1.63 [1.06, 2.53] |
| Yu 2013 | 22245 | 82491 | 7737 | 42804 | 1.0% | 1.67 [1.63, 1.72] |
| Tracy 2019 | 181 | 1086 | 29 | 274 | 0.8% | 1.69 [1.11, 2.56] |
| Orhurhu 2020 | 84 | 520 | 18 | 176 | 0.7% | 1.69 [0.98, 2.90] |
| Holliday 2015 | 6 | 685 | 3 | 582 | 0.2% | 1.71 [0.42, 6.85] |
| Guss 2019 | 60 | 166 | 20 | 81 | 0.6% | 1.73 [0.95, 3.13] |
| Dubosh 2019 | 8 | 77 | 2 | 32 | 0.2% | 1.74 [0.35, 8.68] |
| Jolly 2014 | 5 | 612 | 2 | 437 | 0.2% | 1.79 [0.35, 9.28] |
| Harrington 2018 | 33 | 144 | 10 | 71 | 0.5% | 1.81 [0.84, 3.93] |
| Deipolyi 2020 | 82 | 734 | 7 | 108 | 0.5% | 1.81 [0.82, 4.04] |
| Shollen 2009 | 80 | 244 | 23 | 110 | 0.7% | 1.85 [1.08, 3.14] |
| Ash 2004 | 322 | 941 | 190 | 873 | 1.0% | 1.87 [1.52, 2.31] |
| Carpenter 2018 | 45 | 117 | 7 | 29 | 0.4% | 1.96 [0.78, 4.97] |
| Houtrow 2020 | 5 | 68 | 5 | 130 | 0.3% | 1.98 [0.55, 7.11] |
| Dermody 2019 | 35 | 161 | 6 | 49 | 0.4% | 1.99 [0.78, 5.06] |
| Gerson 2007 | 39 | 195 | 29 | 262 | 0.7% | 2.01 [1.19, 3.38] |
| Niu 2020 | 1497 | 5795 | 332 | 2256 | 1.0% | 2.02 [1.77, 2.30] |
| Morrison 2014 | 116 | 398 | 36 | 213 | 0.8% | 2.02 [1.33, 3.07] |
| Yue 2020 | 98 | 618 | 8 | 95 | 0.5% | 2.05 [0.96, 4.36] |
| Qamar 2018 | 45 | 190 | 11 | 84 | 0.5% | 2.06 [1.01, 4.22] |
| Gambhir 2020 | 65 | 120 | 18 | 50 | 0.6% | 2.10 [1.06, 4.15] |
| Willett 2015 | 54 | 169 | 13 | 72 | 0.6% | 2.13 [1.08, 4.22] |
| Hoof 2020 | 852 | 3889 | 50 | 434 | 0.9% | 2.15 [1.59, 2.92] |
| Kilic 2020 | 1005 | 2436 | 415 | 1755 | 1.0% | 2.27 [1.98, 2.60] |
| Ilin 2020 | 55 | 164 | 4 | 22 | 0.3% | 2.27 [0.73, 7.04] |
| Fritz 1991 | 4 | 22 | 4 | 45 | 0.2% | 2.28 [0.51, 10.13] |
| Wallis 1981 | 142 | 949 | 8 | 112 | 0.5% | 2.29 [1.09, 4.80] |
| Valsangkar 2016 | 1272 | 3087 | 211 | 928 | 1.0% | 2.38 [2.01, 2.82] |
| Villablanca 2017 | 313 | 552 | 114 | 322 | 0.9% | 2.39 [1.80, 3.17] |
| Vydareny 2000 | 176 | 494 | 40 | 213 | 0.8% | 2.39 [1.62, 3.54] |
| Tulunay-Udur 2019 | 72 | 152 | 18 | 66 | 0.6% | 2.40 [1.28, 4.50] |
| Rao 2018 | 316 | 1029 | 70 | 452 | 0.9% | 2.42 [1.81, 3.22] |
| Jena 2016 | 2543 | 6692 | 717 | 3549 | 1.0% | 2.42 [2.20, 2.66] |
| Ingram 2015 | 17 | 50 | 6 | 35 | 0.3% | 2.49 [0.87, 7.16] |
| Qamar 2020 | 89 | 597 | 22 | 335 | 0.7% | 2.49 [1.53, 4.06] |
| Maxwell 2020 | 275 | 1543 | 36 | 450 | 0.8% | 2.49 [1.73, 3.59] |
| Bickel 1991 | 642 | 2918 | 67 | 669 | 0.9% | 2.53 [1.94, 3.31] |
| Woods 2018 | 167 | 1163 | 61 | 988 | 0.9% | 2.55 [1.88, 3.46] |
| Sheikh 2018 | 818 | 3053 | 272 | 2181 | 1.0% | 2.57 [2.21, 2.98] |
| Penfold 1987 | 158 | 1348 | 11 | 225 | 0.6% | 2.58 [1.38, 4.84] |
| Riaz 2020 | 506 | 1310 | 142 | 726 | 0.9% | 2.59 [2.09, 3.21] |
| Yang 2019 | 129 | 653 | 34 | 392 | 0.8% | 2.59 [1.74, 3.87] |
| Bissing 2019 | 906 | 5236 | 218 | 2945 | 1.0% | 2.62 [2.24, 3.06] |
| Hill 2018 | 134 | 420 | 39 | 257 | 0.8% | 2.62 [1.76, 3.90] |
| John 2016 | 182 | 493 | 86 | 474 | 0.9% | 2.64 [1.96, 3.55] |
| Baptiste 2017 | 28 | 91 | 4 | 28 | 0.3% | 2.67 [0.85, 8.41] |
| Khan 2019 | 151 | 349 | 14 | 63 | 0.6% | 2.67 [1.42, 5.01] |
| Cheng 2019 | 178 | 534 | 17 | 108 | 0.7% | 2.68 [1.55, 4.63] |
| Chang 2016 | 2925 | 43189 | 492 | 18642 | 1.0% | 2.68 [2.43, 2.95] |

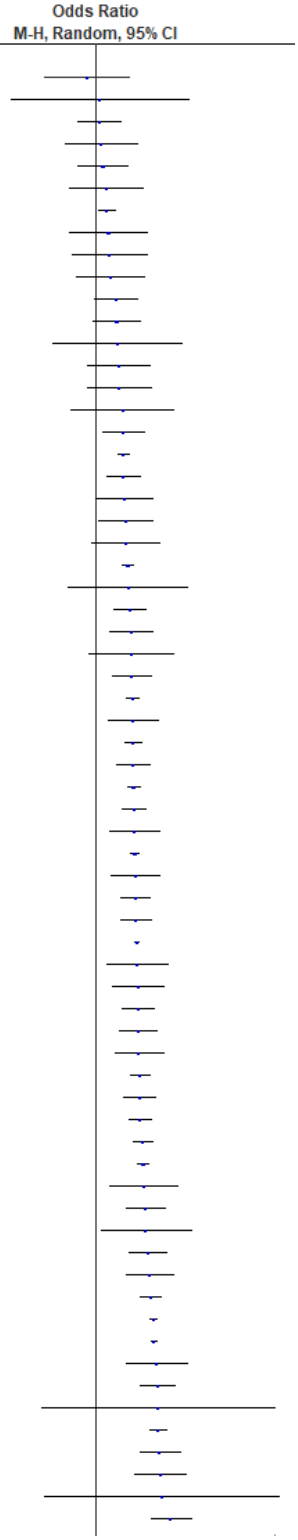
| Study or Subgroup | Men Events | Men Total | Women Events | Women Total | Weight | Odds Ratio M-H, Random, 95% CI |
|---|---|---|---|---|---|---|
| Chaudhary 2020 | 1351 | 5283 | 652 | 5807 | 1.0% | 2.72 [2.45, 3.01] |
| Okafor 2020 | 402 | 1342 | 56 | 412 | 0.9% | 2.72 [2.00, 3.69] |
| Eloy 2013 | 2057 | 6819 | 429 | 3133 | 1.0% | 2.72 [2.43, 3.05] |
| Duke 2019 | 26 | 75 | 14 | 86 | 0.5% | 2.73 [1.30, 5.74] |
| Girod 2011 | 524 | 1397 | 194 | 1077 | 1.0% | 2.73 [2.26, 3.30] |
| Muffly 2016 | 138 | 760 | 43 | 575 | 0.8% | 2.74 [1.91, 3.94] |
| Florence 2011 | 71 | 154 | 4 | 17 | 0.3% | 2.78 [0.87, 8.91] |
| O'Neill 2019 | 501 | 1592 | 31 | 219 | 0.8% | 2.78 [1.88, 4.13] |
| Moghimi 2019 | 62 | 179 | 9 | 58 | 0.5% | 2.89 [1.33, 6.26] |
| Ngaage 2019 | 447 | 1350 | 19 | 131 | 0.7% | 2.92 [1.77, 4.81] |
| Khurshid 2018 | 24 | 112 | 22 | 258 | 0.6% | 2.93 [1.56, 5.48] |
| Madsen 2017 | 156 | 924 | 29 | 447 | 0.8% | 2.93 [1.94, 4.43] |
| Wasserstein 2007 | 355 | 1062 | 54 | 370 | 0.9% | 2.94 [2.14, 4.03] |
| Stewart 2020 | 67 | 178 | 35 | 206 | 0.7% | 2.95 [1.84, 4.73] |
| Cuoco 2020 | 59 | 74 | 4 | 7 | 0.2% | 2.95 [0.60, 14.62] |
| Carnevale 2020 | 203 | 774 | 19 | 177 | 0.7% | 2.96 [1.79, 4.88] |
| Jena 2015 | 17354 | 60609 | 3623 | 30464 | 1.0% | 2.97 [2.86, 3.09] |
| Mayer 2017 | 441 | 1686 | 25 | 236 | 0.8% | 2.99 [1.95, 4.59] |
| Wu 2019 | 122 | 220 | 14 | 48 | 0.6% | 3.02 [1.54, 5.95] |
| Shah 2018 | 310 | 852 | 124 | 784 | 0.9% | 3.04 [2.40, 3.86] |
| Holliday 2014 | 211 | 738 | 34 | 293 | 0.8% | 3.05 [2.06, 4.51] |
| Chauvin 2019 | 251 | 1923 | 68 | 1456 | 0.9% | 3.06 [2.32, 4.04] |
| Nowling 2018 | 490 | 1898 | 117 | 1195 | 0.9% | 3.21 [2.58, 3.98] |
| Hoover 2005 | 23373 | 77947 | 4076 | 35471 | 1.0% | 3.30 [3.18, 3.42] |
| Schnapp 2003 | 1047 | 3364 | 81 | 674 | 0.9% | 3.31 [2.59, 4.22] |
| Epperson 2020 | 185 | 343 | 16 | 62 | 0.6% | 3.37 [1.83, 6.18] |
| Cydulka 2000 | 65 | 923 | 6 | 274 | 0.4% | 3.38 [1.45, 7.90] |
| Garstka 2019 | 210 | 579 | 22 | 153 | 0.7% | 3.39 [2.09, 5.49] |
| McDonald 2017 | 162 | 519 | 22 | 192 | 0.7% | 3.51 [2.17, 5.67] |
| Woodward 2017 | 45 | 118 | 4 | 27 | 0.3% | 3.54 [1.15, 10.92] |
| Dossani 2019 | 298 | 761 | 12 | 80 | 0.6% | 3.65 [1.94, 6.85] |
| Wright 2007 | 123 | 283 | 19 | 110 | 0.7% | 3.68 [2.13, 6.37] |
| Lewis-Stevenson 2001 | 129 | 747 | 21 | 395 | 0.7% | 3.72 [2.30, 6.00] |
| Mascarenhas 2017 | 25 | 54 | 3 | 16 | 0.2% | 3.74 [0.95, 14.62] |
| Fassiotto 2018 | 194 | 599 | 53 | 467 | 0.9% | 3.74 [2.68, 5.22] |
| Monroe 2015 | 86 | 293 | 18 | 181 | 0.7% | 3.76 [2.18, 6.51] |
| Santosa 2019 | 115 | 241 | 8 | 41 | 0.5% | 3.76 [1.67, 8.49] |
| Paik 2014 | 172 | 426 | 12 | 79 | 0.6% | 3.78 [1.99, 7.20] |
| Gawad 2020 | 75 | 294 | 9 | 111 | 0.5% | 3.88 [1.87, 8.06] |
| Rotbart 2012 | 62 | 145 | 19 | 118 | 0.6% | 3.89 [2.16, 7.03] |
| Higgins 1979 | 7406 | 23561 | 279 | 2648 | 1.0% | 3.89 [3.43, 4.42] |
| Breyer 2020 | 1092 | 2597 | 51 | 329 | 0.9% | 3.96 [2.91, 5.38] |
| McDermott 2018 | 475 | 1184 | 76 | 528 | 0.9% | 3.98 [3.04, 5.22] |
| Pace 2020 | 509 | 1433 | 27 | 224 | 0.8% | 4.02 [2.65, 6.10] |
| Herrick-Reynolds 2019 | 48 | 914 | 2 | 149 | 0.2% | 4.07 [0.98, 16.94] |
| Waljee 2015 | 154 | 420 | 35 | 283 | 0.8% | 4.10 [2.73, 6.16] |
| Selter 2020 | 71 | 129 | 30 | 134 | 0.7% | 4.24 [2.49, 7.24] |
| AAMC 1992 | 15941 | 51647 | 1256 | 13580 | 1.0% | 4.38 [4.12, 4.66] |
| Chandrasekar 2020 | 545 | 1894 | 27 | 320 | 0.8% | 4.38 [2.92, 6.58] |
| Johnson 1992 | 17190 | 53891 | 1495 | 15568 | 1.0% | 4.41 [4.17, 4.66] |
| Bickel 1993 | 17653 | 57049 | 1622 | 17682 | 1.0% | 4.44 [4.20, 4.68] |
| Bickel 1988 | 15285 | 48535 | 1094 | 11673 | 1.0% | 4.45 [4.17, 4.74] |
| Awad 2017 | 16 | 137 | 1 | 36 | 0.1% | 4.63 [0.59, 36.13] |
| Wagner 2007 | 49 | 164 | 9 | 107 | 0.5% | 4.64 [2.17, 9.92] |
| Waisbren 2008 | 486 | 3557 | 62 | 1884 | 0.9% | 4.65 [3.55, 6.10] |
| Kirchner 1984 | 626 | 2407 | 23 | 333 | 0.8% | 4.74 [3.07, 7.31] |
| Battaglia 2019 | 7 | 77 | 0 | 22 | 0.1% | 4.79 [0.26, 87.16] |
| Levey 1990 | 2552 | 7015 | 99 | 932 | 0.9% | 4.81 [3.88, 5.96] |
| Sperling 2019 | 174 | 312 | 72 | 347 | 0.8% | 4.82 [3.42, 6.78] |
| Day 2010 | 550 | 2119 | 22 | 327 | 0.8% | 4.86 [3.12, 7.57] |
| Bernardi 2020 | 6 | 22 | 1 | 14 | 0.1% | 4.88 [0.52, 45.79] |
| Reed 2011 | 33 | 50 | 7 | 25 | 0.3% | 4.99 [1.75, 14.28] |
| Whitley 1987 | 332 | 999 | 18 | 200 | 0.7% | 5.03 [3.05, 8.31] |
| Smith 2019 | 217 | 663 | 14 | 159 | 0.6% | 5.04 [2.84, 8.93] |
| Sadeghpour 2012 | 75 | 159 | 15 | 100 | 0.6% | 5.06 [2.69, 9.51] |
| Schroen 2004 | 64 | 168 | 16 | 149 | 0.6% | 5.12 [2.79, 9.37] |
| Karimian 2012 | 11 | 121 | 2 | 106 | 0.2% | 5.20 [1.13, 24.02] |



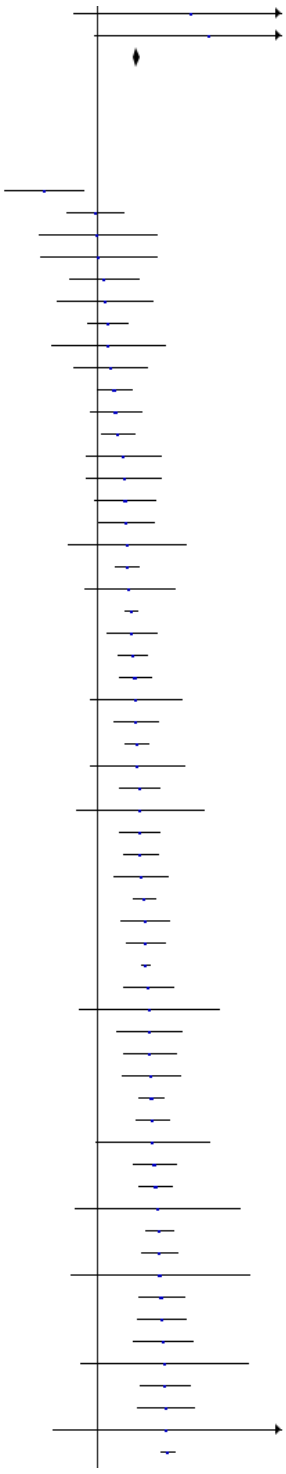


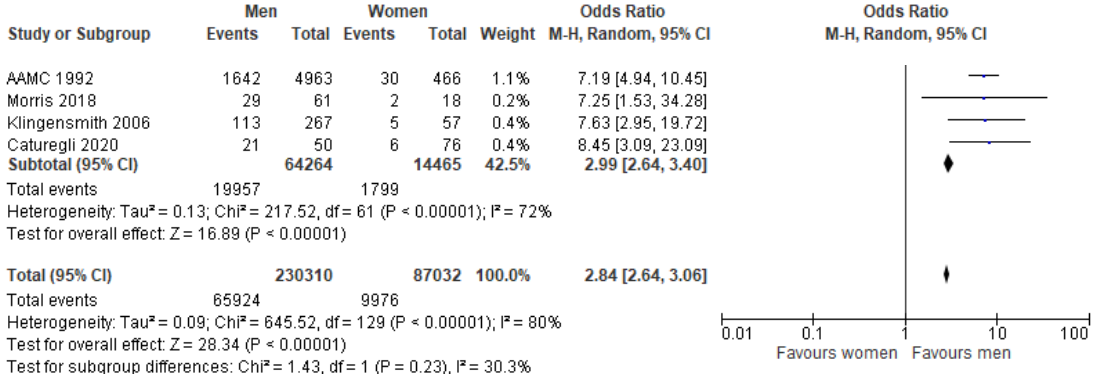

**eFigure 3.** Subgroup analysis of full professorship by geographic region (within North America versus outside of North America). M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).

| Study or Subgroup | Men Events | Men Total | Women Events | Women Total | Weight | Odds Ratio M-H, Random, 95% CI |
|---|---|---|---|---|---|---|
| **4.1.1 Non-surgical specialties** | | | | | | |
| Wilson 2013 | 19 | 66 | 7 | 21 | 0.4% | 0.81 [0.28, 2.32] |
| Zelnik Yovel 2018 | 8 | 58 | 1 | 8 | 0.1% | 1.12 [0.12, 10.36] |
| Ahmadi 2018 | 77 | 358 | 21 | 107 | 0.8% | 1.12 [0.65, 1.92] |
| Kenevan 2018 | 20 | 101 | 8 | 46 | 0.5% | 1.17 [0.47, 2.90] |
| Beeler 2020 | 73 | 135 | 27 | 55 | 0.7% | 1.22 [0.65, 2.29] |
| Winch 2017 | 32 | 60 | 12 | 26 | 0.4% | 1.33 [0.53, 3.36] |
| Aliano 2020 | 272 | 654 | 334 | 965 | 1.4% | 1.35 [1.10, 1.65] |
| Jacquet 2014 | 21 | 81 | 7 | 35 | 0.4% | 1.40 [0.53, 3.68] |
| Burns 2019 | 22 | 192 | 6 | 73 | 0.4% | 1.45 [0.56, 3.72] |
| Long 2010 | 33 | 94 | 10 | 37 | 0.5% | 1.46 [0.63, 3.38] |
| Orhurhu 2020 | 84 | 520 | 18 | 176 | 0.8% | 1.69 [0.98, 2.90] |
| Guss 2019 | 60 | 166 | 20 | 81 | 0.8% | 1.73 [0.95, 3.13] |
| Dubosh 2019 | 8 | 77 | 2 | 32 | 0.2% | 1.74 [0.35, 8.68] |
| Harrington 2018 | 33 | 144 | 10 | 71 | 0.6% | 1.81 [0.84, 3.93] |
| Deipolyi 2020 | 82 | 734 | 7 | 108 | 0.5% | 1.81 [0.82, 4.04] |
| Houtrow 2020 | 5 | 68 | 5 | 130 | 0.3% | 1.98 [0.55, 7.11] |
| Gerson 2007 | 39 | 195 | 29 | 262 | 0.9% | 2.01 [1.19, 3.38] |
| Niu 2020 | 1497 | 5795 | 332 | 2256 | 1.5% | 2.02 [1.77, 2.30] |
| Morrison 2014 | 116 | 398 | 36 | 213 | 1.0% | 2.02 [1.33, 3.07] |
| Qamar 2018 | 45 | 190 | 11 | 84 | 0.6% | 2.06 [1.01, 4.22] |
| Willett 2015 | 54 | 169 | 13 | 72 | 0.7% | 2.13 [1.08, 4.22] |
| Soothill 2010 | 42 | 102 | 9 | 37 | 0.5% | 2.18 [0.93, 5.09] |
| Kilic 2020 | 1005 | 2436 | 415 | 1755 | 1.5% | 2.27 [1.98, 2.60] |
| Fritz 1991 | 4 | 22 | 4 | 45 | 0.2% | 2.28 [0.51, 10.13] |
| Vydareny 2000 | 176 | 494 | 40 | 213 | 1.1% | 2.39 [1.62, 3.54] |
| Waseem 2019 | 175 | 221 | 54 | 89 | 0.9% | 2.47 [1.44, 4.21] |
| Ingram 2015 | 17 | 50 | 6 | 35 | 0.4% | 2.49 [0.87, 7.16] |
| Qamar 2020 | 89 | 597 | 22 | 335 | 0.9% | 2.49 [1.53, 4.06] |
| Sheikh 2018 | 818 | 3053 | 272 | 2181 | 1.5% | 2.57 [2.21, 2.98] |
| Penfold 1987 | 158 | 1348 | 11 | 225 | 0.7% | 2.58 [1.38, 4.84] |
| Riaz 2020 | 506 | 1310 | 142 | 726 | 1.4% | 2.59 [2.09, 3.21] |
| Yang 2019 | 129 | 653 | 34 | 392 | 1.1% | 2.59 [1.74, 3.87] |
| Bissing 2019 | 906 | 5236 | 218 | 2945 | 1.5% | 2.62 [2.24, 3.06] |
| John 2016 | 182 | 493 | 86 | 474 | 1.3% | 2.64 [1.96, 3.55] |
| Khan 2019 | 151 | 349 | 14 | 63 | 0.7% | 2.67 [1.42, 5.01] |
| Chaudhary 2020 | 1351 | 5283 | 652 | 5807 | 1.5% | 2.72 [2.45, 3.01] |
| Rosso 2019 | 53 | 114 | 23 | 95 | 0.8% | 2.72 [1.50, 4.94] |
| Muffly 2016 | 138 | 760 | 43 | 575 | 1.1% | 2.74 [1.91, 3.94] |
| O'Neill 2019 | 501 | 1592 | 31 | 219 | 1.1% | 2.78 [1.88, 4.13] |
| Jena 2015 | 10729 | 39823 | 2604 | 22687 | 1.6% | 2.84 [2.72, 2.98] |
| Moghimi 2019 | 62 | 179 | 9 | 58 | 0.6% | 2.89 [1.33, 6.26] |
| Khurshid 2018 | 24 | 112 | 22 | 258 | 0.7% | 2.93 [1.56, 5.48] |
| Madsen 2017 | 156 | 924 | 29 | 447 | 1.0% | 2.93 [1.94, 4.43] |
| Stewart 2020 | 67 | 178 | 35 | 206 | 0.9% | 2.95 [1.84, 4.73] |
| Killaspy 2003 | 135 | 265 | 17 | 66 | 0.8% | 2.99 [1.64, 5.46] |
| Shah 2018 | 310 | 852 | 124 | 784 | 1.4% | 3.04 [2.40, 3.86] |
| Holliday 2014 | 211 | 738 | 34 | 293 | 1.1% | 3.05 [2.06, 4.51] |
| Chauvin 2019 | 251 | 1923 | 68 | 1456 | 1.3% | 3.06 [2.32, 4.04] |
| Schnapp 2003 | 1047 | 3364 | 81 | 674 | 1.3% | 3.31 [2.59, 4.22] |
| Eloy 2013 | 1237 | 3539 | 290 | 2082 | 1.5% | 3.32 [2.88, 3.83] |
| Cydulka 2000 | 65 | 923 | 6 | 274 | 0.5% | 3.38 [1.45, 7.90] |
| McDonald 2017 | 162 | 519 | 22 | 192 | 0.9% | 3.51 [2.17, 5.67] |
| Woodward 2017 | 45 | 118 | 4 | 27 | 0.3% | 3.54 [1.15, 10.92] |
| Lewis-Stevenson 2001 | 129 | 747 | 21 | 395 | 0.9% | 3.72 [2.30, 6.00] |
| Rotbart 2012 | 62 | 145 | 19 | 118 | 0.8% | 3.89 [2.16, 7.03] |
| McDermott 2018 | 475 | 1184 | 76 | 528 | 1.3% | 3.98 [3.04, 5.22] |
| AAMC 1992 | 8913 | 32259 | 747 | 9163 | 1.6% | 4.30 [3.98, 4.65] |
| Johnson 1992 | 8880 | 31538 | 796 | 9672 | 1.6% | 4.37 [4.05, 4.72] |
| Wagner 2007 | 49 | 164 | 9 | 107 | 0.6% | 4.64 [2.17, 9.92] |
| Kirchner 1984 | 626 | 2407 | 23 | 333 | 1.0% | 4.74 [3.07, 7.31] |
| Battaglia 2019 | 7 | 77 | 0 | 22 | 0.1% | 4.79 [0.26, 87.16] |
| Levey 1990 | 2552 | 7015 | 99 | 932 | 1.4% | 4.81 [3.88, 5.96] |
| Whitley 1987 | 332 | 999 | 18 | 200 | 0.9% | 5.03 [3.05, 8.31] |
| Sadeghpour 2012 | 75 | 159 | 15 | 100 | 0.7% | 5.06 [2.69, 9.51] |
| Tsukada 2009 | 5 | 58 | 0 | 25 | 0.1% | 5.24 [0.28, 98.51] |
| Wilkinson 1986 | 332 | 1388 | 17 | 374 | 0.9% | 6.60 [4.00, 10.91] |

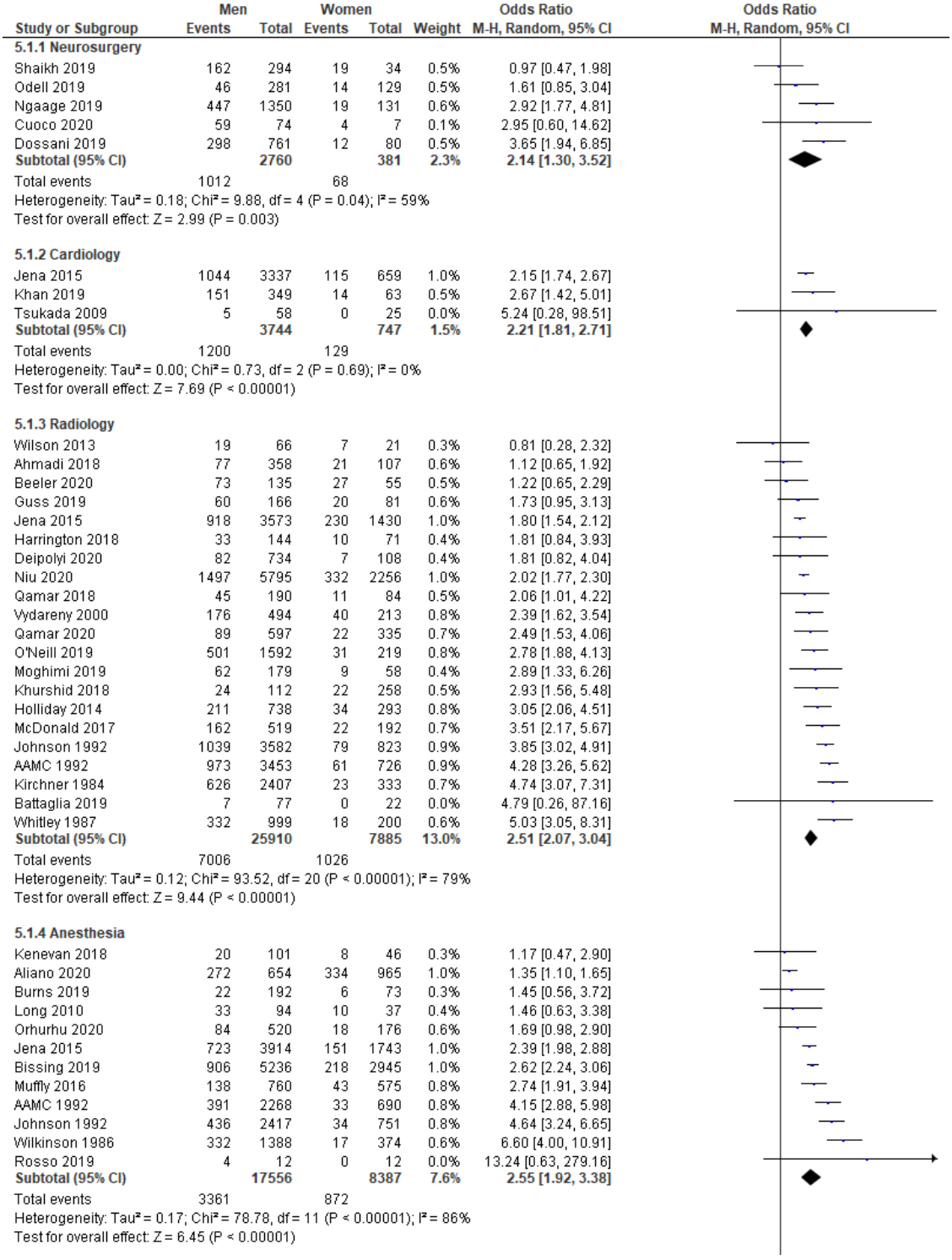


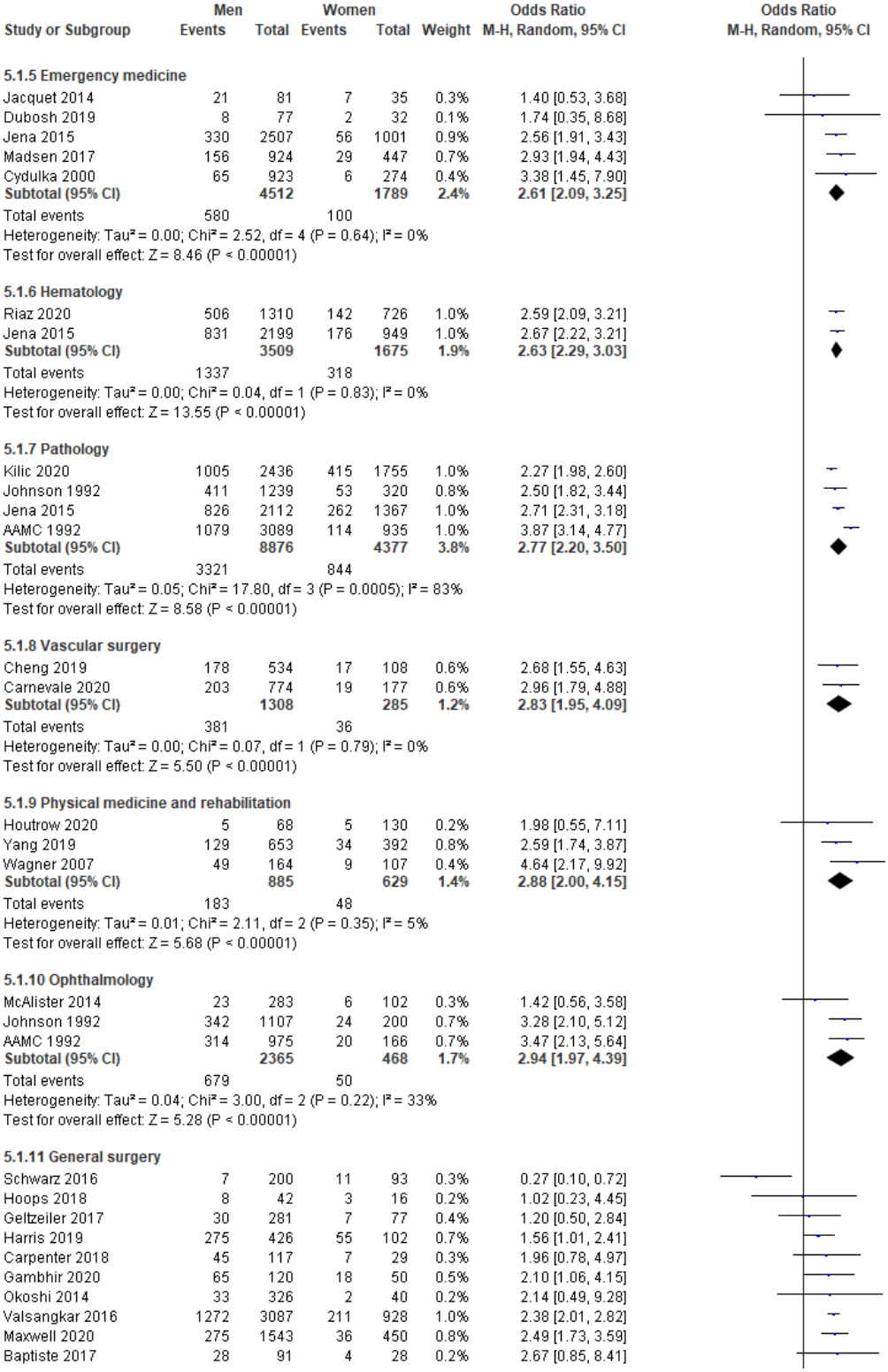
| Study or Subgroup | Men Events | Men Total | Women Events | Women Total | Weight | Odds Ratio M-H, Random, 95% CI |
|---|---|---|---|---|---|---|
| Gjerde 1994 | 9 | 29 | 0 | 11 | 0.1% | 10.66 [0.57, 200.42] |
| Herzke 2020 | 19 | 52 | 0 | 14 | 0.1% | 16.88 [0.95, 298.90] |
| **Subtotal (95% CI)** | | **166046** | | **72567** | **57.5%** | **2.72 [2.48, 2.98]** |
| Total events | 45967 | | 8177 | | | |
| Heterogeneity: Tau² = 0.08; Chi² = 423.11, df = 67 (P < 0.00001); I² = 84% | | | | | | |
| Test for overall effect: Z = 21.33 (P < 0.00001) | | | | | | |
| **4.1.2 Surgical specialties** | | | | | | |
| Schwarz 2016 | 7 | 200 | 11 | 93 | 0.4% | 0.27 [0.10, 0.72] |
| Shaikh 2019 | 162 | 294 | 19 | 34 | 0.6% | 0.97 [0.47, 1.98] |
| Hoops 2018 | 8 | 42 | 3 | 16 | 0.2% | 1.02 [0.23, 4.45] |
| Gerety 2013 | 36 | 104 | 3 | 9 | 0.2% | 1.06 [0.25, 4.49] |
| Geltzeiler 2017 | 30 | 281 | 7 | 77 | 0.5% | 1.20 [0.50, 2.84] |
| Dorismond 2020 | 12 | 47 | 5 | 23 | 0.3% | 1.23 [0.38, 4.05] |
| Higgins 2020 | 108 | 287 | 28 | 90 | 0.9% | 1.34 [0.81, 2.22] |
| Morgan 2019 | 3 | 25 | 8 | 87 | 0.2% | 1.35 [0.33, 5.51] |
| McAlister 2014 | 23 | 283 | 6 | 102 | 0.4% | 1.42 [0.56, 3.58] |
| Harris 2019 | 275 | 426 | 55 | 102 | 1.0% | 1.56 [1.01, 2.41] |
| Odell 2019 | 46 | 281 | 14 | 129 | 0.7% | 1.61 [0.85, 3.04] |
| Tracy 2019 | 181 | 1086 | 29 | 274 | 1.0% | 1.69 [1.11, 2.56] |
| Carpenter 2018 | 45 | 117 | 7 | 29 | 0.4% | 1.96 [0.78, 4.97] |
| Dermody 2019 | 35 | 161 | 6 | 49 | 0.4% | 1.99 [0.78, 5.06] |
| Yue 2020 | 98 | 618 | 8 | 95 | 0.6% | 2.05 [0.96, 4.36] |
| Gambhir 2020 | 65 | 120 | 18 | 50 | 0.7% | 2.10 [1.06, 4.15] |
| Okoshi 2014 | 33 | 326 | 2 | 40 | 0.2% | 2.14 [0.49, 9.28] |
| Hoof 2020 | 852 | 3889 | 50 | 434 | 1.2% | 2.15 [1.59, 2.92] |
| Ilin 2020 | 55 | 164 | 4 | 22 | 0.3% | 2.27 [0.73, 7.04] |
| Valsangkar 2016 | 1272 | 3087 | 211 | 928 | 1.5% | 2.38 [2.01, 2.82] |
| Tulunay-Udur 2019 | 72 | 152 | 18 | 66 | 0.7% | 2.40 [1.28, 4.50] |
| Maxwell 2020 | 275 | 1543 | 36 | 450 | 1.1% | 2.49 [1.73, 3.59] |
| Hill 2018 | 134 | 420 | 39 | 257 | 1.1% | 2.62 [1.76, 3.90] |
| Baptiste 2017 | 28 | 91 | 4 | 28 | 0.3% | 2.67 [0.85, 8.41] |
| Cheng 2019 | 178 | 534 | 17 | 108 | 0.8% | 2.68 [1.55, 4.63] |
| Okafor 2020 | 402 | 1342 | 56 | 412 | 1.2% | 2.72 [2.00, 3.69] |
| Florence 2011 | 71 | 154 | 4 | 17 | 0.3% | 2.78 [0.87, 8.91] |
| Ngaage 2019 | 447 | 1350 | 19 | 131 | 0.9% | 2.92 [1.77, 4.81] |
| Cuoco 2020 | 59 | 74 | 4 | 7 | 0.2% | 2.95 [0.60, 14.62] |
| Carnevale 2020 | 203 | 774 | 19 | 177 | 0.9% | 2.96 [1.79, 4.88] |
| Mayer 2017 | 441 | 1686 | 25 | 236 | 1.0% | 2.99 [1.95, 4.59] |
| Wu 2019 | 122 | 220 | 14 | 48 | 0.7% | 3.02 [1.54, 5.95] |
| Eloy 2013 | 553 | 1740 | 66 | 534 | 1.3% | 3.30 [2.50, 4.36] |
| Epperson 2020 | 185 | 343 | 16 | 62 | 0.7% | 3.37 [1.83, 6.18] |
| Garstka 2019 | 210 | 579 | 22 | 153 | 0.9% | 3.39 [2.09, 5.49] |
| Jena 2015 | 3925 | 13010 | 436 | 3873 | 1.5% | 3.41 [3.06, 3.79] |
| Dossani 2019 | 298 | 761 | 12 | 80 | 0.7% | 3.65 [1.94, 6.85] |
| Rosso 2019 | 10 | 22 | 2 | 11 | 0.2% | 3.75 [0.65, 21.51] |
| Santosa 2019 | 115 | 241 | 8 | 41 | 0.5% | 3.76 [1.67, 8.49] |
| Paik 2014 | 172 | 426 | 12 | 79 | 0.7% | 3.78 [1.99, 7.20] |
| Gawad 2020 | 75 | 294 | 9 | 111 | 0.6% | 3.88 [1.87, 8.06] |
| Breyer 2020 | 1092 | 2597 | 51 | 329 | 1.2% | 3.96 [2.91, 5.38] |
| Pace 2020 | 509 | 1433 | 27 | 224 | 1.0% | 4.02 [2.65, 6.10] |
| Herrick-Reynolds 2019 | 48 | 914 | 2 | 149 | 0.2% | 4.07 [0.98, 16.94] |
| Selter 2020 | 71 | 129 | 30 | 134 | 0.9% | 4.24 [2.49, 7.24] |
| Chandrasekar 2020 | 545 | 1894 | 27 | 320 | 1.1% | 4.38 [2.92, 6.58] |
| Awad 2017 | 16 | 137 | 1 | 36 | 0.1% | 4.63 [0.59, 36.13] |
| Sperling 2019 | 174 | 312 | 72 | 347 | 1.2% | 4.82 [3.42, 6.78] |
| Day 2010 | 550 | 2119 | 22 | 327 | 1.0% | 4.86 [3.12, 7.57] |
| Bernardi 2020 | 6 | 22 | 1 | 14 | 0.1% | 4.88 [0.52, 45.79] |
| Smith 2019 | 217 | 663 | 14 | 159 | 0.8% | 5.04 [2.84, 8.93] |
| Schroen 2004 | 64 | 168 | 16 | 149 | 0.8% | 5.12 [2.79, 9.37] |
| Wise 2004 | 59 | 187 | 9 | 112 | 0.6% | 5.28 [2.50, 11.14] |
| Inam 2020 | 10 | 67 | 1 | 32 | 0.1% | 5.44 [0.66, 44.49] |
| LaRocca 2018 | 108 | 216 | 14 | 92 | 0.7% | 5.57 [2.97, 10.44] |
| Sasor 2018 | 194 | 678 | 9 | 136 | 0.6% | 5.66 [2.82, 11.35] |
| Nakata 2017 | 21 | 121 | 0 | 13 | 0.1% | 5.78 [0.33, 100.94] |
| Johnson 1992 | 3147 | 9672 | 128 | 1711 | 1.4% | 5.96 [4.96, 7.18] |

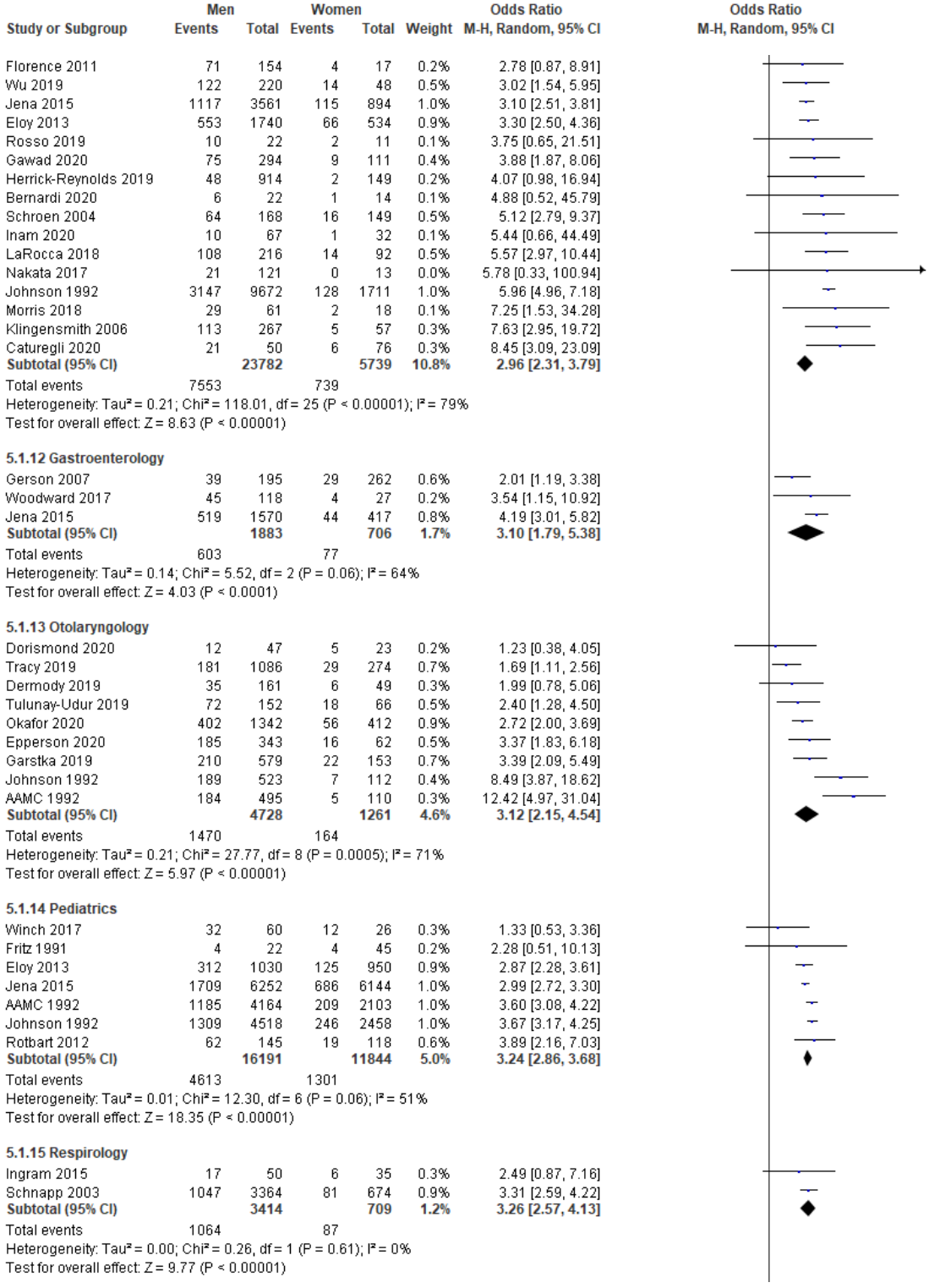


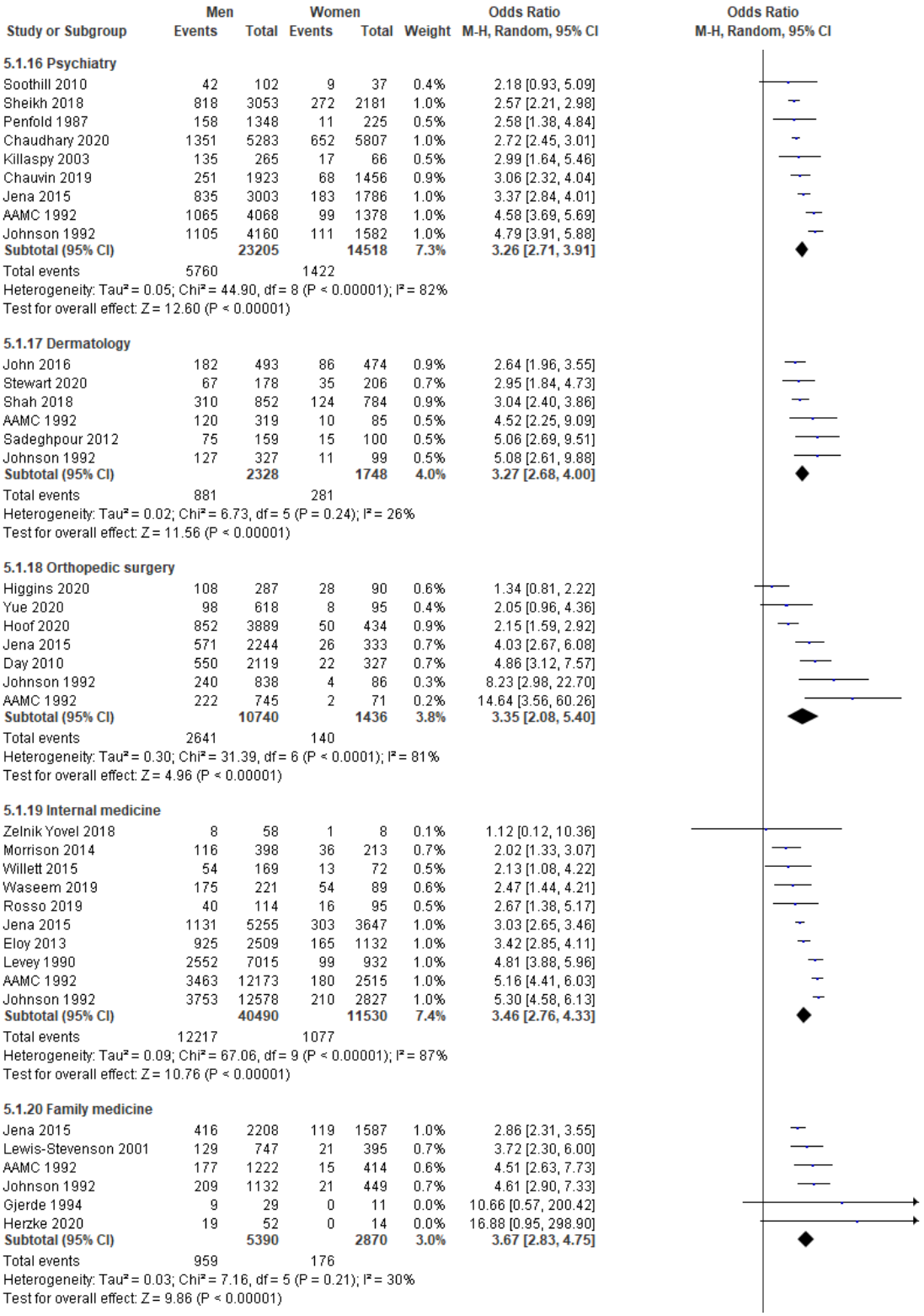


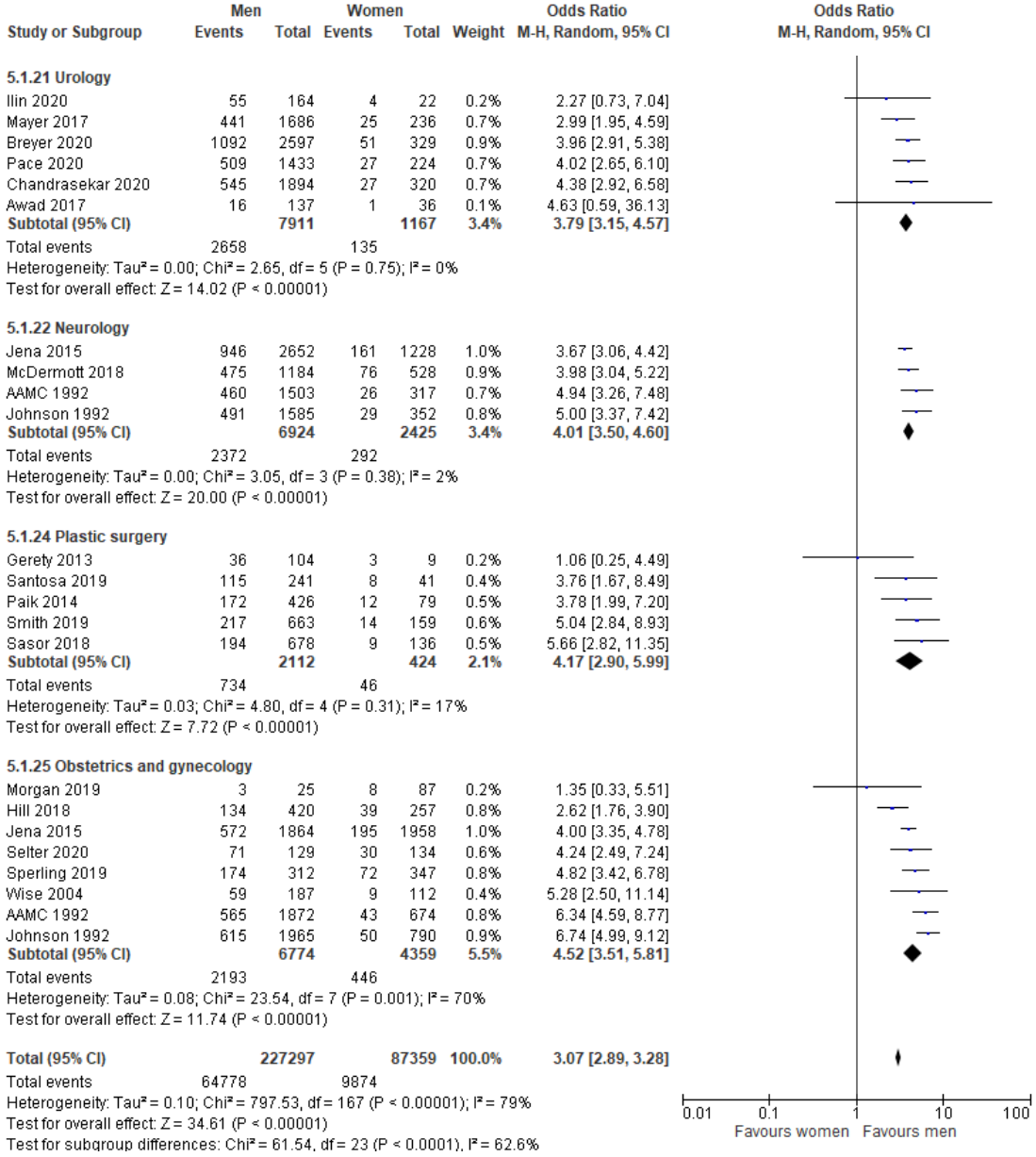

**eFigure 4.** Subgroup analysis of full professorship for surgical versus non-surgical specialties. M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).


Men
Women
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
5.1.1 Neurosurgery
Shaikh 2019 162 294 19 34 0.5% 0.97 [0.47, 1.98]
Odell 2019 46 281 14 129 0.5% 1.61 [0.85, 3.04]
Ngaage 2019 447 1350 19 131 0.6% 2.92 [1.77, 4.81]
Cuoco 2020 59 74 4 7 0.1% 2.95 [0.60, 14.62]
Dossani 2019 298 761 12 80 0.5% 3.65 [1.94, 6.85]
Subtotal (95% CI) 2760 381 2.3% 2.14 [1.30, 3.52]
Total events 1012 68
Heterogeneity: Tau² = 0.18; Chi² = 9.88, df = 4 (P = 0.04); I² = 59%
Test for overall effect: Z = 2.99 (P = 0.003)
5.1.2 Cardiology
Jena 2015 1044 3337 115 659 1.0% 2.15 [1.74, 2.67]
Khan 2019 151 349 14 63 0.5% 2.67 [1.42, 5.01]
Tsukada 2009 5 58 0 25 0.0% 5.24 [0.28, 98.51]
Subtotal (95% CI) 3744 747 1.5% 2.21 [1.81, 2.71]
Total events 1200 129
Heterogeneity: Tau² = 0.00; Chi² = 0.73, df = 2 (P = 0.69); I² = 0%
Test for overall effect: Z = 7.69 (P < 0.00001)
5.1.3 Radiology
Wilson 2013 19 66 7 21 0.3% 0.81 [0.28, 2.32]
Ahmadi 2018 77 358 21 107 0.6% 1.12 [0.65, 1.92]
Beeler 2020 73 135 27 55 0.5% 1.22 [0.65, 2.29]
Guss 2019 60 166 20 81 0.6% 1.73 [0.95, 3.13]
Jena 2015 918 3573 230 1430 1.0% 1.80 [1.54, 2.12]
Harrington 2018 33 144 10 71 0.4% 1.81 [0.84, 3.93]
Deipolyi 2020 82 734 7 108 0.4% 1.81 [0.82, 4.04]
Niu 2020 1497 5795 332 2256 1.0% 2.02 [1.77, 2.30]
Qamar 2018 45 190 11 84 0.5% 2.06 [1.01, 4.22]
Vydareny 2000 176 494 40 213 0.8% 2.39 [1.62, 3.54]
Qamar 2020 89 597 22 335 0.7% 2.49 [1.53, 4.06]
O'Neill 2019 501 1592 31 219 0.8% 2.78 [1.88, 4.13]
Moghimi 2019 62 179 9 58 0.4% 2.89 [1.33, 6.26]
Khurshid 2018 24 112 22 258 0.5% 2.93 [1.56, 5.48]
Holliday 2014 211 738 34 293 0.8% 3.05 [2.06, 4.51]
McDonald 2017 162 519 22 192 0.7% 3.51 [2.17, 5.67]
Johnson 1992 1039 3582 79 823 0.9% 3.85 [3.02, 4.91]
AAMC 1992 973 3453 61 726 0.9% 4.28 [3.26, 5.62]
Kirchner 1984 626 2407 23 333 0.7% 4.74 [3.07, 7.31]
Battaglia 2019 7 77 0 22 0.0% 4.79 [0.26, 87.16]
Whitley 1987 332 999 18 200 0.6% 5.03 [3.05, 8.31]
Subtotal (95% CI) 25910 7885 13.0% 2.51 [2.07, 3.04]
Total events 7006 1026
Heterogeneity: Tau² = 0.12; Chi² = 93.52, df = 20 (P < 0.00001); I² = 79%
Test for overall effect: Z = 9.44 (P < 0.00001)
5.1.4 Anesthesia
Kenevan 2018 20 101 8 46 0.3% 1.17 [0.47, 2.90]
Aliano 2020 272 654 334 965 1.0% 1.35 [1.10, 1.65]
Burns 2019 22 192 6 73 0.3% 1.45 [0.56, 3.72]
Long 2010 33 94 10 37 0.4% 1.46 [0.63, 3.38]
Orhurhu 2020 84 520 18 176 0.6% 1.69 [0.98, 2.90]
Jena 2015 723 3914 151 1743 1.0% 2.39 [1.98, 2.88]
Bissing 2019 906 5236 218 2945 1.0% 2.62 [2.24, 3.06]
Muffly 2016 138 760 43 575 0.8% 2.74 [1.91, 3.94]
AAMC 1992 391 2268 33 690 0.8% 4.15 [2.88, 5.98]
Johnson 1992 436 2417 34 751 0.8% 4.64 [3.24, 6.65]
Wilkinson 1986 332 1388 17 374 0.6% 6.60 [4.00, 10.91]
Rosso 2019 4 12 0 12 0.0% 13.24 [0.63, 279.16]
Subtotal (95% CI) 17556 8387 7.6% 2.55 [1.92, 3.38]
Total events 3361 872
Heterogeneity: Tau² = 0.17; Chi² = 78.78, df = 11 (P < 0.00001); I² = 86%
Test for overall effect: Z = 6.45 (P < 0.00001)

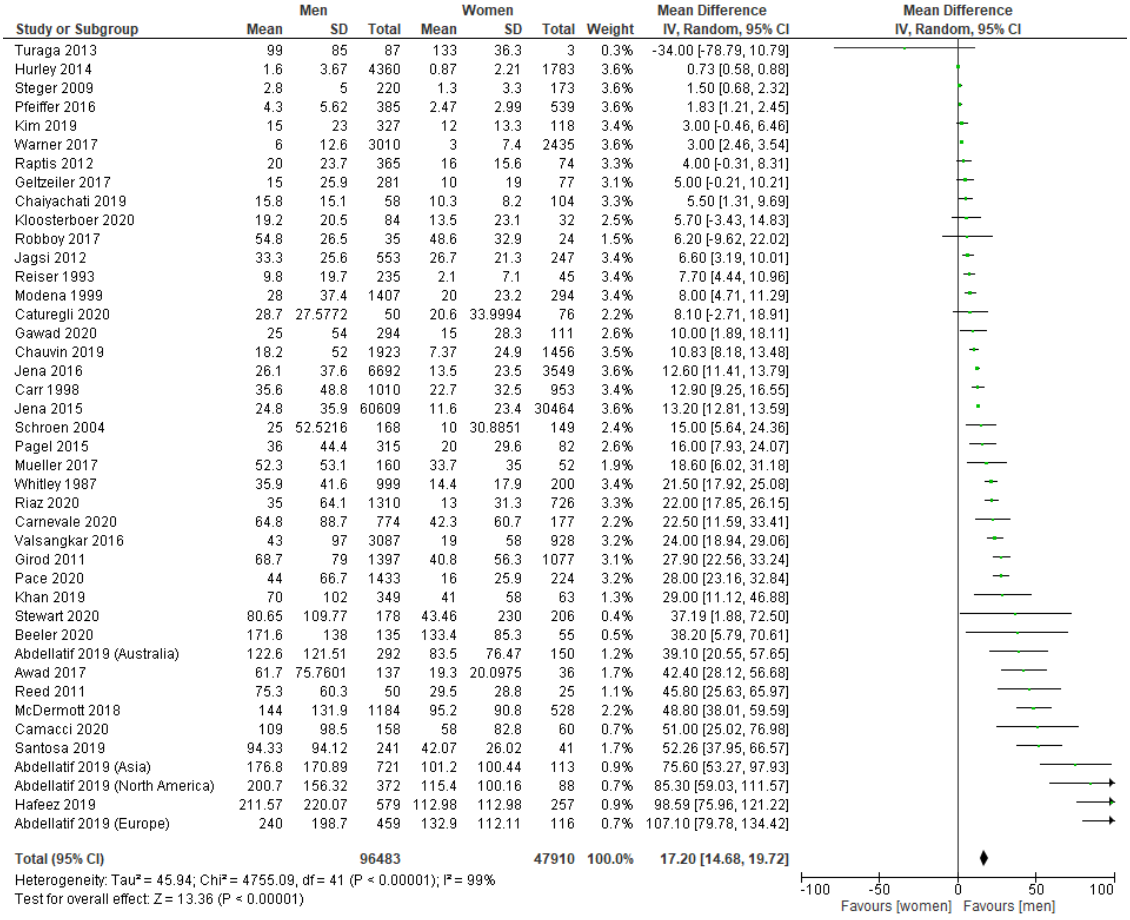

Men
Women
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
5.1.5 Emergency medicine
Jacquet 2014 21 81 7 35 0.3% 1.40 [0.53, 3.68]
Dubosh 2019 8 77 2 32 0.1% 1.74 [0.35, 8.68]
Jena 2015 330 2507 56 1001 0.9% 2.56 [1.91, 3.43]
Madsen 2017 156 924 29 447 0.7% 2.93 [1.94, 4.43]
Cydulka 2000 65 923 6 274 0.4% 3.38 [1.45, 7.90]
Subtotal (95% CI) 4512 1789 2.4% 2.61 [2.09, 3.25]
Total events 580 100
Heterogeneity: Tau² = 0.00; Chi² = 2.52, df = 4 (P = 0.64); I² = 0%
Test for overall effect: Z = 8.46 (P < 0.00001)
5.1.6 Hematology
Riaz 2020 506 1310 142 726 1.0% 2.59 [2.09, 3.21]
Jena 2015 831 2199 176 949 1.0% 2.67 [2.22, 3.21]
Subtotal (95% CI) 3509 1675 1.9% 2.63 [2.29, 3.03]
Total events 1337 318
Heterogeneity: Tau² = 0.00; Chi² = 0.04, df = 1 (P = 0.83); I² = 0%
Test for overall effect: Z = 13.55 (P < 0.00001)
5.1.7 Pathology
Kilic 2020 1005 2436 415 1755 1.0% 2.27 [1.98, 2.60]
Johnson 1992 411 1239 53 320 0.8% 2.50 [1.82, 3.44]
Jena 2015 826 2112 262 1367 1.0% 2.71 [2.31, 3.18]
AAMC 1992 1079 3089 114 935 1.0% 3.87 [3.14, 4.77]
Subtotal (95% CI) 8876 4377 3.8% 2.77 [2.20, 3.50]
Total events 3321 844
Heterogeneity: Tau² = 0.05; Chi² = 17.80, df = 3 (P = 0.0005); I² = 83%
Test for overall effect: Z = 8.58 (P < 0.00001)
5.1.8 Vascular surgery
Cheng 2019 178 534 17 108 0.6% 2.68 [1.55, 4.63]
Carnevale 2020 203 774 19 177 0.6% 2.96 [1.79, 4.88]
Subtotal (95% CI) 1308 285 1.2% 2.83 [1.95, 4.09]
Total events 381 36
Heterogeneity: Tau² = 0.00; Chi² = 0.07, df = 1 (P = 0.79); I² = 0%
Test for overall effect: Z = 5.50 (P < 0.00001)
5.1.9 Physical medicine and rehabilitation
Houtrow 2020 5 68 5 130 0.2% 1.98 [0.55, 7.11]
Yang 2019 129 653 34 392 0.8% 2.59 [1.74, 3.87]
Wagner 2007 49 164 9 107 0.4% 4.64 [2.17, 9.92]
Subtotal (95% CI) 885 629 1.4% 2.88 [2.00, 4.15]
Total events 183 48
Heterogeneity: Tau² = 0.01; Chi² = 2.11, df = 2 (P = 0.35); I² = 5%
Test for overall effect: Z = 5.68 (P < 0.00001)
5.1.10 Ophthalmology
McAlister 2014 23 283 6 102 0.3% 1.42 [0.56, 3.58]
Johnson 1992 342 1107 24 200 0.7% 3.28 [2.10, 5.12]
AAMC 1992 314 975 20 166 0.7% 3.47 [2.13, 5.64]
Subtotal (95% CI) 2365 468 1.7% 2.94 [1.97, 4.39]
Total events 679 50
Heterogeneity: Tau² = 0.04; Chi² = 3.00, df = 2 (P = 0.22); I² = 33%
Test for overall effect: Z = 5.28 (P < 0.00001)
5.1.11 General surgery
Schwarz 2016 7 200 11 93 0.3% 0.27 [0.10, 0.72]
Hoops 2018 8 42 3 16 0.2% 1.02 [0.23, 4.45]
Geltzeiler 2017 30 281 7 77 0.4% 1.20 [0.50, 2.84]
Harris 2019 275 426 55 102 0.7% 1.56 [1.01, 2.41]
Carpenter 2018 45 117 7 29 0.3% 1.96 [0.78, 4.97]
Gambhir 2020 65 120 18 50 0.5% 2.10 [1.06, 4.15]
Okoshi 2014 33 326 2 40 0.2% 2.14 [0.49, 9.28]
Valsangkar 2016 1272 3087 211 928 1.0% 2.38 [2.01, 2.82]
Maxwell 2020 275 1543 36 450 0.8% 2.49 [1.73, 3.59]
Baptiste 2017 28 91 4 28 0.2% 2.67 [0.85, 8.41]


Men
Women
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
Florence 2011 71 154 4 17 0.2% 2.78 [0.87, 8.91]
Wu 2019 122 220 14 48 0.5% 3.02 [1.54, 5.95]
Jena 2015 1117 3561 115 894 1.0% 3.10 [2.51, 3.81]
Eloy 2013 553 1740 66 534 0.9% 3.30 [2.50, 4.36]
Rosso 2019 10 22 2 11 0.1% 3.75 [0.65, 21.51]
Gawad 2020 75 294 9 111 0.4% 3.88 [1.87, 8.06]
Herrick-Reynolds 2019 48 914 2 149 0.2% 4.07 [0.98, 16.94]
Bernardi 2020 6 22 1 14 0.1% 4.88 [0.52, 45.79]
Schroen 2004 64 168 16 149 0.5% 5.12 [2.79, 9.37]
Inam 2020 10 67 1 32 0.1% 5.44 [0.66, 44.49]
LaRocca 2018 108 216 14 92 0.5% 5.57 [2.97, 10.44]
Nakata 2017 21 121 0 13 0.0% 5.78 [0.33, 100.94]
Johnson 1992 3147 9672 128 1711 1.0% 5.96 [4.96, 7.18]
Morris 2018 29 61 2 18 0.1% 7.25 [1.53, 34.28]
Klingensmith 2006 113 267 5 57 0.3% 7.63 [2.95, 19.72]
Caturegli 2020 21 50 6 76 0.3% 8.45 [3.09, 23.09]
Subtotal (95% CI) 23782 5739 10.8% 2.96 [2.31, 3.79]
Total events 7553 739
Heterogeneity: Tau² = 0.21; Chi² = 118.01, df = 25 (P < 0.00001); I² = 79%
Test for overall effect: Z = 8.63 (P < 0.00001)
5.1.12 Gastroenterology
Gerson 2007 39 195 29 262 0.6% 2.01 [1.19, 3.38]
Woodward 2017 45 118 4 27 0.2% 3.54 [1.15, 10.92]
Jena 2015 519 1570 44 417 0.8% 4.19 [3.01, 5.82]
Subtotal (95% CI) 1883 706 1.7% 3.10 [1.79, 5.38]
Total events 603 77
Heterogeneity: Tau² = 0.14; Chi² = 5.52, df = 2 (P = 0.06); I² = 64%
Test for overall effect: Z = 4.03 (P < 0.0001)
5.1.13 Otolaryngology
Dorismond 2020 12 47 5 23 0.2% 1.23 [0.38, 4.05]
Tracy 2019 181 1086 29 274 0.7% 1.69 [1.11, 2.56]
Dermody 2019 35 161 6 49 0.3% 1.99 [0.78, 5.06]
Tulunay-Udur 2019 72 152 18 66 0.5% 2.40 [1.28, 4.50]
Okafor 2020 402 1342 56 412 0.9% 2.72 [2.00, 3.69]
Epperson 2020 185 343 16 62 0.5% 3.37 [1.83, 6.18]
Garstka 2019 210 579 22 153 0.7% 3.39 [2.09, 5.49]
Johnson 1992 189 523 7 112 0.4% 8.49 [3.87, 18.62]
AAMC 1992 184 495 5 110 0.3% 12.42 [4.97, 31.04]
Subtotal (95% CI) 4728 1261 4.6% 3.12 [2.15, 4.54]
Total events 1470 164
Heterogeneity: Tau² = 0.21; Chi² = 27.77, df = 8 (P = 0.0005); I² = 71%
Test for overall effect: Z = 5.97 (P < 0.00001)
5.1.14 Pediatrics
Winch 2017 32 60 12 26 0.3% 1.33 [0.53, 3.36]
Fritz 1991 4 22 4 45 0.2% 2.28 [0.51, 10.13]
Eloy 2013 312 1030 125 950 0.9% 2.87 [2.28, 3.61]
Jena 2015 1709 6252 686 6144 1.0% 2.99 [2.72, 3.30]
AAMC 1992 1185 4164 209 2103 1.0% 3.60 [3.08, 4.22]
Johnson 1992 1309 4518 246 2458 1.0% 3.67 [3.17, 4.25]
Rotbart 2012 62 145 19 118 0.6% 3.89 [2.16, 7.03]
Subtotal (95% CI) 16191 11844 5.0% 3.24 [2.86, 3.68]
Total events 4613 1301
Heterogeneity: Tau² = 0.01; Chi² = 12.30, df = 6 (P = 0.06); I² = 51%
Test for overall effect: Z = 18.35 (P < 0.00001)
5.1.15 Respirology
Ingram 2015 17 50 6 35 0.3% 2.49 [0.87, 7.16]
Schnapp 2003 1047 3364 81 674 0.9% 3.31 [2.59, 4.22]
Subtotal (95% CI) 3414 709 1.2% 3.26 [2.57, 4.13]
Total events 1064 87
Heterogeneity: Tau² = 0.00; Chi² = 0.26, df = 1 (P = 0.61); I² = 0%
Test for overall effect: Z = 9.77 (P < 0.00001)

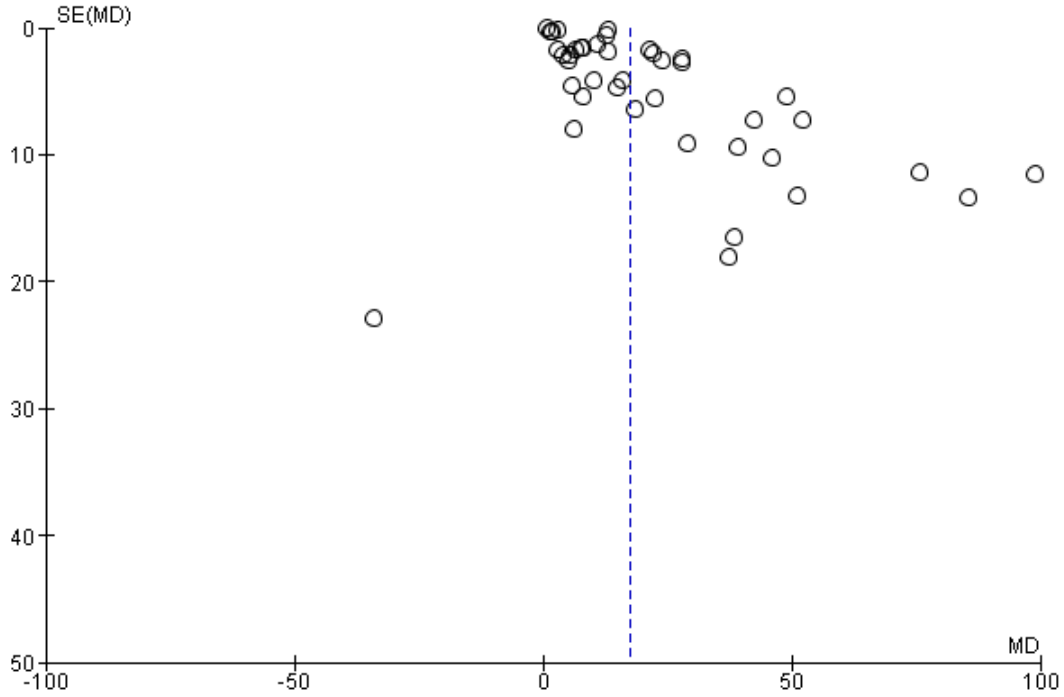

Men
Women
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
5.1.16 Psychiatry
Soothill 2010 42 102 9 37 0.4% 2.18 [0.93, 5.09]
Sheikh 2018 818 3053 272 2181 1.0% 2.57 [2.21, 2.98]
Penfold 1987 158 1348 11 225 0.5% 2.58 [1.38, 4.84]
Chaudhary 2020 1351 5283 652 5807 1.0% 2.72 [2.45, 3.01]
Killaspy 2003 135 265 17 66 0.5% 2.99 [1.64, 5.46]
Chauvin 2019 251 1923 68 1456 0.9% 3.06 [2.32, 4.04]
Jena 2015 835 3003 183 1786 1.0% 3.37 [2.84, 4.01]
AAMC 1992 1065 4068 99 1378 1.0% 4.58 [3.69, 5.69]
Johnson 1992 1105 4160 111 1582 1.0% 4.79 [3.91, 5.88]
Subtotal (95% CI) 23205 14518 7.3% 3.26 [2.71, 3.91]
Total events 5760 1422
Heterogeneity: Tau² = 0.05; Chi² = 44.90, df = 8 (P < 0.00001); I² = 82%
Test for overall effect: Z = 12.60 (P < 0.00001)
5.1.17 Dermatology
John 2016 182 493 86 474 0.9% 2.64 [1.96, 3.55]
Stewart 2020 67 178 35 206 0.7% 2.95 [1.84, 4.73]
Shah 2018 310 852 124 784 0.9% 3.04 [2.40, 3.86]
AAMC 1992 120 319 10 85 0.5% 4.52 [2.25, 9.09]
Sadeghpour 2012 75 159 15 100 0.5% 5.06 [2.69, 9.51]
Johnson 1992 127 327 11 99 0.5% 5.08 [2.61, 9.88]
Subtotal (95% CI) 2328 1748 4.0% 3.27 [2.68, 4.00]
Total events 881 281
Heterogeneity: Tau² = 0.02; Chi² = 6.73, df = 5 (P = 0.24); I² = 26%
Test for overall effect: Z = 11.56 (P < 0.00001)
5.1.18 Orthopedic surgery
Higgins 2020 108 287 28 90 0.6% 1.34 [0.81, 2.22]
Yue 2020 98 618 8 95 0.4% 2.05 [0.96, 4.36]
Hoof 2020 852 3889 50 434 0.9% 2.15 [1.59, 2.92]
Jena 2015 571 2244 26 333 0.7% 4.03 [2.67, 6.08]
Day 2010 550 2119 22 327 0.7% 4.86 [3.12, 7.57]
Johnson 1992 240 838 4 86 0.3% 8.23 [2.98, 22.70]
AAMC 1992 222 745 2 71 0.2% 14.64 [3.56, 60.26]
Subtotal (95% CI) 10740 1436 3.8% 3.35 [2.08, 5.40]
Total events 2641 140
Heterogeneity: Tau² = 0.30; Chi² = 31.39, df = 6 (P < 0.0001); I² = 81%
Test for overall effect: Z = 4.96 (P < 0.00001)
5.1.19 Internal medicine
Zelnik Yovel 2018 8 58 1 8 0.1% 1.12 [0.12, 10.36]
Morrison 2014 116 398 36 213 0.7% 2.02 [1.33, 3.07]
Willett 2015 54 169 13 72 0.5% 2.13 [1.08, 4.22]
Waseem 2019 175 221 54 89 0.6% 2.47 [1.44, 4.21]
Rosso 2019 40 114 16 95 0.5% 2.67 [1.38, 5.17]
Jena 2015 1131 5255 303 3647 1.0% 3.03 [2.65, 3.46]
Eloy 2013 925 2509 165 1132 1.0% 3.42 [2.85, 4.11]
Levey 1990 2552 7015 99 932 1.0% 4.81 [3.88, 5.96]
AAMC 1992 3463 12173 180 2515 1.0% 5.16 [4.41, 6.03]
Johnson 1992 3753 12578 210 2827 1.0% 5.30 [4.58, 6.13]
Subtotal (95% CI) 40490 11530 7.4% 3.46 [2.76, 4.33]
Total events 12217 1077
Heterogeneity: Tau² = 0.09; Chi² = 67.06, df = 9 (P < 0.00001); I² = 87%
Test for overall effect: Z = 10.76 (P < 0.00001)
5.1.20 Family medicine
Jena 2015 416 2208 119 1587 1.0% 2.86 [2.31, 3.55]
Lewis-Stevenson 2001 129 747 21 395 0.7% 3.72 [2.30, 6.00]
AAMC 1992 177 1222 15 414 0.6% 4.51 [2.63, 7.73]
Johnson 1992 209 1132 21 449 0.7% 4.61 [2.90, 7.33]
Gjerde 1994 9 29 0 11 0.0% 10.66 [0.57, 200.42]
Herzke 2020 19 52 0 14 0.0% 16.88 [0.95, 298.90]
Subtotal (95% CI) 5390 2870 3.0% 3.67 [2.83, 4.75]
Total events 959 176
Heterogeneity: Tau² = 0.03; Chi² = 7.16, df = 5 (P = 0.21); I² = 30%
Test for overall effect: Z = 9.86 (P < 0.00001)



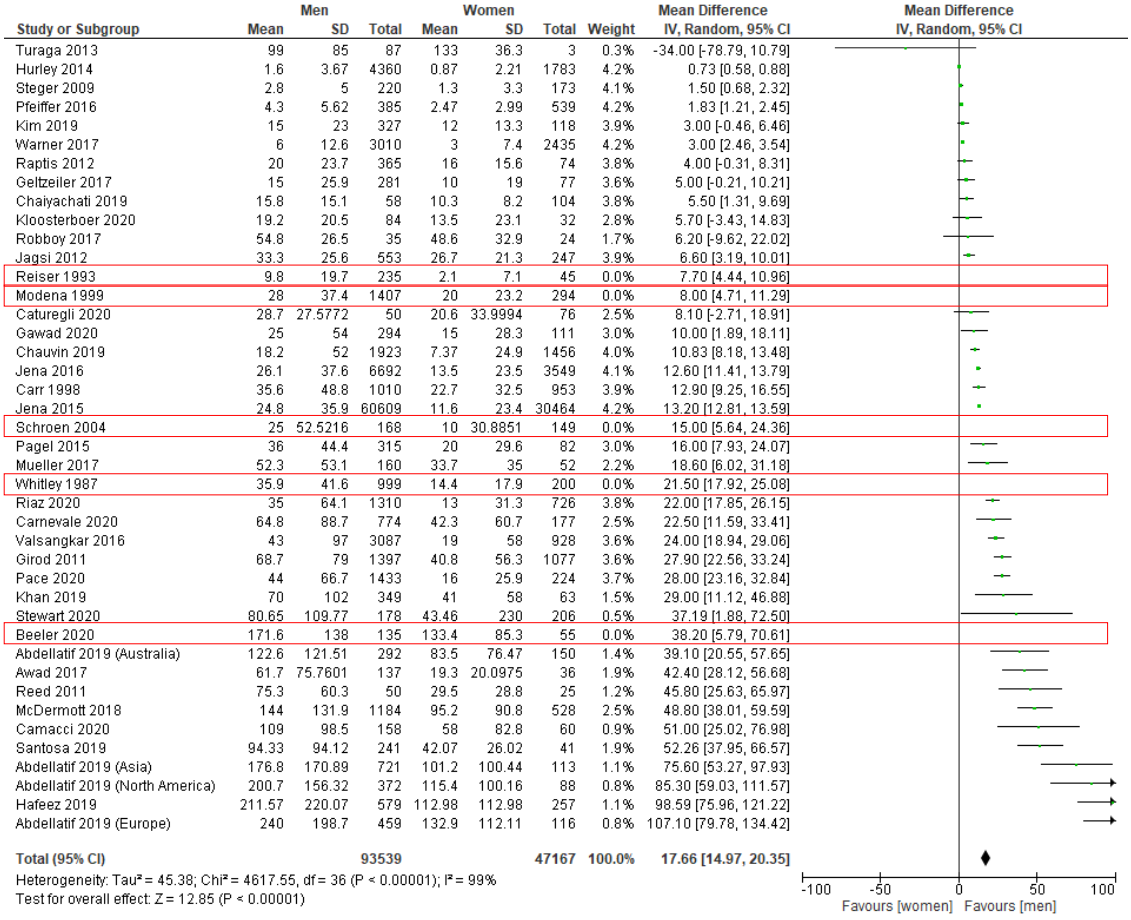

**eFigure 5.** Subgroup analysis of full professorship by specialty. M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).



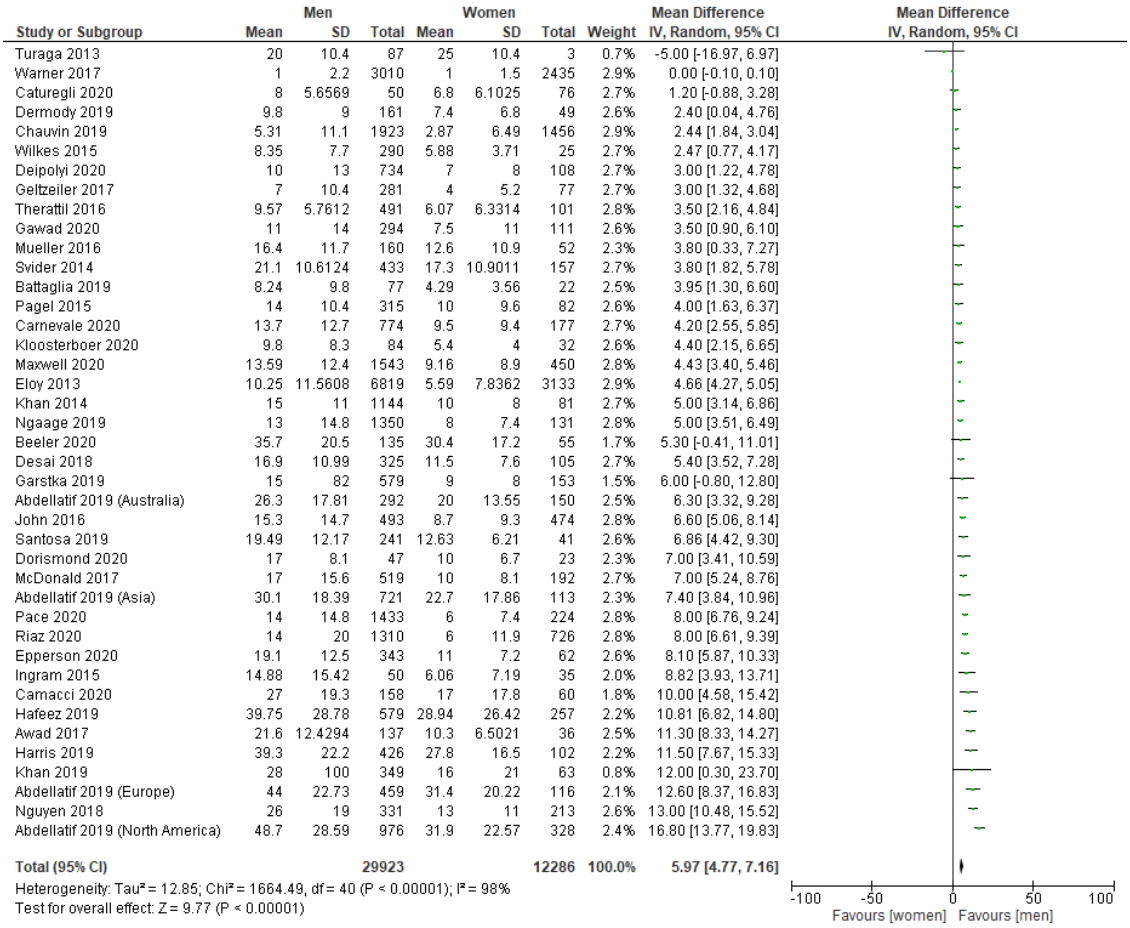

**eFigure 6.** Forest plot of total publications for men vs. women. SD (standard deviation), Total (sample size), IV (inverse variance), Random (random effects model), CI (confidence interval).



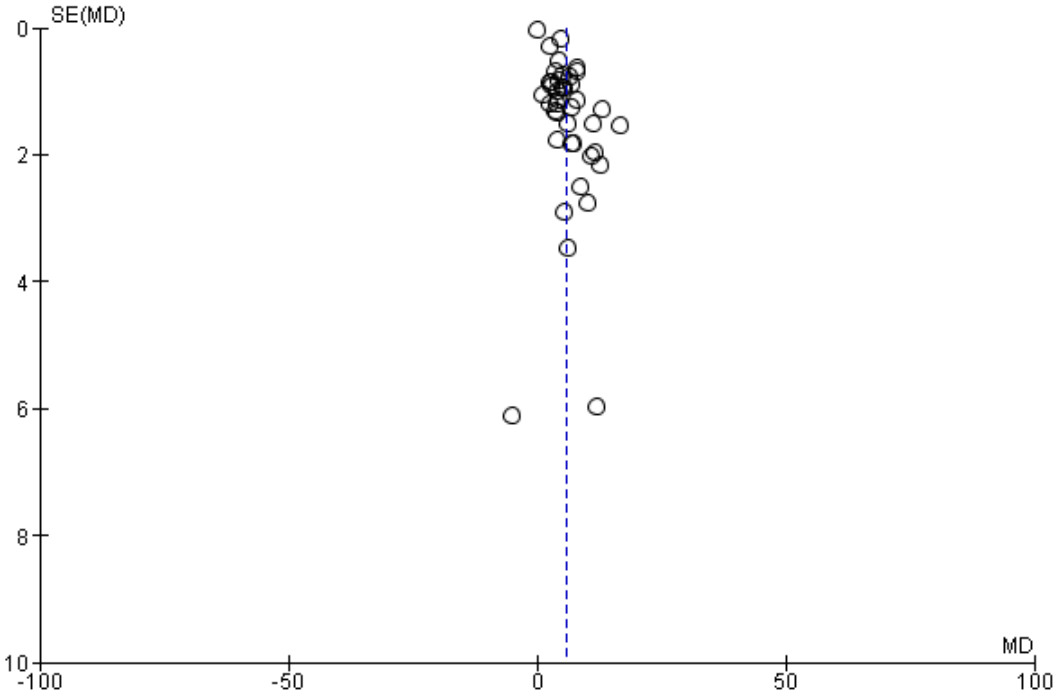

**eFigure 7.** Funnel plot of total publications for men vs. women. SE (standard error), MD (mean difference).



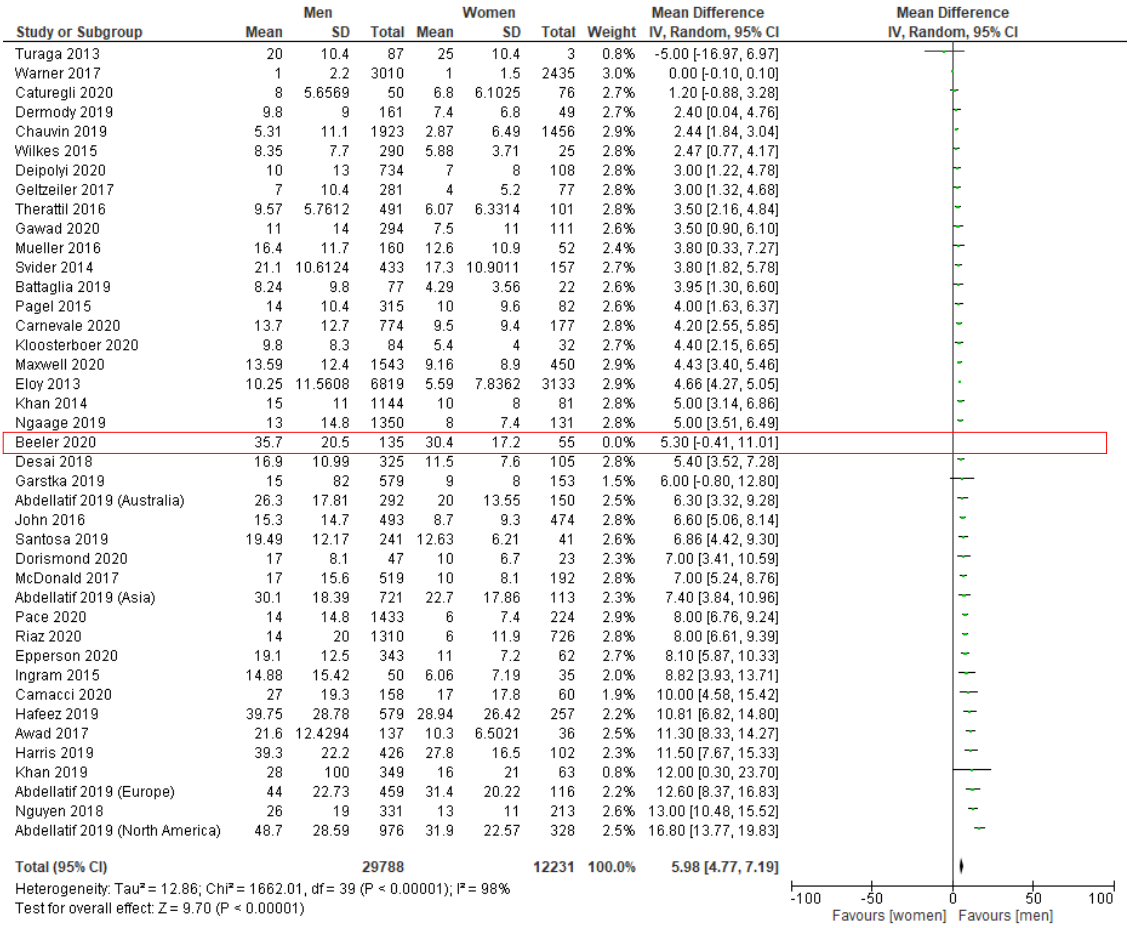

**eFigure 8.** Sensitivity analysis of total publications removing 5 poor quality studies (red box, NOS score ≤ 5). SD (standard deviation), Total (sample size), IV (inverse variance), Random (random effects model), CI (confidence interval).



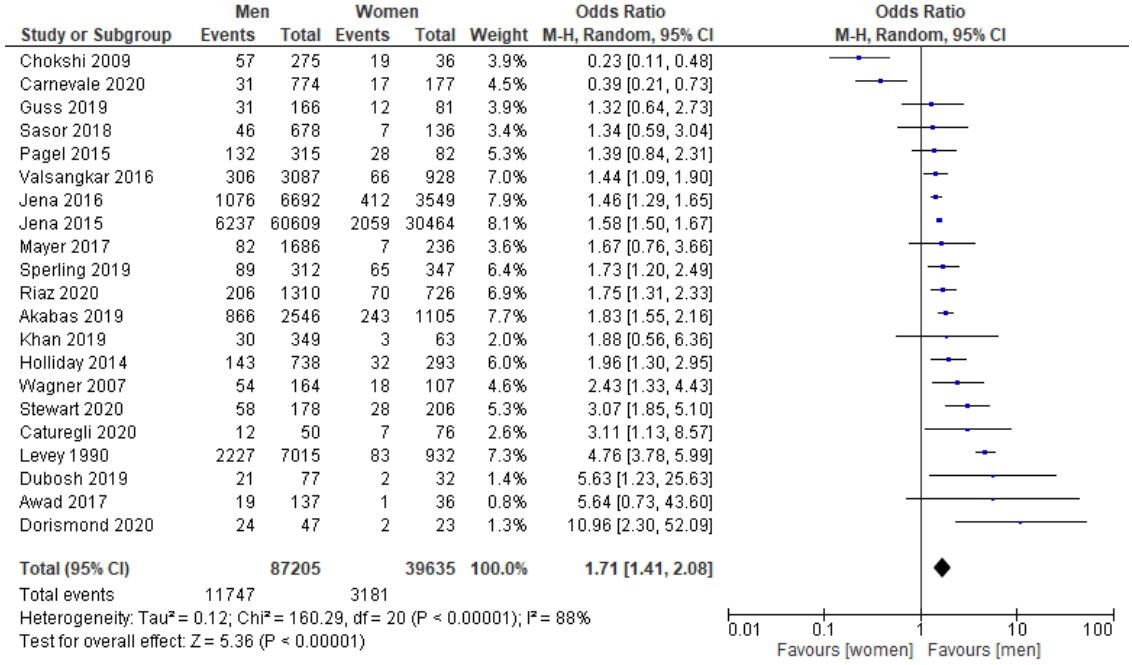

**eFigure 9.** Forest plot of h-index for men vs. women. SD (standard deviation), Total (sample size), IV (inverse variance), Random (random effects model), CI (confidence interval).



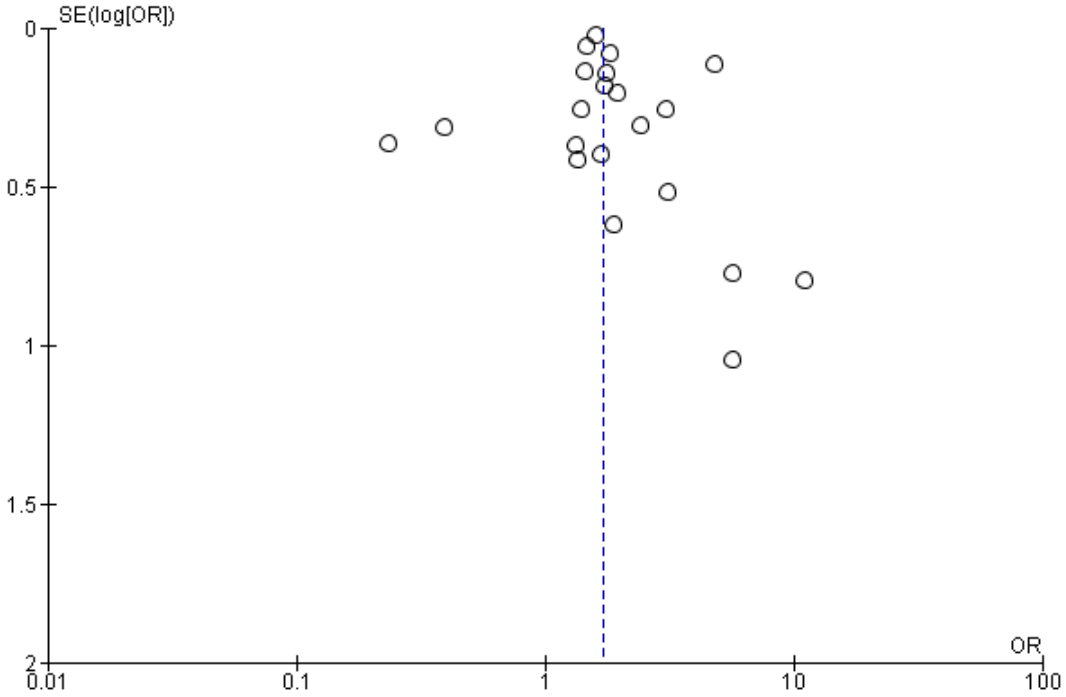

**eFigure 10.** Funnel plot of h-index for men vs. women. SE (standard error), MD (mean difference).



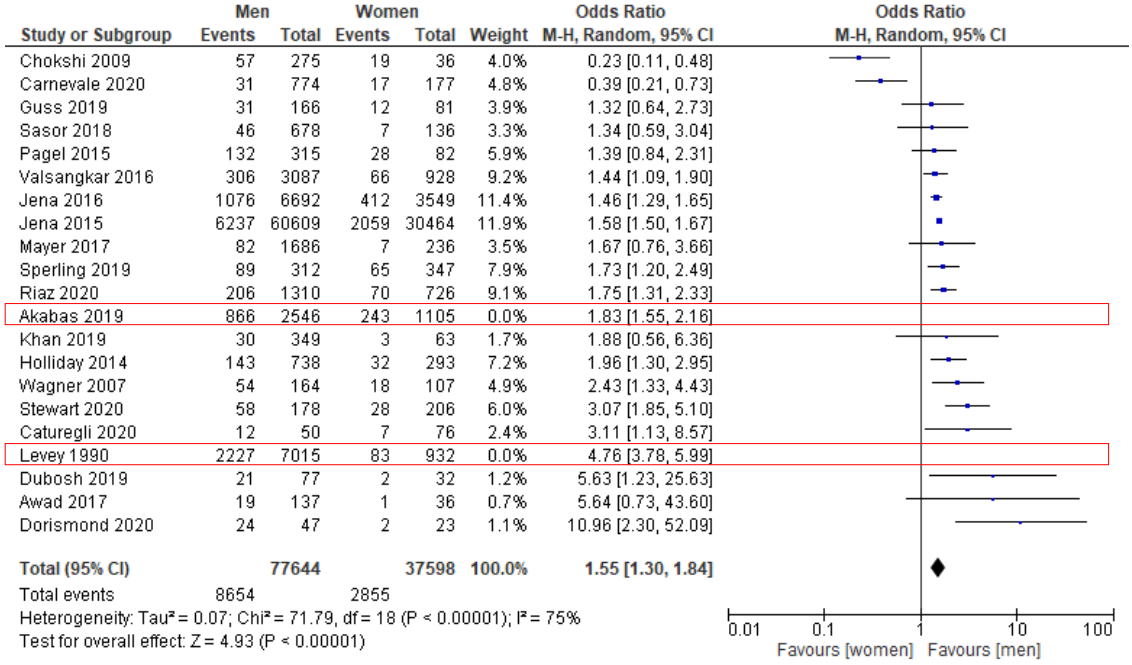

**eFigure 11.** Sensitivity analysis of h-index removing 1 poor quality study (red box, NOS score ≤ 5). SD (standard deviation), Total (sample size), IV (inverse variance), Random (random effects model), CI (confidence interval).



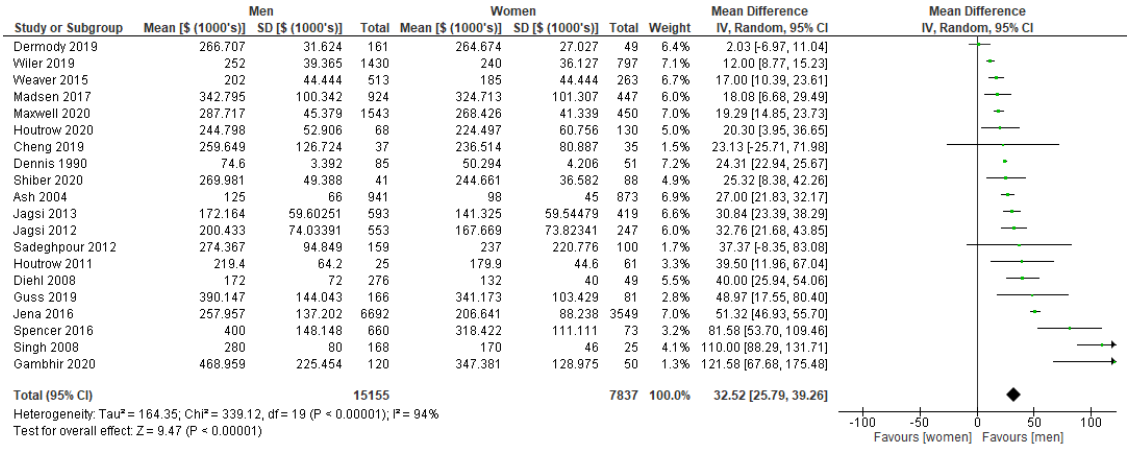

**eFigure 12.** Forest plot of federal research funding for men vs. women. M-H (Mantel-Haenszel), Random (random effects), CI (confidence interval).



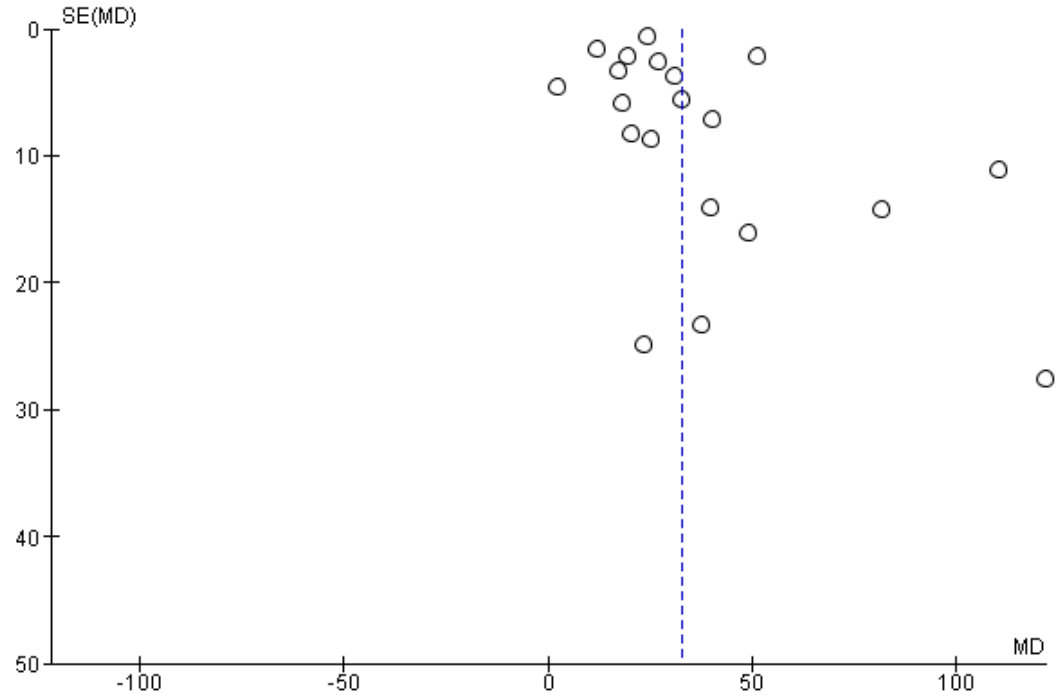

**eFigure 13.** Funnel plot of federal research funding for men vs. women. SE (standard error), OR (odds ratio).



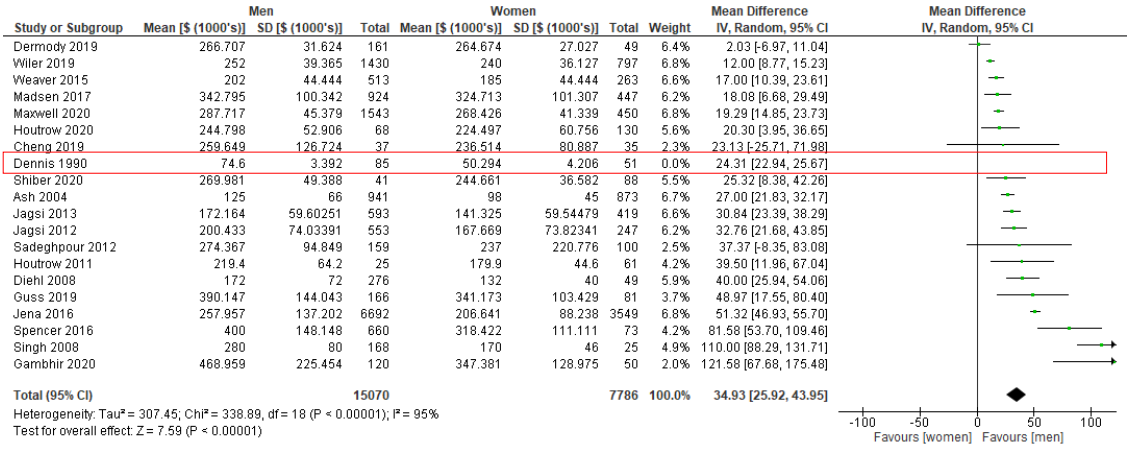

**eFigure 14.** Sensitivity analysis of federal research funding removing 2 poor quality studies (red box, NOS score ≤ 5). M-H (Mantel-Haenszel), Random (random effects), CI (confidence interval).



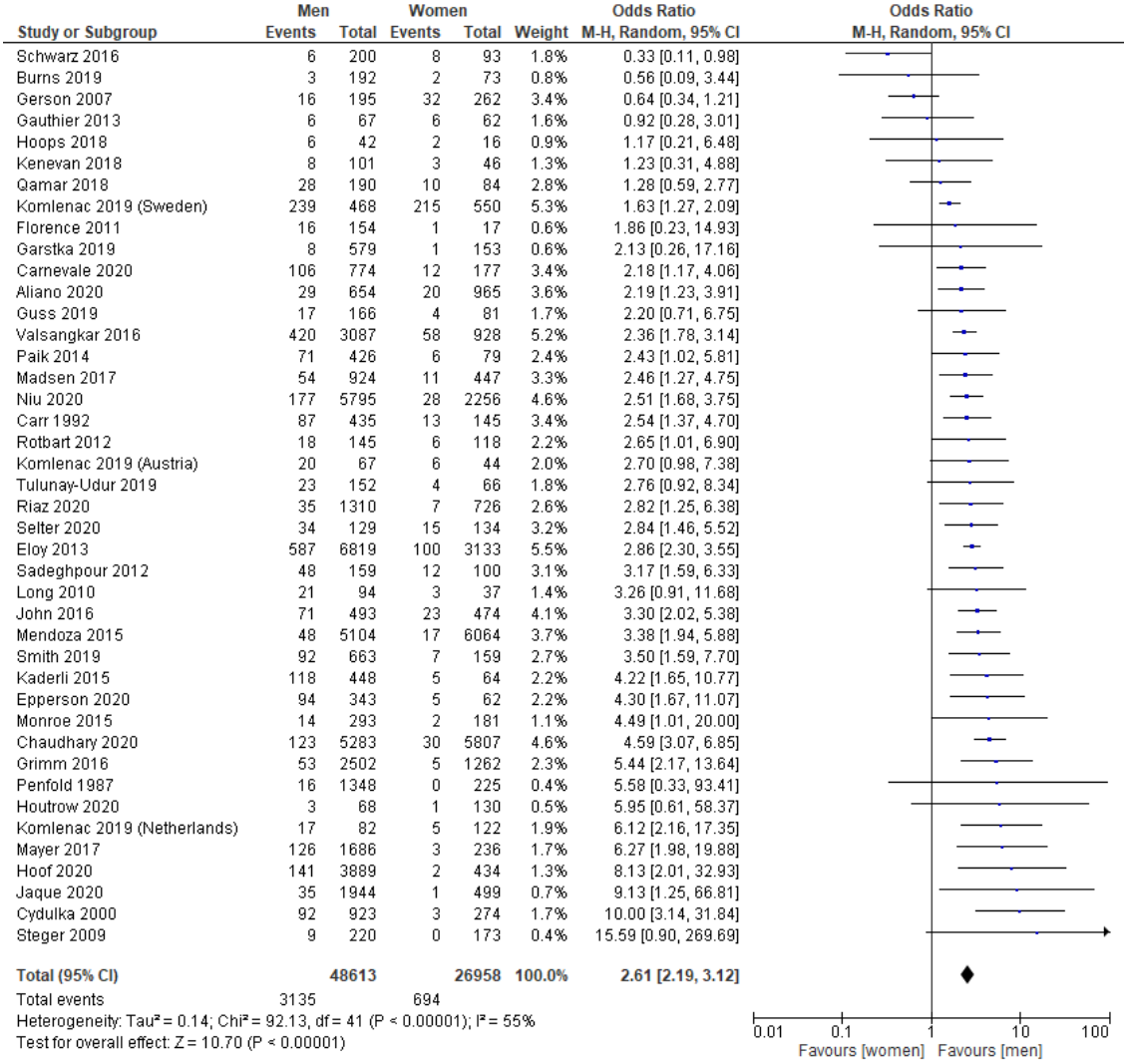

**eFigure 15.** Forest plot of salary for men vs. women. Means and standard deviations (SD) are presented in thousands of dollars. Total (sample size), IV (inverse variance), Random (random effects), CI (confidence interval).



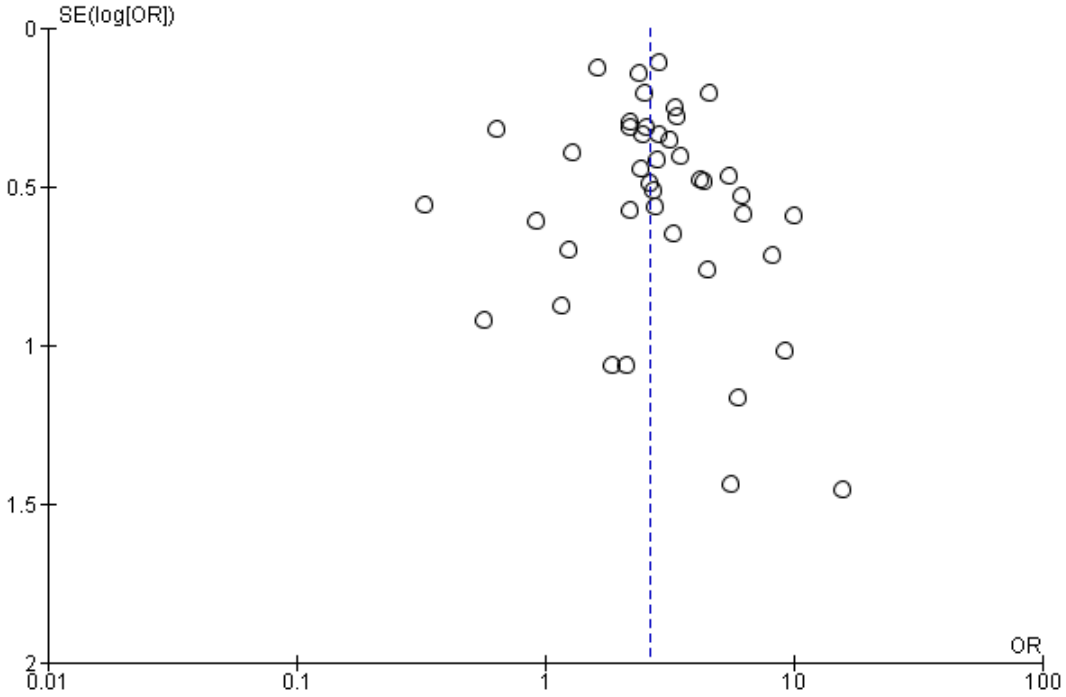

**eFigure 16.** Funnel plot of salary for men vs. women. SE (standard error), MD (mean difference).



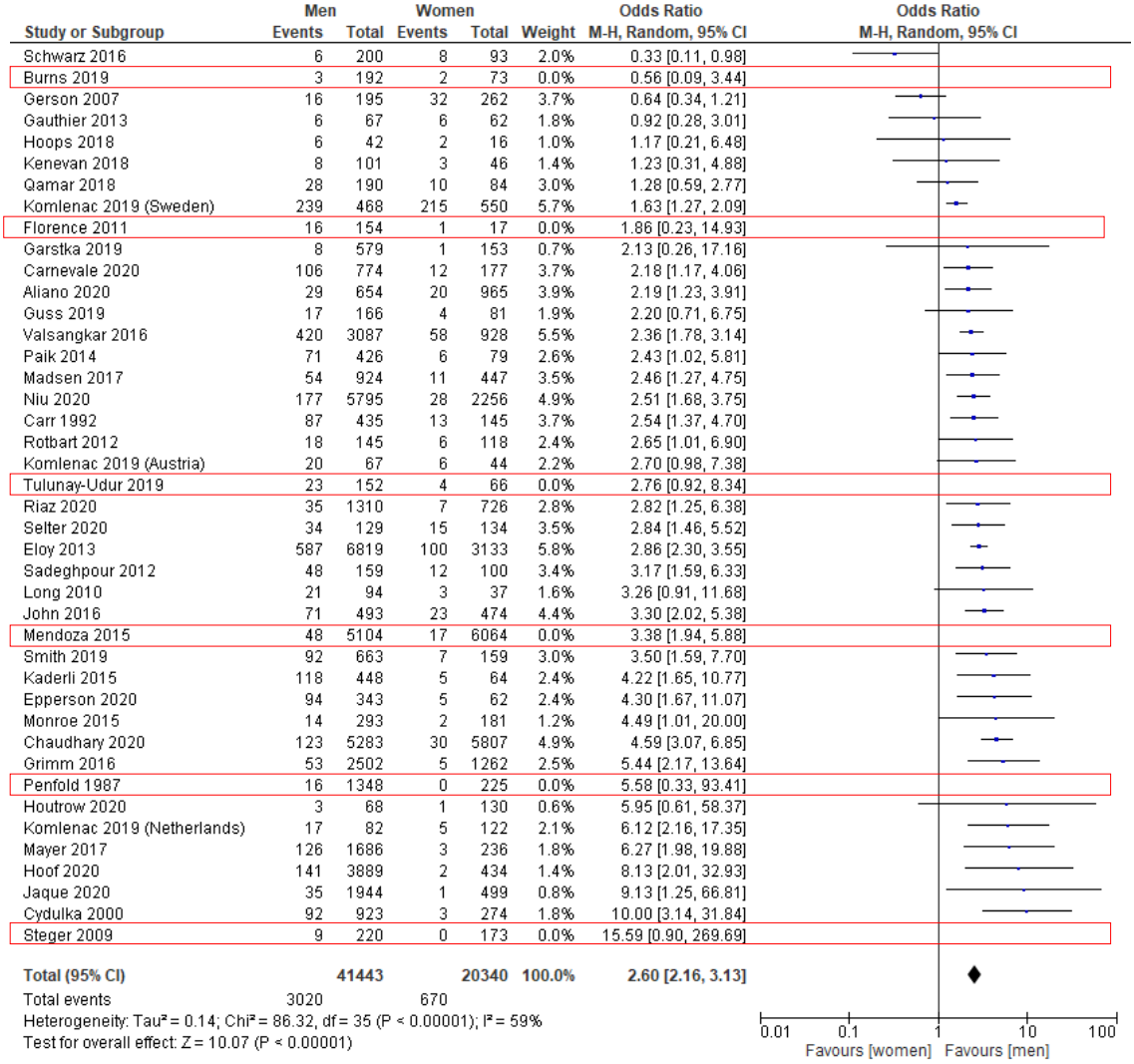

**eFigure 17.** Sensitivity analysis of salary removing 1 poor quality study (red box, NOS score ≤ 5). Means and standard deviations (SD) are presented in thousands of dollars. Total (sample size), IV (inverse variance), Random (random effects), CI (confidence interval).




**eFigure 18.** Forest plot of leadership (departmental chair) for men vs. women. M-H (Mantel-Haenszel), Random (random effects), CI (confidence interval).




**eFigure 19.** Funnel plot of leadership (departmental chair) for men vs. women. M-H (Mantel-Haenszel), Random (random effects), CI (confidence interval).




**eFigure 20.** Sensitivity analysis of leadership (departmental chair) removing 6 poor quality studies (red box, NOS score ≤ 5). M-H (Mantel-Haenszel), Random (random effects), CI (confidence interval).

## eReferences


1. Elizabeth A. Sherman. *Participation of Women and Minorities on U.S. Medical School Faculties, 1980 - 1990*.; 1993. https://books.google.ca/books/about/Participation_of_Women_and_Minorities_on.html?id=VnhrAAAAMAAJ&hl=en&output=html_text&redir_esc=y. Accessed December 25, 2020.
2. Abdellatif W, Ding J, Jalal S, et al. Leadership Gender Disparity Within Research-Intensive Medical Schools. *J Contin Educ Health Prof*. 2019;39(4):243-250. doi:10.1097/CEH.0000000000000270
3. Ahmadi M, Khurshid K, Sanelli PC, et al. Influences for gender disparity in academic neuroradiology. *Am J Neuroradiol*. 2018;39(1):18-23. doi:10.3174/ajnr.A5443
4. Akabas MH, Brass LF. The national MD-PhD program outcomes study: Outcomes variation by sex, race, and ethnicity. *JCI Insight*. 2019;4(19). doi:10.1172/jci.insight.133010
5. Aliaño M, Franco G, Gilsanz F. Gender differences in Anaesthesiology. At what point do we find ourselves in Spain? Results from a Spanish survey. *Rev Española Anestesiol y Reanim (English Ed*. 2020;67(7):374-380. doi:10.1016/j.redare.2019.10.011
6. Al-Tamimi D. Saudi women in academic medicine. Are they succeeding? *Saudi Med J*. 2004. https://pubmed.ncbi.nlm.nih.gov/15573179/. Accessed December 25, 2020.
7. Amering M, Schrank B, Sibitz I. The gender gap in high-impact psychiatry journals. *Acad Med*. 2011;86(8):946-952. doi:10.1097/ACM.0b013e3182222887
8. Ash AS, Carr PL, Goldstein R, Friedman RH. Compensation and advancement of women in academic medicine: Is there equity? *Ann Intern Med*. 2004;141(3):205-212. doi:10.7326/0003-4819-141-3-200408030-00009
9. Awad MA, Gaither TW, Osterberg CE, et al. Gender differences in promotions and scholarly productivity in academic urology. *Can J Urol*. 2017. https://pubmed.ncbi.nlm.nih.gov/28971789/. Accessed December 25, 2020.
10. Baptiste D, Fecher AM, Dolejs SC, et al. Gender differences in academic surgery, work-life balance, and satisfaction. *J Surg Res*. 2017;218:99-107. doi:10.1016/j.jss.2017.05.075
11. Battaglia F, Shah S, Jalal S, et al. Gender disparity in academic emergency radiology. *Emerg Radiol*. 2019;26(1):21-28. doi:10.1007/s10140-018-1642-7
12. Beeler WH, Griffith KA, Evans SB, Golden DW, Jagsi R. Visiting Professorship in Academic Radiation Oncology. *Int J Radiat Oncol Biol Phys*. 2020;108(3):824-829. doi:10.1016/j.ijrobp.2020.05.004
13. Bernardi K, Shah P, Lyons NB, et al. Perceptions on gender disparity in surgery and surgical leadership: A multicenter mixed methods study. In: *Surgery (United States)*. Vol 167. Mosby Inc.; 2020:743-750. doi:10.1016/j.surg.2019.12.004
14. Bickel J. Women in Medical Education. *N Engl J Med*. 1988;319(24):1579-1584. doi:10.1056/nejm198812153192405
15. Bickel J. Women in Academic Medicine: Statistics. *AAMC*. 1993.
16. Bickel J, Whiting BE. Comparing the representation and promotion of men and women faculty at U.S. medical schools. *Acad Med*. 1991;66(8):497. doi:10.1097/00001888-199108000-00025
17. Bissing MA, Lange EMS, Davila WF, et al. Status of Women in Academic Anesthesiology. *Anesth Analg*. 2019;128(1):137-143. doi:10.1213/ANE.0000000000003691
18. Breyer BN, Butler C, Fang R, et al. Promotion Disparities in Academic Urology. *Urology*. 2020;138:16-23. doi:10.1016/j.urology.2019.10.042

19. Brüggmann D, Groneberg DA. An index to characterize female career promotion in academic medicine. *J Occup Med Toxicol*. 2017;12(1):18. doi:10.1186/s12995-017-0164-7

20. Buckley LM, Sanders K, Shih M, Kallar S, Hampton C. Obstacles to promotion? Values of women faculty about career success and recognition. *Acad Med*. 2000;75(3):283-288. doi:10.1097/00001888-200003000-00021

21. Burns KEA, Fox-Robichaud A, Lorens E, Martin CM. Gender differences in career satisfaction, moral distress, and incivility: a national, cross-sectional survey of Canadian critical care physicians. *Can J Anesth*. 2019;66(5):503-511. doi:10.1007/s12630-019-01321-y

22. Camacci ML, Lu A, Lehman EB, et al. Association between Sex Composition and Publication Productivity of Journal Editorial and Professional Society Board Members in Ophthalmology. *JAMA Ophthalmol*. 2020;138(5):451-458. doi:10.1001/jamaophthalmol.2020.0164

23. Carnevale M, Phair J, Batarseh P, LaFontaine S, Koelling E, Koleilat I. Gender disparities in academic vascular surgeons. *J Vasc Surg*. 2020;72(4):1445-1450. doi:10.1016/j.jvs.2019.12.042

24. Carpenter AM, Tan SA, Costopoulos K, Cooper LA, Sarosi GA, Shaw CM. Gender Diversity in General Surgery Residency Leadership. *J Surg Educ*. 2018;75(6):e68-e71. doi:10.1016/j.jsurg.2018.07.021

25. Carr P, Friedman RH, Moskowitz MA, Kazis LE, Weed HG. Research, academic rank, and compensation of women and men faculty in academic general internal medicine. *J Gen Intern Med*. 1992;7(4):418-423. doi:10.1007/BF02599159

26. Carr PL, Ash AS, Friedman RH, et al. Relation of family responsibilities and gender to the productivity and career satisfaction of medical faculty. *Ann Intern Med*. 1998;129(7):532-538. http://www.ncbi.nlm.nih.gov/pubmed/9758572. Accessed January 12, 2019.

27. Caturegli I, Caturegli G, Hayes N, et al. Trends in female surgeon authorship – The role of the middle author. *Am J Surg*. 2020;220(6):1541-1548. doi:10.1016/j.amjsurg.2020.04.025

28. Chaiyachati KH, Liao JM, Weissman GE, et al. Gender Differences in Retention and Promotion Among Generalists Who Graduated From Research-Intensive Fellowships. *J Grad Med Educ*. 2019;11(5):535-542. doi:10.4300/JGME-D-19-00016.1

29. Chandrasekar T, Han TM, Glick L, et al. Setting the Standards: Examining Research Productivity Among Academic Urologists in the USA and Canada in 2019. *Eur Urol Focus*. 2020. doi:10.1016/j.euf.2020.02.003

30. Chang S, Morahan PS, Magrane D, et al. Retaining faculty in academic medicine: The impact of career development programs for women. *J Women's Heal*. 2016;25(7):687-696. doi:10.1089/jwh.2015.5608

31. Chaudhary AMD, Naveed S, Siddiqi J, Mahmood A, Khosa F. US Psychiatry Faculty: Academic Rank, Gender and Racial Profile. *Acad Psychiatry*. 2020;44(3):260-266. doi:10.1007/s40596-020-01192-2

32. Chauvin S, Mulsant BH, Sockalingam S, Stergiopoulos V, Taylor VH, Vigod SN. Gender Differences in Research Productivity among Academic Psychiatrists in Canada. *Can J Psychiatry*. 2019;64(6):415-422. doi:10.1177/0706743718802798

33. Cheng TW, Farber A, Rajani RR, et al. National criteria for academic appointment in vascular surgery. *J Vasc Surg*. 2019;69(5):1559-1565. doi:10.1016/j.jvs.2018.08.178

34. Cheng Y, Mohanty AF, Ogunyemi OI, Smith CA, Leroy G, Zeng QT. 2018 Salary Survey of AMIA Members: Factors Associated with Higher Salaries. *AMIA . Annu Symp proceedings AMIA Symp*. 2019;2019:275-284. /pmc/articles/PMC7153054/?report=abstract. Accessed December 25, 2020.

35. Chevreul K, Gandré C, Gervaix J, Thillard J, Alberti C, Meurs D. Gender and access to professorships in academic medical settings in France. *Med Educ*. 2018;52(10):1073-1082. doi:10.1111/medu.13633

36. Chokshi NK, Simeone DM, Chari RS, Dorey F, Guner YS, Upperman JS. A survey of academic surgeons: Work, stress, and research. *Surgery*. 2009;146(3):462-468. doi:10.1016/j.surg.2009.02.015

37. Colletti LM, Mulholland MW, Sonnad SS. Perceived obstacles to career success for women in academic surgery. *Arch Surg*. 2000;135(8):972-977. doi:10.1001/archsurg.135.8.972

38. Cuoco JA, Klein BJ, Kar A, et al. Factors affecting R01 grant funding among academic neurosurgeons over the last decade: Factors Affecting R01 Grant Funding. *Ann Med Surg*. 2020;55:260-264. doi:10.1016/j.amsu.2020.06.002

39. Cydulka RK, D'Onofrio G, Schneider S, Emerman CL, Sullivan LM. Women in academic emergency medicine. *Acad Emerg Med*. 2000;7(9):999-1007. doi:10.1111/j.1553-2712.2000.tb02091.x

40. Danic A, Hadzibegovic I, Loparic M. Status of women in small academic medical communities: case study of the Zagreb University School of Medicine. *Croat Med J*. 2003. https://pubmed.ncbi.nlm.nih.gov/12590426/. Accessed December 25, 2020.

41. Day CS, Lage DE, Ahn CS. Diversity based on race, ethnicity, and sex between academic orthopaedic surgery and other specialties: A comparative study. *J Bone Jt Surg - Ser A*. 2010;92(13):2328-2335. doi:10.2106/JBJS.I.01482

42. Deipolyi AR, Becker AS, Covey AM, et al. Gender disparity in industry relationships with academic interventional radiology physicians. *Am J Roentgenol*. 2020;215(2):494-501. doi:10.2214/AJR.19.22176

43. Dennis T, Harris I, Petzel R, et al. Influences of marital status and parental status on the professional choices of physicians about to enter practice. *Acad Med*. 1990;65(12):775-777. doi:10.1097/00001888-199012000-00015

44. Dermody SM, Litvack JR, Randall JA, Malekzadeh S, Maxwell JH. Compensation of Otolaryngologists in the Veterans Health Administration: Is There a Gender Gap? *Laryngoscope*. 2019;129(1):113-118. doi:10.1002/lary.27311

45. Desai N, Veras L V., Gosain A. Using bibliometrics to analyze the state of academic productivity in US pediatric surgery training programs. *J Pediatr Surg*. 2018;53(6):1098-1104. doi:10.1016/j.jpedsurg.2018.02.063

46. Diehi JJ, Pirozzolo JJ, Best TM. The practice of primary care sports medicine in the USA. *Br J Sports Med*. 2008;42(10):506-508. doi:10.1136/bjsm.2007.044487

47. Dorismond C, Prince AC, Farzal Z, Zanation AM. Long-Term Academic Outcomes of Triological Society Research Career Development Award Recipients. *Laryngoscope*. 2020. doi:10.1002/lary.28714

48. Dossani RH, Terrell D, Kosty JA, et al. Gender disparities in academic rank achievement in neurosurgery: a critical assessment. *J Neurosurg*. 2019;133(6):1922-1927. doi:10.3171/2019.8.jns191219

49. Dubosh NM, Boyle KL, Carreiro S, Yankama T, Landry AM. Gender differences in funding among grant recipients in emergency medicine: A multicenter analysis. *Am J Emerg Med*. 2020;38(7):1357-1361. doi:10.1016/j.ajem.2019.11.006

50. Duke NN, Gross A, Moran A, et al. Institutional Factors Associated With Burnout Among Assistant Professors. *Teach Learn Med*. 2020;32(1):61-70. doi:10.1080/10401334.2019.1638263

51. Eloy JA, Svider PF, Cherla D V., et al. Gender disparities in research productivity among 9952 academic physicians. *Laryngoscope*. 2013;123(8):1865-1875. doi:10.1002/lary.24039

52. Eloy JA, Svider PF, Kovalerchik O, Baredes S, Kalyoussef E, Chandrasekhar SS. Gender differences in successful NIH grant funding in otolaryngology. *Otolaryngol - Head Neck Surg (United States)*. 2013;149(1):77-83. doi:10.1177/0194599813486083

53. Epperson M, Gouveia CJ, Tabangin ME, et al. Female Representation in Otolaryngology Leadership Roles. *Laryngoscope*. 2020;130(7):1664-1669. doi:10.1002/lary.28308

54. Fassiotto M, Li J, Maldonado Y, Kothary N. Female Surgeons as Counter Stereotype: The Impact of Gender Perceptions on Trainee Evaluations of Physician Faculty. *J Surg Educ*. 2018;75(5):1140-1148.

doi:10.1016/j.jsurg.2018.01.011

55. Florence LS, Feng S, Foster CE, et al. Academic careers and lifestyle characteristics of 171 transplant surgeons in the ASTS. *Am J Transplant*. 2011;11(2):261-271. doi:10.1111/j.1600-6143.2010.03381.x

56. Fitz NE, Lantos JD. Pediatrician's practice choices: differences between part-time and full-time practice. *Pediatrics*. 1991. https://pubmed.ncbi.nlm.nih.gov/1896280/. Accessed December 25, 2020.

57. Gambhir S, Daly SC, Elfenbein D, et al. The effect of transparency on the gender-based compensation gap in surgical disciplines within a large academic healthcare system. *Surg Endosc*. 2020. doi:10.1007/s00464-020-07679-1

58. Garstka ME, Randolph GW, Haddad AB, et al. Gender disparities are present in academic rank and leadership positions despite overall equivalence in research productivity indices among senior members of American Head and Neck Society (AHNS) Fellowship Faculty. *Head Neck*. 2019;41(11):3818-3825. doi:10.1002/hed.25913

59. Gauthier S, Buddeberg-Fischer B, Bucher M, Thali M, Bartsch C. Pilot study on doctors working in departments of forensic medicine in German-speaking areas. *J Forensic Leg Med*. 2013;20(8):1069-1074. doi:10.1016/j.jflm.2013.09.029

60. Gawad N, Tran A, Martel AB, et al. Gender and academic promotion of Canadian general surgeons: a cross-sectional study. *C Open*. 2020;8(1):E34-E40. doi:10.9778/cmajo.20190090

61. Geltzeiler CB, Kelley KA, Srikanth P, et al. Does sex influence publication productivity among colorectal surgeons participating in fellowship training programs? In: *Diseases of the Colon and Rectum*. Vol 60. Lippincott Williams and Wilkins; 2017:537-543. doi:10.1097/DCR.0000000000000746

62. Gerety PA, Serletti JM, Taylor JA. An international survey of craniofacial surgeons: Current trends in practice. *J Craniofac Surg*. 2013;24(1):136-140. doi:10.1097/SCS.0b013e318267bcde

63. Gerson LB, Twomey K, Hecht G, et al. Does Gender Affect Career Satisfaction and Advancement in Gastroenterology? Results of an AGA Institute-Sponsored Survey. *Gastroenterology*. 2007;132(4):1598-1606. doi:10.1053/j.gastro.2007.02.045

64. Girod S, Gilmartin SK, Valantine H, Schiebinger L. Academic couples: Implications for medical school faculty recruitment and retention. *J Am Coll Surg*. 2011;212(3):310-319. doi:10.1016/j.jamcollsurg.2010.11.005

65. Gjerde C. Faculty promotion and publication rates in family medicine: 1981 versus 1989. *Fam Med*. 1994. https://pubmed.ncbi.nlm.nih.gov/8050657/. Accessed December 25, 2020.

66. Grimm LJ, Ngo J, Pisano ED, Yoon S. Men (and Women) in Academic Radiology: How Can We Reduce the Gender Discrepancy? *Am J Roentgenol*. 2016;206(4):678-680. doi:10.2214/AJR.15.15277

67. Guss ZD, Chen Q, Hu C, Guss LG, DeWeese TL, Terezakis SA. Differences in Physician Compensation Between Men and Women at United States Public Academic Radiation Oncology Departments. *Int J Radiat Oncol Biol Phys*. 2019;103(2):314-319. doi:10.1016/j.ijrobp.2018.09.042

68. Hafeez DM, Waqas A, Majeed S, et al. Gender distribution in psychiatry journals' editorial boards worldwide. *Compr Psychiatry*. 2019;94:152119. doi:10.1016/j.comppsych.2019.152119

69. Harrington KA, Chang G. Women in academic musculoskeletal radiology. *Clin Imaging*. 2018;52:180-182. doi:10.1016/j.clinimag.2018.07.022

70. Harris CA, Banerjee T, Cramer M, et al. Editorial (Spring) Board? Gender Composition in High-impact General Surgery Journals over 20 Years. *Ann Surg*. 2019;269(3):582-588. doi:10.1097/SLA.0000000000002667

71. Herrick-Reynolds K, Brooks D, Wind G, Jackson P, Latham K. Military Medicine and the Academic Surgery Gender Gap. *Mil Med*. 2019;184(9-10):383-387. doi:10.1093/milmed/usz083

72. Herzke C, Bonsall J, Bertram A, Yeh HC, Apfel A, Cofrancesco J. Gender Issues in Academic Hospital Medicine: a National Survey of Hospitalist Leaders. *J Gen Intern Med*. 2020;35(6):1641-1646. doi:10.1007/s11606-019-05527-0

73. Higgins MJ, Kale NN, Brown SM, Mulcahey MK. Taking Family Call: Understanding How Orthopaedic Surgeons Manage Home, Family, and Life Responsibilities. *J Am Acad Orthop Surg*. 2020. https://journals.lww.com/jaaos/Abstract/2021/01010/Taking_Family_Call__Understanding_How_Orthopaedic.10.aspx.

74. Higgins EJ. COMPARISON OF CHARACTERISTICS OF U.S. MEDICAL SCHOOL SALARIED FACULTY. 1979.

75. Hill EK, Stuckey A, Fiascone S, et al. Gender and the Balance of Parenting and Professional Life among Gynecology Subspecialists. *J Minim Invasive Gynecol*. 2019;26(6):1088-1094. doi:10.1016/j.jmig.2018.10.020

76. Holliday E, Griffith KA, De Castro R, Stewart A, Ubel P, Jagsi R. Gender Differences in Resources and Negotiation Among Highly Motivated Physician-Scientists. *J Gen Intern Med*. 2015;30(4):401-407. doi:10.1007/s11606-014-2988-5

77. Holliday EB, Jagsi R, Wilson LD, Choi M, Thomas CR, Fuller CD. Gender differences in publication productivity, academic position, career duration, and funding among U.S. Academic radiation oncology faculty. *Acad Med*. 2014;89(5):767-773. doi:10.1097/ACM.0000000000000229

78. Hoof MA, Sommi C, Meyer LE, Bird ML, Brown SM, Mulcahey MK. Gender-related Differences in Research Productivity, Position, and Advancement Among Academic Orthopaedic Faculty Within the United States. *J Am Acad Orthop Surg*. 2020;28(21):893-899. doi:10.5435/JAAOS-D-19-00408

79. Hoops HE, Brasel KJ, Dewey E, et al. Analysis of Gender-based Differences in Surgery Faculty Compensation, Promotion, and Retention: Establishing Equity. *Ann Surg*. 2018;268(3):479-487. doi:10.1097/SLA.0000000000002920

80. Hoover E. An analysis of the association of American medical colleges' review of minorities in medical education. *J Natl Med Assoc*. 2005. https://pubmed.ncbi.nlm.nih.gov/16296215/. Accessed December 25, 2020.

81. Houtrow AJ, Zigler CK, Pruitt DW. The State of the Field: Results from the 2014 and 2017 Pediatric Rehabilitation Medicine Practice Surveys. *PM&R*. 2020;12(2):168-179. doi:10.1002/pmrj.12235

82. Houtrow A. Results From the 2009 Pediatric Rehabilitation Practice Survey of the AAPM&R Pediatric Rehabilitation/Developmental Disabilities Council. *PM R*. 2011;3(1):45-52. doi:10.1016/j.pmrj.2010.08.013

83. Hurley RW, Zhao K, Tighe PJ, Ko PS, Pronovost PJ, Wu CL. Examination of publications from academic anesthesiology faculty in the United States. *Anesth Analg*. 2014;118(1):192-199. doi:10.1213/ANE.0b013e3182a91aa9

84. Ahmed M, Ibn Auf A, Awadalla H, Ahmed M. Comparing the participation of men and women in academic medicine in medical colleges in Sudan: A cross-sectional survey. *J Educ Health Promot*. 2019;8(1). doi:10.4103/jehp.jehp_225_18

85. Ilin J, Langlois E, Jalal S, Khosa F. Gender disparity within academic Canadian urology. *Can Urol Assoc J*. 2020;14(4):107-111. doi:10.5489/cuaj.6117

86. Inam H, Janjua M, Martins RS, et al. Cultural Barriers for Women in Surgery: How Thick is the Glass Ceiling? An Analysis from a Low Middle-Income Country. *World J Surg*. 2020;44(9). doi:10.1007/s00268-020-05544-9

87. Ingram DG. Publication Measures of Scholarly Productivity: Norms for Academic Pediatric Pulmonologists. *Pediatr Allergy Immunol Pulmonol*. 2015;28(3):183-186. doi:10.1089/ped.2015.0511

88. Jacquet GA, Hansoti B, Levine AC, Martin IBK. Career paths and practice patterns of global emergency

medicine physicians: A cross-sectional survey. *J Emerg Med*. 2014;47(3):348-354. doi:10.1016/j.jemermed.2014.01.029

89. Jagsi R, Griffith KA, Stewart A, Sambuco D, Decastro R, Ubel PA. Gender differences in salary in a recent cohort of early-career physician-researchers. *Acad Med*. 2013;88(11):1689-1699. doi:10.1097/ACM.0b013e3182a71519

90. Jagsi R, Griffith KA, Stewart A, Sambuco D, DeCastro R, Ubel PA. Gender differences in the salaries of physician researchers. *JAMA - J Am Med Assoc*. 2012;307(22):2410-2417. doi:10.1001/jama.2012.6183

91. Jaque J, Wanke EM, Müller R, et al. Gender and workforce in urology: Use of the BG index to assess female career promotion in academic urology. *Urol J*. 2020;17(1):86-90. doi:10.22037/uj.v0i0.4116

92. Jena AB, Olenski AR, Blumenthal DM. Sex differences in physician salary in US public medical schools. *JAMA Intern Med*. 2016;176(9):1294-1304. doi:10.1001/jamainternmed.2016.3284

93. Jena AB, Khullar D, Ho O, Olenski AR, Blumenthal DM. Sex differences in academic rank in US medical schools in 2014. *JAMA - J Am Med Assoc*. 2015;314(11):1149-1158. doi:10.1001/jama.2015.10680

94. John AM, Gupta AB, John ES, Lopez SA, Lambert WC. A gender-based comparison of promotion and research productivity in academic dermatology. *Dermatol Online J*. 2016. https://pubmed.ncbi.nlm.nih.gov/27617455/. Accessed December 25, 2020.

95. Johnson D. U.S. Medical School Faculty, 1992. Faculty Roster System. *AAMC*. 1992. http://www.eric.ed.gov/contentdelivery/servlet/ERICServlet?accno=ED354826. Accessed December 25, 2020.

96. Jolly S, Griffith KA, DeCastro R, Stewart A, Ubel P, Jagsi R. Gender differences in time spent on parenting and domestic responsibilities by high-achieving young physician-researchers. *Ann Intern Med*. 2014;160(5):344-353. doi:10.7326/m13-0974

97. Kaderli RM, Klasen JM, Businger AP. Mentoring in general surgery in Switzerland. *Med Educ Online*. 2015;20(1). doi:10.3402/meo.v20.27528

98. Karimian Z, Sabbaghian Z, Salehi A, Sedghpour BS. Obstacles to undertaking research and their effect on research output: a survey of faculty members' views at Shiraz University of Medical Sciences. *East Mediterr Heal J*. 2012;18(11):1143-1150. doi:10.26719/2012.18.11.1143

99. Kenevan MR, Stewart TM, Warner MA, Rose SH, Long TR. The Changing Characteristics of Anesthesiology Program Directors. *J Educ Perioper Med*. 2018;20(3):1. doi:10.46374/volxx-issue3-kenevan

100. Khan MS, Usman MS, Siddiqi TJ, et al. Women in leadership positions in Academic cardiology: A study of program directors and division chiefs. *J Women's Heal*. 2019;28(2):225-232. doi:10.1089/jwh.2018.7135

101. Khan NR, Thompson CJ, Taylor DR, et al. An analysis of publication productivity for 1225 academic neurosurgeons and 99 departments in the United States: Clinical article. *J Neurosurg*. 2014;120(3):746-755. doi:10.3171/2013.11.JNS131708

102. Khurshid K, Shah S, Ahmadi M, et al. Gender differences in the publication rate among breast imaging radiologists in the United States and Canada. *Am J Roentgenol*. 2018;210(1):2-7. doi:10.2214/AJR.17.18303

103. Kilic I, Nigdelioglu R, Lapadat R, et al. Current state of Grand Rounds in U.S. pathology training programs. *Ann Diagn Pathol*. 2020;46:151484. doi:10.1016/j.anndiagpath.2020.151484

104. Killaspy H, Johnson S, Livingston G, Hassiotis A, Robertson M. Women in academic psychiatry in the United Kingdom. *Psychiatr Bull*. 2003;27(9):323-326. doi:10.1192/pb.27.9.323

105. Kim S, Li B, Rosoklija I, Johnson EK, Yerkes E, Chu DI. Federal research funding and academic productivity in pediatric urology: from early career to research independence. *J Pediatr Urol*. 2019;15(3):233-239. doi:10.1016/j.jpurol.2019.02.018

106. Kirchner SG, Kossoff J, Pickens DR. Women in radiology in the United States: 1982 survey of their professional practices. *AJR Am J Roentgenol*. 1984;142(5):1055-1059. doi:10.2214/ajr.142.5.1055

107. Klingensmith ME, Anderson KD. Educational scholarship as a route to academic promotion: A depiction of surgical education scholars. *Am J Surg*. 2006;191(4):533-537. doi:10.1016/j.amjsurg.2006.01.007

108. Kloosterboer A, Yannuzzi NA, Gedde SJ, Sridhar J. Residency Program Directors of United States Ophthalmology Programs: A Descriptive Analysis. *Am J Ophthalmol*. 2020;209:71-76. doi:10.1016/j.ajo.2019.09.003

109. Komlenac N, Gustafsson Sendén M, Verdonk P, Hochleitner M, Siller H. Parenthood does not explain the gender difference in clinical position in academic medicine among Swedish, Dutch and Austrian physicians. *Adv Heal Sci Educ*. 2019;24(3):539-557. doi:10.1007/s10459-019-09882-9

110. Filon FL, Paniz E, Prodi A. The gender gap in italian academic medicine from 2005 to 2015: Still a glass ceiling. *Med del Lav*. 2019;110(1):29-36. doi:10.23749/mdl.v110i1.7617

111. LaRocca CJ, Wong P, Eng OS, Raoof M, Warner SG, Melstrom LG. Academic productivity in surgical oncology: Where is the bar set for those training the next generation? *J Surg Oncol*. 2018;118(3):397-402. doi:10.1002/jso.25143

112. Levey BA, Gentile NO, Jolly HP, Beaty HN, Levey GS. Comparing research activities of women and men faculty in departments of internal medicine. *Acad Med*. 1990. https://journals.lww.com/academicmedicine/Abstract/1990/02000/Comparing_research_activities_of_women_and_men.11.aspx. Accessed December 25, 2020.

113. Lewis-Stevenson S, Hueston WJ, Mainous AG, Bazell PC, Ye X. Female and underrepresented minority faculty in academic departments of family medicine: are women and minorities better off in family medicine? *Fam Med*. 2001. https://pubmed.ncbi.nlm.nih.gov/11411975/. Accessed December 25, 2020.

114. Long TR, Brown MJ, Elliott BA, Rose SH. Characteristics of anesthesiology residency program directors. *J Clin Anesth*. 2010;22(8):583-586. doi:10.1016/j.jclinane.2010.05.005

115. Madsen TE, Linden JA, Rounds K, et al. Current Status of Gender and Racial/Ethnic Disparities Among Academic Emergency Medicine Physicians. *Acad Emerg Med*. 2017;24(10):1182-1192. doi:10.1111/acem.13269

116. Mascarenhas A, Moore JE, Tricco AC, et al. Perceptions and experiences of a gender gap at a Canadian research institute and potential strategies to mitigate this gap: a sequential mixed-methods study. *C Open*. 2017;5(1):E144-E151. doi:10.9778/cmajo.20160114

117. Maxwell JH, Randall JA, Dermody SM, et al. Gender and compensation among surgical specialties in the Veterans Health Administration. In: *American Journal of Surgery*. Vol 220. Elsevier Inc.; 2020:256-261. doi:10.1016/j.amjsurg.2020.02.045

118. Mayer EN, Lenherr SM, Hanson HA, Jessop TC, Lowrance WT. Gender Differences in Publication Productivity Among Academic Urologists in the United States. *Urology*. 2017;103:39-46. doi:10.1016/j.urology.2016.12.064

119. McAlister C, Jin YP, Braga-Mele R, DesMarchais BF, Buys YM. Comparison of lifestyle and practice patterns between male and female Canadian ophthalmologists. *Can J Ophthalmol*. 2014;49(3):287-290. doi:10.1016/j.jcjo.2014.02.007

120. McDermott M, Gelb DJ, Wilson K, et al. Sex differences in academic rank and publication rate at top-ranked US neurology programs. *JAMA Neurol*. 2018;75(8):956-961. doi:10.1001/jamaneurol.2018.0275

121. McDonald JS, McDonald RJ, Davenport MS, et al. Gender and Radiology Publication Productivity: An Examination of Academic Faculty From Four Health Systems in the United States. *J Am Coll Radiol*. 2017;14(8):1100-1108. doi:10.1016/j.jacr.2017.04.017

122. Mendoza FS, Walker LR, Stoll BJ, et al. Diversity and inclusion training in pediatric departments.

*Pediatrics*. 2015;135(4):707-713. doi:10.1542/peds.2014-1653

123. Modena MG, Lalla M, Molinari R. Determinants of career structure and advancement among Italian cardiologists. An example of segregation and discrimination against women? *Eur Heart J*. 1999;20(17):1276-1284. doi:10.1053/euhj.1999.1579

124. Moghimi S, Khurshid K, Jalal S, et al. Gender differences in leadership positions among academic nuclear medicine specialists in Canada and the United States. *Am J Roentgenol*. 2019;212(1):146-150. doi:10.2214/AJR.18.20062

125. Monroe AK, Levine RB, Clark JM, Bickel J, Macdonald SM, Resar LMS. Through a Gender Lens: A View of Gender and Leadership Positions in a Department of Medicine. *J Women's Heal*. 2015;24(10):837-842. doi:10.1089/jwh.2014.5054

126. Morgan HK, Graziano SC, Craig LB, et al. A National Survey of Profiles of Clerkship Directors in Obstetrics and Gynecology. *Obstet Gynecol*. 2019;134(4):869-873. doi:10.1097/AOG.0000000000003471

127. Morris M, Chen H, Heslin MJ, Krontiras H. A Structured Compensation Plan Improves But Does Not Erase the Sex Pay Gap in Surgery. *Ann Surg*. 2018;268(3):442-448. doi:10.1097/SLA.0000000000002928

128. Morrison LJ, Lorens E, Bandiera G, et al. Impact of a formal mentoring program on academic promotion of Department of Medicine faculty: A comparative study. *Med Teach*. 2014;36(7):608-614. doi:10.3109/0142159X.2014.899683

129. Mueller C, Wright R, Girod S. The publication gender gap in US academic surgery. *BMC Surg*. 2017;17(1). doi:10.1186/s12893-017-0211-4

130. Muffly MK, Muffly TM, Weterings R, Singleton M, Honkanen A. The Current Landscape of US Pediatric Anesthesiologists. *Anesth Analg*. 2016;123(1):179-185. doi:10.1213/ANE.0000000000001266

131. Nakata Y, Watanabe Y, Narimatsu H, Yoshimura T, Otake H, Sawa T. Predictors of surgeons' efficiency in the operating rooms. *Heal Serv Manag Res*. 2017;30(1):16-21. doi:10.1177/0951484816678012

132. Ngaage LM, Harris C, Gao C, et al. Investigating the Gender Pay Gap in Industry Contributions to Academic Neurosurgeons. *World Neurosurg*. 2019;130:516-522.e1. doi:10.1016/j.wneu.2019.06.145

133. Nguyen V, Marmor RA, Ramamoorthy SL, Blair SL, Clary BM, Sicklick JK. Academic Surgical Oncologists' Productivity Correlates with Gender, Grant Funding, and Institutional NCI Comprehensive Cancer Center Affiliation. *Ann Surg Oncol*. 2018;25(7):1852-1859. doi:10.1245/s10434-018-6450-5

134. Nickerson KG, Bennett NM, Estes D, Shea S. The Status of Women at One Academic Medical Center: Breaking Through the Glass Ceiling. *JAMA J Am Med Assoc*. 1990;264(14):1813-1817. doi:10.1001/jama.1990.03450140035030

135. Niu BT, Nicolaou S, Levine D, et al. Trends in Gender and Racial Profiles of US Academic Radiology Faculty. *J Am Coll Radiol*. 2020;17(10):1337-1343. doi:10.1016/j.jacr.2020.03.019

136. Nowling TK, McClure E, Simpson A, Sheidow AJ, Shaw D, Feghali-Bostwick C. A Focused Career Development Program for Women Faculty at an Academic Medical Center. *J Women's Heal*. 2018;27(12):1474-1481. doi:10.1089/jwh.2018.6937

137. Odell T, Toor H, Takayanagi A, et al. Gender Disparity in Academic Neurosurgery. *Cureus*. 2019;11(5). doi:10.7759/cureus.4628

138. Okafor S, Tibbetts K, Shah G, Tillman B, Agan A, Halderman AA. Is the gender gap closing in otolaryngology subspecialties? An analysis of research productivity. *Laryngoscope*. 2020;130(5):1144-1150. doi:10.1002/lary.28189

139. Okoshi K, Nomura K, Fukami K, et al. Gender inequality in career advancement for females in japanese academic surgery. *Tohoku J Exp Med*. 2014;234(3):221-227. doi:10.1620/tjem.234.221

140. O'Neill SB, Maddu K, Jalal S, et al. Gender Disparity in Chest Radiology in North America. *Curr Probl Diagn Radiol*. 2019;50(1). doi:10.1067/j.cpradiol.2019.10.001

141. Orhurhu MS, Orhurhu V, Salisu B, Abimbola A, Cohen SP. Factors associated with academic rank among chronic pain medicine faculty in the USA. *Reg Anesth Pain Med*. 2020;45(8):589-596. doi:10.1136/rapm-2019-101139

142. Pace NM, Maganty A, Siripong N, et al. Gender Gap in Industry Relationships and Scholarly Impact Among Academic Urologists in the United States. *Urology*. 2020;139:90-96. doi:10.1016/j.urology.2020.01.022

143. Pagel PS, Hudetz JA. Scholarly productivity and National Institutes of Health funding of Foundation for Anesthesia Education and Research grant recipients: Insights from a bibliometric analysis. *Anesthesiology*. 2015;123(3):683-691. doi:10.1097/ALN.0000000000000737

144. Paik AM, Mady LJ, Villanueva NL, et al. Research productivity and gender disparities: A look at academic plastic surgery. *J Surg Educ*. 2014;71(4):593-600. doi:10.1016/j.jsurg.2014.01.010

145. Paulus JK, Switkowski KM, Allison GM, et al. Where is the leak in the pipeline? Investigating gender differences in academic promotion at an academic medical centre. *Perspect Med Educ*. 2016;5(2):125-128. doi:10.1007/s40037-016-0263-7

146. Penfold PS. Women in academic psychiatry in Canada. *Can J Psychiatry*. 1987;32(8):660-665. doi:10.1177/070674378703200804

147. Pfeiffer M, Fischer MR, Bauer D. Publication activities of German junior researchers in academic medicine: Which factors impact impact factors? *BMC Med Educ*. 2016;16(1). doi:10.1186/s12909-016-0712-3

148. Pololi LH, Civian JT, Brennan RT, Dottolo AL, Krupat E. Experiencing the culture of academic medicine: Gender matters, a national study. *J Gen Intern Med*. 2013;28(2):201-207. doi:10.1007/s11606-012-2207-1

149. Qamar SR, Khurshid K, Jalal S, et al. Gender disparity among leaders of Canadian academic radiology departments. *Am J Roentgenol*. 2020;214(1):3-9. doi:10.2214/AJR.18.20992

150. Qamar SR, Khurshid K, Jalal S, et al. Academic musculoskeletal radiology: influences for gender disparity. *Skeletal Radiol*. 2018;47(3):381-387. doi:10.1007/s00256-017-2836-x

151. Rao AD, Nicholas SE, Kachniarz B, et al. Association of a Simulated Institutional Gender Equity Initiative With Gender-Based Disparities in Medical School Faculty Salaries and Promotions. *JAMA Netw open*. 2018;1(8):e186054. doi:10.1001/jamanetworkopen.2018.6054

152. Raptis DA, Schlegel A, Tschuor C, Clavien PA. Job satisfaction among young board-certified surgeons at academic centers in Europe and North America. *Ann Surg*. 2012;256(5). doi:10.1097/sla.0b013e31827386eb

153. Rayburn WF, Schrader RM, Fullilove AM, Rutledge TL, Phelan ST, Gener Y. Promotion rates for assistant and associate professors in obstetrics and gynecology. *Obstet Gynecol*. 2012;119(5):1023-1029. doi:10.1097/AOG.0b013e31824fc980

154. Reed DA, Enders F, Lindor R, McClees M, Lindor KD. Gender differences in academic productivity and leadership appointments of physicians throughout academic careers. *Acad Med*. 2011;86(1):43-47. doi:10.1097/ACM.0b013e3181ff9ff2

155. Reiser LW, Sledge WH, Fenton W, Leaf P. Beginning careers in academic psychiatry for women - "Bermuda triangle"? *Am J Psychiatry*. 1993;150(9):1392-1397. doi:10.1176/ajp.150.9.1392

156. Riaz I Bin, Siddiqi R, Zahid U, et al. Gender Differences in Faculty Rank and Leadership Positions Among Hematologists and Oncologists in the United States. *JCO Oncol Pract*. 2020;16(6):e507-e516. doi:10.1200/op.19.00255

157. Robboy SJ, McLendon R. Structured Annual Faculty Review Program Accelerates Professional Development and Promotion: Long-Term Experience of the Duke University Medical Center's Pathology

Department. *Acad Pathol*. 2017;4. doi:10.1177/2374289516689471

158. Rosso C, Leger A, Steichen O. Glass ceiling for women in academic medicine in France. *Rev Med Interne*. 2019;40(2):82-87. doi:10.1016/j.revmed.2018.05.007

159. Rotbart HA, McMillen D, Taussig H, Daniels SR. Assessing gender equity in a large academic department of pediatrics. *Acad Med*. 2012;87(1):98-104. doi:10.1097/ACM.0b013e31823be028

160. Sadeghpour M, Bernstein I, Ko C, Jacobe H. Role of sex in academic dermatology: Results from a national survey. *Arch Dermatol*. 2012;148(7):809-814. doi:10.1001/archdermatol.2011.3617

161. Sandhu B, Margerison C, Holdcroft A. Women in the UK academic medicine workforce. *Med Educ*. 2007;41(9):909-914. doi:10.1111/j.1365-2923.2007.02825.x

162. Santosa KB, Larson EL, Vannucci B, et al. Gender Imbalance at Academic Plastic Surgery Meetings. *Plast Reconstr Surg*. 2019;143(6):1798-1806. doi:10.1097/PRS.0000000000005672

163. Sasor SE, Cook JA, Duquette SP, et al. Scholarly activity in academic plastic surgery: the gender difference. *J Surg Res*. 2018;229:332-336. doi:10.1016/j.jss.2018.04.031

164. Schnapp LM, Matosian M, Weisman I, Welsh CH. A snapshot of pulmonary medicine at the turn of the century: The American Thoracic Society membership. *Am J Respir Crit Care Med*. 2003;167(9):1176-1180. doi:10.1164/rccm.200203-186OC

165. Schroen AT, Brownstein MR, Sheldon GF. Women in Academic General Surgery. *Acad Med*. 2004;79(4):310-318. doi:10.1097/00001888-200404000-00006

166. Schwarz L, Sippel S, Entwistle A, Hell AK, Koenig S. Biographic Characteristics and Factors Perceived as Affecting Female and Male Careers in Academic Surgery: The Tenured Gender Battle to Make It to the Top. *Eur Surg Res*. 2016;57(3-4):139-154. doi:10.1159/000446874

167. Selter JH, Spurlin EE, Brady PC. Gender inequality in leadership and academic rank in academic reproductive endocrinology programs. *J Assist Reprod Genet*. 2020;37(8):1959-1962. doi:10.1007/s10815-020-01866-8

168. Shah A, Jalal S, Khosa F. Influences for gender disparity in dermatology in North America. *Int J Dermatol*. 2018;57(2):171-176. doi:10.1111/ijd.13875

169. Shaikh AT, Farhan SA, Siddiqi R, Fatima K, Siddiqi J, Khosa F. Disparity in Leadership in Neurosurgical Societies: A Global Breakdown. *World Neurosurg*. 2019;123:95-102. doi:10.1016/j.wneu.2018.11.145

170. Sheikh MH, Chaudhary AMD, Khan AS, et al. Influences for Gender Disparity in Academic Psychiatry in the United States. *Cureus*. 2018;10(4). doi:10.7759/cureus.2514

171. Shiber LDJ, Yao M, Adedayo P, Dassel M. Current Trends in Compensation for Fellowship in Minimally Invasive Gynecologic Surgery Graduates: A 6-Year Follow-Up. *J Minim Invasive Gynecol*. 2020. doi:10.1016/j.jmig.2020.05.008

172. Shollen SL, Bland CJ, Finstad DA, Taylor AL. Organizational climate and family life: How these factors affect the status of women faculty at one medical school. *Acad Med*. 2009;84(1):87-94. doi:10.1097/ACM.0b013e3181900edf

173. Singh A, Burke CA, Larive B, Sastri S V. Do gender disparities persist in gastroenterology after 10 years of practice? *Am J Gastroenterol*. 2008;103(7):1589-1595. doi:10.1111/j.1572-0241.2008.01976.x

174. Smith BT, Egro FM, Murphy CP, Stavros AG, Kenny EM, Nguyen VT. Change Is Happening: An Evaluation of Gender Disparities in Academic Plastic Surgery. *Plast Reconstr Surg*. 2019;144(4):1001-1009. doi:10.1097/PRS.0000000000006086

175. Soothill K. The qualifications and careers of psychiatrists, psychologists, social scientists and nurses interested in forensic mental health research. *J Forensic Psychiatry Psychol*. 2010;21(5):697-710.

doi:10.1080/14789949.2010.488696

176. Speck RM, Sammel MD, Troxel AB, et al. Factors impacting the departure rates of female and male junior medical school faculty: Evidence from a longitudinal analysis. *J Women's Heal*. 2012;21(10):1059-1065. doi:10.1089/jwh.2011.3394

177. Spencer ES, Deal AM, Pruthi NR, et al. Gender Differences in Compensation, Job Satisfaction and Other Practice Patterns in Urology. *J Urol*. 2016;195(2):450-455. doi:10.1016/j.juro.2015.08.100

178. Sperling JD, Shulman R, Blat C, et al. Gender Differences in Academic Rank and NIH Funding among Academic Maternal-Fetal Medicine Physicians in the United States. *Am J Perinatol*. 2019;36(5):443-448. doi:10.1055/s-0038-1675332

179. Stadler DJ, Archuleta S, Ibrahim H, Shah NG, Al-Mohammed AA, Cofrancesco J. Gender and international clinician educators. *Postgrad Med J*. 2017;93(1106):719-724. doi:10.1136/postgradmedj-2016-134599

180. Steger B, Colvin HP, Rieder J. Scientific activity and working hours of physicians in university hospitals: Results from the Innsbruck and Salzburg physician lifestyle assessment (TISPLA). *Wien Klin Wochenschr*. 2009;121(21-22):685-689. doi:10.1007/s00508-009-1226-3

181. Stewart C, Lipner SR. Gender and race trends in academic rank of dermatologists at top U.S. institutions: A cross-sectional study. *Int J Women's Dermatology*. 2020;6(4):283-285. doi:10.1016/j.ijwd.2020.04.010

182. Svider PF, D'Aguillo CM, White PE, et al. Gender differences in successful national institutes of health funding in ophthalmology. *J Surg Educ*. 2014;71(5):680-688. doi:10.1016/j.jsurg.2014.01.020

183. Tekerek M, Karakaya F, Tekerek B. Ethical Reasoning in STEM Disciplines. *J Educ Pract*. 2016;7. https://files.eric.ed.gov/fulltext/EJ1122533.pdf. Accessed December 25, 2020.

184. Therattil PJ, Hoppe IC, Granick MS, Lee ES. Application of the h-index in academic plastic surgery. *Ann Plast Surg*. 2016;76(5):545-549. doi:10.1097/SAP.0000000000000382

185. Tracy LF, Jabbour N, Rubin BR, et al. Satisfaction in Academic Otolaryngology: Do Physician Demographics Impact Press Ganey Survey Scores? *Laryngoscope*. 2020;130(8):1902-1906. doi:10.1002/lary.28335

186. Tsukada YT, Tokita M, Kato K, et al. Solutions for Retention of Female Cardiologists from the Survey of Gender Differences in the Work and Life of Cardiologists. *Circ J*. 2009;73(11):2076-2083. doi:10.1253/circj.CJ-09-0206

187. Tulunay-Ugur OE, Sinclair CF, Chen AY. Assessment of Gender Differences in Perceptions of Work-Life Integration among Head and Neck Surgeons. In: *JAMA Otolaryngology - Head and Neck Surgery*. Vol 145. American Medical Association; 2019:453-458. doi:10.1001/jamaoto.2019.0104

188. Turaga KK, Green DE, Jayakrishnan TT, Hwang M, Clark Gamblin T. Attributes of a surgical chairperson associated with extramural funding of a department of surgery. *J Surg Res*. 2013;185(2):549-554. doi:10.1016/j.jss.2013.07.019

189. Valsangkar N, Fecher AM, Rozycki GS, et al. Understanding the Barriers to Hiring and Promoting Women in Surgical Subspecialties. In: *Journal of the American College of Surgeons*. Vol 223. Elsevier Inc.; 2016:387-398.e2. doi:10.1016/j.jamcollsurg.2016.03.042

190. Villablanca AC, Li Y, Beckett LA, Howell LP. Evaluating a Medical School's Climate for Women's Success: Outcomes for Faculty Recruitment, Retention, and Promotion. *J Women's Heal*. 2017;26(5):530-539. doi:10.1089/jwh.2016.6018

191. Vydareny KH, Waldrop SM, Jackson VP, et al. Career advancement of men and women in academic radiology: Is the playing field level? *Acad Radiol*. 2000;7(7):493-501. doi:10.1016/S1076-6332(00)80321-9

192. Wagner AK, McElligott J, Chan L, Wagner EP, Segal NA, Gerber LH. How Gender Impacts Career Development and Leadership in Rehabilitation Medicine: A Report From the AAPM&R Research

Committee. *Arch Phys Med Rehabil*. 2007;88(5):560-568. doi:10.1016/j.apmr.2007.01.014

193. Waisbren SE, Bowles H, Hasan T, et al. Gender differences in research grant applications and funding outcomes for medical school faculty. *J Women's Heal*. 2008;17(2):207-214. doi:10.1089/jwh.2007.0412

194. Waljee JF, Chang KWC, Kim HM, et al. Gender disparities in academic practice. *Plast Reconstr Surg*. 2015;136(3):380e-387e. doi:10.1097/PRS.0000000000001530

195. Wallis LA, Gilder H, Thaler H. Advancement of Men and Women in Medical Academia: A Pilot Study. *JAMA J Am Med Assoc*. 1981;246(20):2350-2353. doi:10.1001/jama.1981.03320200036022

196. Warner ET, Carapinha R, Weber GM, Hill E V., Reede JY. Gender Differences in Receipt of National Institutes of Health R01 Grants among Junior Faculty at an Academic Medical Center: The Role of Connectivity, Rank, and Research Productivity. *J Women's Heal*. 2017;26(10):1086-1093. doi:10.1089/jwh.2016.6102

197. Waseem Y, Mahmood S, Siddiqi R, et al. Gender differences amongst board members of endocrinology and diabetes societies. *Endocrine*. 2019;64(3):496-499. doi:10.1007/s12020-019-01861-9

198. Wasserstein AG, Quistberg DA, Shea JA. Mentoring at the University of Pennsylvania: Results of a faculty survey. *J Gen Intern Med*. 2007;22(2):210-214. doi:10.1007/s11606-006-0051-x

199. Weaver AC, Wetterneck TB, Whelan CT, Hinami K. A matter of priorities? Exploring the persistent gender pay gap in hospital medicine. *J Hosp Med*. 2015;10(8):486-490. doi:10.1002/jhm.2400

200. Whitley NO, Evens RG, Moody MA, Putman CE, Sackett JF, Vydareny KH. Advancement of Women in Academic Radiology. *Invest Radiol*. 1987. https://journals.lww.com/investigativeradiology/Citation/1987/05000/Advancement_of_Women_in_Academic_Radiology.15.aspx. Accessed December 25, 2020.

201. Wiler JL, Rounds K, McGowan B, Baird J. Continuation of Gender Disparities in Pay Among Academic Emergency Medicine Physicians. Choo EK, ed. *Acad Emerg Med*. 2019;26(3):286-292. doi:10.1111/acem.13694

202. Wilkes FA, Akram H, Hyam JA, Kitchen ND, Hariz MI, Zrinzo L. Publication productivity of neurosurgeons in Great Britain and Ireland. *J Neurosurg*. 2015;122(4):948-954. doi:10.3171/2014.11.JNS14856

203. Wilkinson CJ, Linde HW. Status of women in academic anesthesiology. *Anesthesiology*. 1986;64(4):496-500. doi:10.1097/00000542-198604000-00014

204. Willett LL, Halvorsen AJ, McDonald FS, Chaudhry SI, Arora VM. Gender differences in salary of internal medicine residency directors: A national survey. *Am J Med*. 2015;128(6):659-665. doi:10.1016/j.amjmed.2015.02.002

205. Wilson LD, Haffty BG, Smith BD. A profile of academic training program directors and chairs in radiation oncology. *Int J Radiat Oncol Biol Phys*. 2013;85(5):1168-1171. doi:10.1016/j.ijrobp.2012.10.035

206. Winch R, McColgan M, Modi N, Greenough A. Comparison of UK paediatric consultants' participation in child health research between 2011 and 2015. *Arch Dis Child*. 2017;102(8):702-706. doi:10.1136/archdischild-2016-312129

207. Wise MR, Shapiro H, Bodley J, et al. Factors Affecting Academic Promotion in Obstetrics and Gynaecology in Canada. *J Obstet Gynaecol Canada*. 2004;26(2):127-136. doi:10.1016/S1701-2163(16)30488-1

208. Allison Woods L, Fox Wetle T, Sharkey KM. Why Aren't More Women in Academic Medicine Reaching the Top? doi:10.1186/s12889-017-4937-9

209. Woodward Z, Rodriguez Z, Jou JH, et al. Gender disparities in gastroenterology fellowship director positions in the United States. *Gastrointest Endosc*. 2017;86(4):595-599. doi:10.1016/j.gie.2017.01.019

210. Wright AL, Ryan K, St. Germain P, Schwindt L, Sager R, Reed KL. Compensation in academic medicine: Progress toward gender equity. *J Gen Intern Med*. 2007;22(10):1398-1402. doi:10.1007/s11606-007-0329-7

211. Wu B, Bhulani N, Jalal S, Ding J, Khosa F. Gender Disparity in Leadership Positions of General Surgical Societies in North America, Europe, and Oceania. *Cureus*. 2019;11(12). doi:10.7759/cureus.6285

212. Xierali IM, Nivet MA, Syed ZA, Shakil A, Schneider FD. Trends in Tenure Status in Academic Family Medicine, 1977-2017: Implications for Recruitment, Retention, and the Academic Mission. *Acad Med*. 2020;95(2):241-247. doi:10.1097/ACM.0000000000002890

213. Yang HY, Rhee G, Xuan L, Silver JK, Jalal S, Khosa F. Analysis of H-index in Assessing Gender Differences in Academic Rank and Leadership in Physical Medicine and Rehabilitation in the United States and Canada. *Am J Phys Med Rehabil*. 2019;98(6):479-483. doi:10.1097/PHM.0000000000001129

214. Yasukawa K, Nomura K. The perception and experience of gender-based discrimination related to professional advancement among Japanese physicians. *Tohoku J Exp Med*. 2014;232(1):35-42. doi:10.1620/tjem.232.35

215. Yu PT, Parsa P V., Hassanein O, Rogers SO, Chang DC. Minorities struggle to advance in academic medicine: A 12-y review of diversity at the highest levels of America's teaching institutions. *J Surg Res*. 2013;182(2):212-218. doi:10.1016/j.jss.2012.06.049

216. Yue T, Khosa F. Academic Gender Disparity in Orthopedic Surgery in Canadian Universities. *Cureus*. 2020;12(3). doi:10.7759/cureus.7205

217. Zelnik Yovel D, Tal O, Dicker D, Elis A, Rapoport M. Chairing the internal medicine department- analysis of current state and future trends in Israel. *Isr J Health Policy Res*. 2018;7(1):73. doi:10.1186/s13584-018-0267-3

218. Zivin K, Brower KJ, Sen S, Brownlee RM, Gold KJ. Relationship Between Faculty Characteristics and Emotional Exhaustion in a Large Academic Medical Center. *J Occup Environ Med*. 2020;62(8):611-617. doi:10.1097/JOM.0000000000001898